FUNCTIONAL SPECIFICATION DOCUMENT

E-skolarian: PUPSRC Organization Document Management System (ODMS) - Dashboard and Settings Module

DOCUMENT VERSION 3.0

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DOCUMENT HISTORY

Date	Version	Document Revision Description	Document Author
April 20, 2025	1.0	Initial FSD Document Mostly Introduction	Jonlee Altar
April 21, 2025	1.1	System Configurations, Change of Impacts and Dependencies, and Scope of the Documents has been added to the FSD	Jonlee Altar, Earl Lance Dakila, Eurika Angela Bialen
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April 23, 2025	1.3	Added mostly of the diagrams (System Flow, Business Flow, Data Flow, Context Flow, Application Flow, Sitemap)	Jamil Aaron Carapatan, Jeruh Fornal, Jonlee Altar
April 25, 2025	1.4	Revised some other parts of the contents	Jamil Aaron Carapatan, Jeruh Fornal
April 28, 2025	1.5	Updated use case diagram to include new use cases and Revised the SFDs	Jeruh Fornal
May 10, 2025	2.0	Added Settings Module, Revised Data Flow Diagrams (DFDs) and Business Process Flows (BPFs), Added DFD and BPF for the Settings Module, Revised the Entity Relationship Diagram (ERD), Replaced the module-specific sitemap with a system-wide sitemap	Jonlee Altar, Jeruh Fornal

May 11, 2025	2.1	Updated Introduction, Revised Sections 1.1 (Purpose), 1.2 (Project Scope), 1.3 (Scope of the Document), and 1.6 (Risks and Assumptions) to reflect added Settings Module, Revised SFD and added diagrams based on the changes	Jonlee Altar, Jeruh Fornal
May 13, 2025	2.2	Added Dashboard and Setting's Sequence Diagram (7. System Integration	Earl Dakila
May 18, 2025	2.3	Revised SFDs for Settings Processes, Correct SFD Symbols, Updated Section 1.6 (Risks and Assumptions)	Jonlee Altar, Jeruh Fornal
May 23, 2025	2.4	Added Use Case Diagrams (UCDs) and System Flow Diagrams (SFDs) for Archive, Restore, and Delete Announcement processes, Revised Context Flow Diagram (CFD) and Business Process Flow Diagram (BPF), Revised Entity Relationship Diagram (ERD), Adjustments on System Configurations part	Jeruh Fornal, Earl Dakila, Jonlee Alter
May 31, 2025	2.5	Added contents in 7.1 (Exception Handling/Error Reporting)	Eurika Angela Bialen
June 1, 2025	2.6	Revised Context Flow Diagram, Data Flow Diagrams (DFDs), System Flow Diagram, and Use Case Diagram (Section 2.1); added Business Process Flow Diagram for Super Admin; replaced "Student" with "Student Organization"; corrected arrow labels and updated data store numbering and labeling; revised Section 1 including 1.1 Purpose of the Document, 1.2 Project Scope and Scope of the Document, and 1.6 Risk Assessment.	Jeruh Fornal, Jonlee Altar
June 2, 2025	2.7	Revised Application Flow Diagram	Jamil Aaron Carapatan
June 7, 2025	2.8	Revised Context Flow Diagram (CFD) and Data Flow Diagram (DFD) due to the addition of Super Admin as an entity, Revised Business Process Flow Diagram (BPF) and System Flow Diagram (SFD) to reflect added processes and changes based on module access updates.	Jeruh Fornal, Jonlee Altar
June 12, 2025	2.9	Update Integration Requirements Description	Jamil Aaron Carapatan
June 13, 2025	3.0	Update All the Diagrams and Revised Other Sections	Jeruh Fornal, Jonlee Altar

APPROVALS

Approval Date	Approved Version	Approver Role	Approver
April 20, 2025	1.0	Project Manager	Jonlee Altar
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April 25, 2025	1.4	Project Manager	Jonlee Altar
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May 10, 2025	2.0	Project Manager	Jonlee Altar
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June 1, 2025	2.5	Project Manager	Jonlee Altar
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Table of Contents

1. Introduction..... 6
1.1 Purpose of the document.....7
1.2 Project Scope.....7
1.3 Scope of the document..... 7
1.4 Related documents.....8
1.5 Terms/Acronyms and Definitions.....9
1.6 Risks and Assumptions..... 11
2. System/ Solution Overview..... 12
2.1 Business Process Flow, Context Diagram/ Data Flow Diagram,/ Application Screen Flow/, Sitemap, Process Flow.....12
2.2 System Actors.....66
2.3 Dependencies and Change Impacts.....66
3. Functional Specifications..... 67
4. System Configurations.....89
5. Other System Requirements/ Non-Functional Requirements.....92
6. Reporting Requirements..... 93
7. Integration Requirements..... 93
7.1 Exception Handling/ Error Reporting.....97
8. Data Migration/ Conversion Requirements..... 98
8.1 Data Conversion Strategy..... 98
8.2 Data Conversion Preparation.....99
8.3 Data Conversion Specifications.....99
9. References.....100
10. Open Issues.....101
Appendix.....101

1. Introduction

The Dashboard and Settings Module serves as a vital component of the E-skolarian: PUPSRC Organization Document Management System (ODMS) at the Polytechnic University of the Philippines Santa Rosa Campus, providing student Organization, Campus administration, and super admin with a centralized, of document submission activities. Through clearly labeled indicators such as Pending, Under Review, Approved, and Total Documents, the module allows users to monitor the progress of the submissions and stay informed about the compliance status, effectively addressing previous inefficiencies caused by disorganized manual uploads and fragmented tracking methods.

For student organizations either academic or non-academic organizations, the module offers an accessible interface that presents recent document Submitted, and announcement updates in a user-friendly format. By replacing reliance on external platforms like Google Drive with an integrated space for managing requirements, it helps students track deadlines and status updates more efficiently, reducing delays and improving the accuracy of institutional reporting.

Campus Administration benefits from enhanced functionalities, including the ability to post announcements directly through the system, ensuring timely communication of important updates on submission deadlines or procedural changes. Built-in analytics provide insights into submission trends and overall document flow, complementing the existing Electronic Quarterly Accomplishment Report (EQAR) system and supporting more transparent and efficient institutional reporting.

The super admin can also post announcements, just like the campus administration, to remind students Organization about deadlines or documents that need to be submitted. In doing so, the super admin plays a key role in reinforcing communication across all user levels. Their announcements can cover system-wide reminders, urgent notices, or important updates that require immediate attention. This helps ensure that all student organizations are aligned with submission schedules and institutional requirements. The super admin's ability to broadcast announcements enhances coordination, minimizes missed deadlines, and supports the smooth flow of document-related activities within the system.

In addition to its dashboard and analytics capabilities, the module features a Settings section where student Organizations, Campus administration, and super admin can manage their profiles and update account passwords. This addition enhances user control and account security by allowing users to personalize their information and maintain secure access, also by adding recovery email it furthers the security of each user account. By combining monitoring, streamlined communication, analytical insights, and account management tools, the Dashboard and Settings Module fosters a more transparent, efficient, and student-inclusive approach to academic document management.

1.1 Purpose of the document

This Functional Specification Document (FSD) defines the required features, behavior, and design of the Dashboard and Settings Module in the E-skolarian: PUPSRC Organization Document Management System (ODMS). The purpose of this document is to ensure that all stakeholders including developers, testers, and university administrators share a unified understanding of the module's intended functionality and goals.

This module focuses on enabling real-time tracking of document submissions from student organizations and academic departments. It also includes user account management features, such as profile update, password changes and also the recovery email. By documenting these functionalities, the FSD ensures a streamlined development process and supports the improvement of transparency, accountability, and operational efficiency within the university's document workflow.

1.2 Project Scope

The project targets the development and deployment of a Dashboard and Settings Module to enhance the document submission process at the Polytechnic University of the Philippines, Santa Rosa Campus. It addresses the limitations of prior manual systems by offering both student users and administrative personnel a centralized, interactive interface to track document statuses, such as Pending, Under Review, Approved, and Total Documents In real time.

The scope includes key functionalities like viewing submission history, receiving system announcements, and managing user accounts. For student users, the module promotes autonomy and clarity in handling submission responsibilities. For administrative users, it provides tools to communicate updates, track compliance rates, and manage document workflows effectively.

1.3 Scope of the document

This document outlines the technical and functional specifications of the Dashboard and Settings Module, which is a core component of the ODMS. It details the features available to student Organizations, Campus administration, and Super admin at PUP Santa Rosa Campus. The module provides role-based dashboards that allow users to track the status of document submissions in real time. It includes an announcement feature that enables both Campus administration and Super admin to communicate important information such as deadlines, requirements, and system updates.

The Settings section allows users to manage their profiles , add recovery email, and update passwords, supporting secure and personalized access. In addition, super admin are equipped with tools to manage system-wide announcements. This Functional Specification Document serves as a guide for the development and testing of the module, ensuring that it supports a transparent, secure, and organized document management process.

1.4 Related documents

Component	Name (with link to the document)	Description
Concept Paper	*Concept Paper*	Provides the background, objectives, scope, and goals of the ODMS project.
Project Charter	*Project Charter*	Outlines the project's authority, objectives, team roles, and key milestones.
Roadmap	*Roadmap*	High-level overview of feature releases and project development stages
Gantt Chart	*Gantt Chart*	Visual project timeline including tasks, sprints, and deadlines.
UI/UX Wireframes	*UI/UX Wireframes*	Initial layout and design structure of the module's interface.
Bug Testing Worksheets	*Bug Testing Worksheets*	Bug Testing Worksheet is a tool used during software testing to record, track, and manage defects in module features, helping the team

		monitor and resolve issues efficiently.
Tech Stack	*Tech Stack*	Propose and finalize the technologies (frontend, backend, database, tools) that will be used for system development, subject to approval.

1.5 Terms/Acronyms and Definitions

Term/Acronym	Definition	Description
E-skolarian	The name of the system/module – "E-skolarian: PUPSRC ODMS."	A digital platform designed to manage and analyze student organization document submissions at PUP Santa Rosa Campus.
ODMS (Organization Document Management System)	Organization Document Management System	It streamlines the process for student organizations and admin staff, replacing manual uploads with a centralized system.
Dashboard	A graphical interface that displays data in visual formats such as charts and tables.	The core of the module, enabling users to monitor key data points and performance indicators in a user-friendly layout.
Campus Administration	Administrators	Manages users, configures dashboard settings, and has access to all modules and data views.

Stakeholder	A person or group with an interest in the development or outcome of the system.	Includes users, managers, and clients who interact with or are impacted by the analytics dashboard.
Module	A distinct and independent part of a software application.	The analytics dashboard is a module focused on presenting and analyzing key business data.
Analytics Module	A component of a software system that focuses on data collection, processing, and visualization.	Provides users with meaningful insights through interactive and filterable dashboards.
Role-based Access	A security model that assigns permissions based on user roles.	Controls who can view, edit, or manage certain parts of the dashboard, ensuring data integrity and confidentiality.
Real-time Data	Data that is available immediately after it is collected	Allows users to see up-to-date information on the dashboard for timely decision-making.
Mock-up	A visual prototype or draft of an interface design.	Used during the design phase to demonstrate layout and functionality before actual development begins.
EQAR	Electronic Quarterly Accomplishment Report	A digital reporting tool used for submitting organizational accomplishments on a quarterly basis.
Laravel	A PHP-based web application framework.	Used to build the backend of the ODMS, supporting routing, data handling, and logic execution.
Tailwind CSS	A utility-first CSS framework for building custom user interfaces.	It was used to design a responsive front-end for both admin and student dashboards.
MySQL	Standard Query Language	An open-source relational database management system
XAMPP	A free, open-source web server solution stack.	Used locally to develop and test the Laravel application with Apache and MySQL.
UI/UX Wireframes	UI is the visual layout of an application; UX is how a user experiences and interacts with it.	In this system, UI/UX design focuses on creating an intuitive and seamless interface for users to navigate dashboards and interpret analytics.

FSD	Functional Specification Document	It acts as a blueprint for developers and stakeholders to ensure alignment on system expectations, user needs, and technical specifications.
PUPSRC	Polytechnic University of the Philippines- Santa Rosa Campus	The Polytechnic University of the Philippines (PUP) is a government educational institution governed by Republic Act Number 8292 known as the Higher Education Modernization Act of 1997, and its Implementing Rules and Regulations contained in the Commission on Higher Education Memorandum Circular No. 4, series 1997.

1.6 Risks and Assumptions

The implementation of the E-skolarian Dashboard and Settings Module comes with several key assumptions and potential risks that may affect its performance, reliability, and user experience. It is assumed that the system will consistently have access to real-time data, stable internet connectivity, and adequate server resources to handle multiple users accessing the dashboard simultaneously. The module also assumes that user roles student, administrator, and super administrator are correctly configured and maintained to ensure that each user can only access features appropriate to their role. Successful implementation also relies on the ongoing cooperation of administrative staff to accurately manage, review, and validate submitted document records.

For the Settings section, it is assumed that users will actively maintain their profile information and practice secure account management, such as remembering credentials and using strong, unique passwords when updating accounts. The effectiveness of the module depends in part on the users' adherence to these best practices.

However, several risks may arise. A primary concern is the potential for delayed or inaccurate updates to document statuses, which can confuse student organization leaders and administrators who depend on timely updates to meet submission deadlines. There is also a risk of information overload on the admin dashboard if too much data is displayed at once, possibly making it difficult for users to focus on relevant tasks. During periods of high usage, the system may experience slower performance, leading to delays in dashboard loading or accessing analytics features.

In the Settings module, risks include users entering incorrect or incomplete profile information, which may reduce data accuracy. Forgotten or mishandled login credentials may cause access issues or increase the number of support requests. Lastly, weak password practices or insufficient security measures could result in unauthorized access attempts, posing a potential security threat to user accounts and institutional data. These risks highlight the need for secure, responsive design and clear user guidelines throughout system deployment and use.

2. System/ Solution Overview

The E-skolarian Organization Document Management System improves workflow throughout the entire document management process, providing the organization representative and admin facilitators with real-time access to relevant information throughout the document management process. The dashboard optimizes this process for both student organization representatives and administrative facilitators, enhancing their efficiency and providing comprehensive information on document management performance indicators, including pending document count, documents under review, approved documents, and total documents. Furthermore, the announcement feature assists in coordinating student organization representatives with the administrator's announcement. This feature enables the administrator to publish formal announcements within the organization. Finally, the recent documents feature enables the user to view their latest documents that require updating or completion.

2.1 Business Process Flow, Context Diagram/ Data Flow Diagram,/ Application Screen Flow/, Sitemap, Process Flow

Figure 1. Context Flow Diagram Proposed

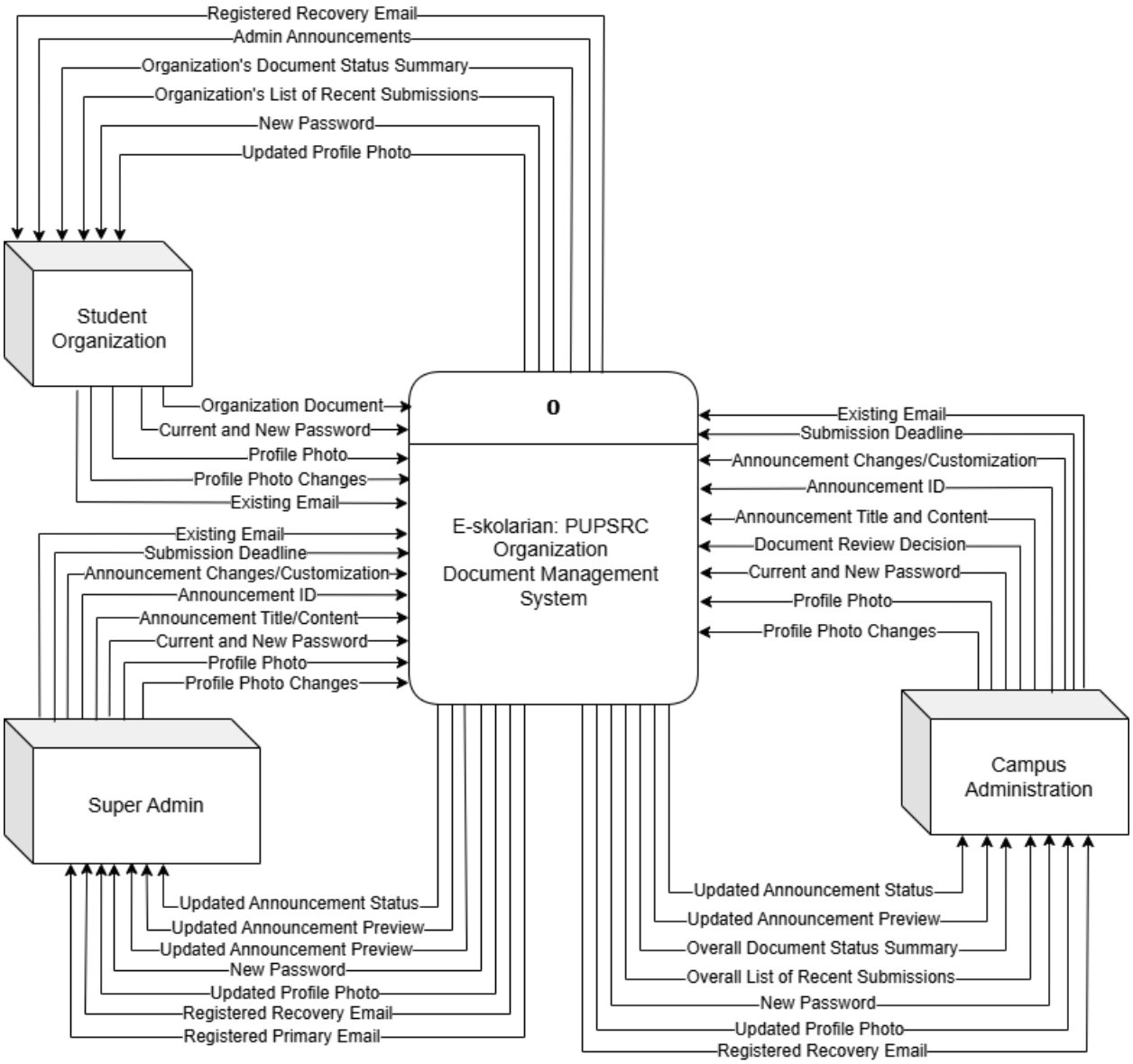


Figure 2. Data Flow Diagram Proposed – Dashboard Level 0

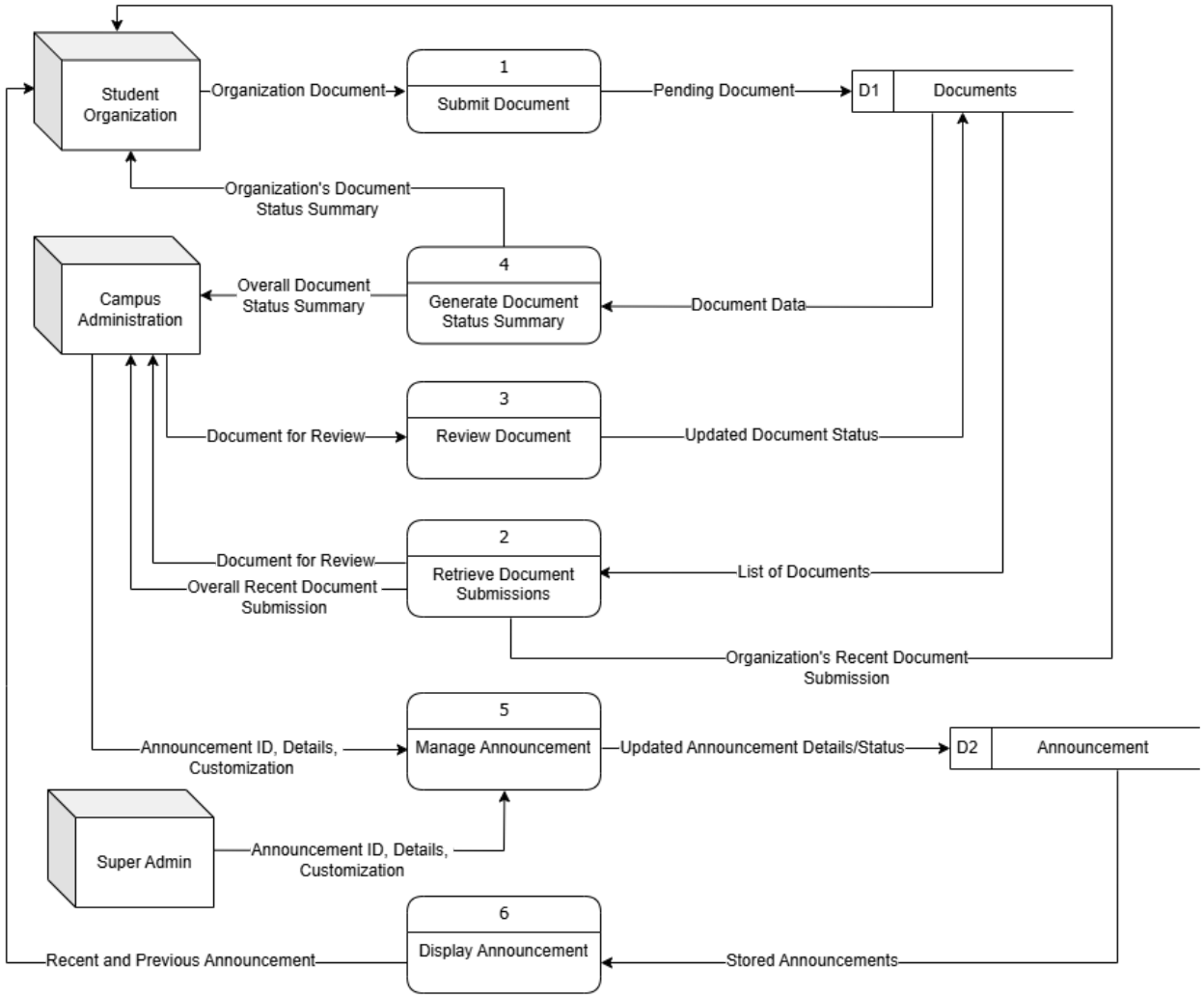


Figure 3. Data Flow Diagram Proposed – Generate Document Status Summary Level 1

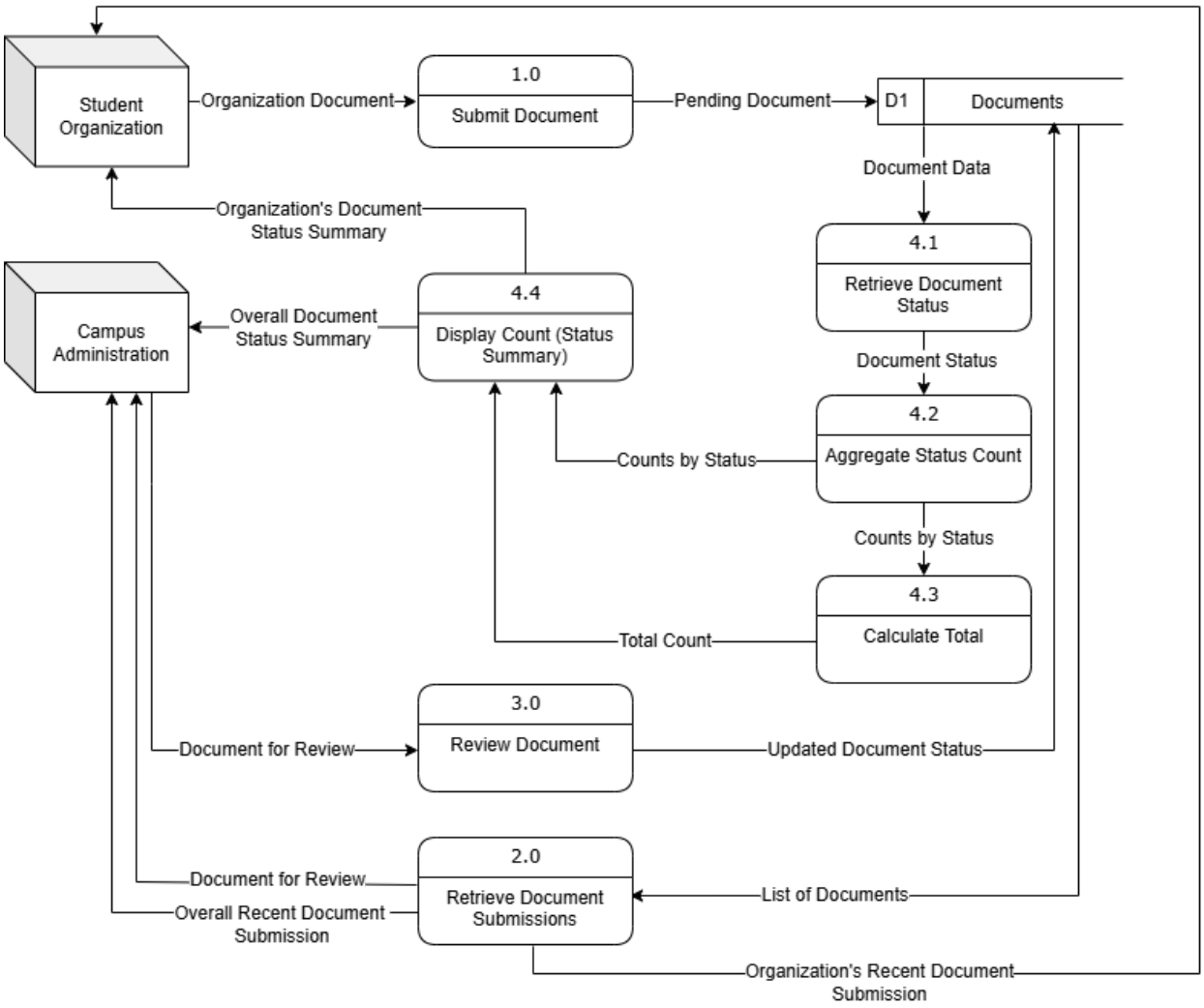


Figure 4. Data Flow Diagram Proposed – Super Admin Manage Announcement Level 1

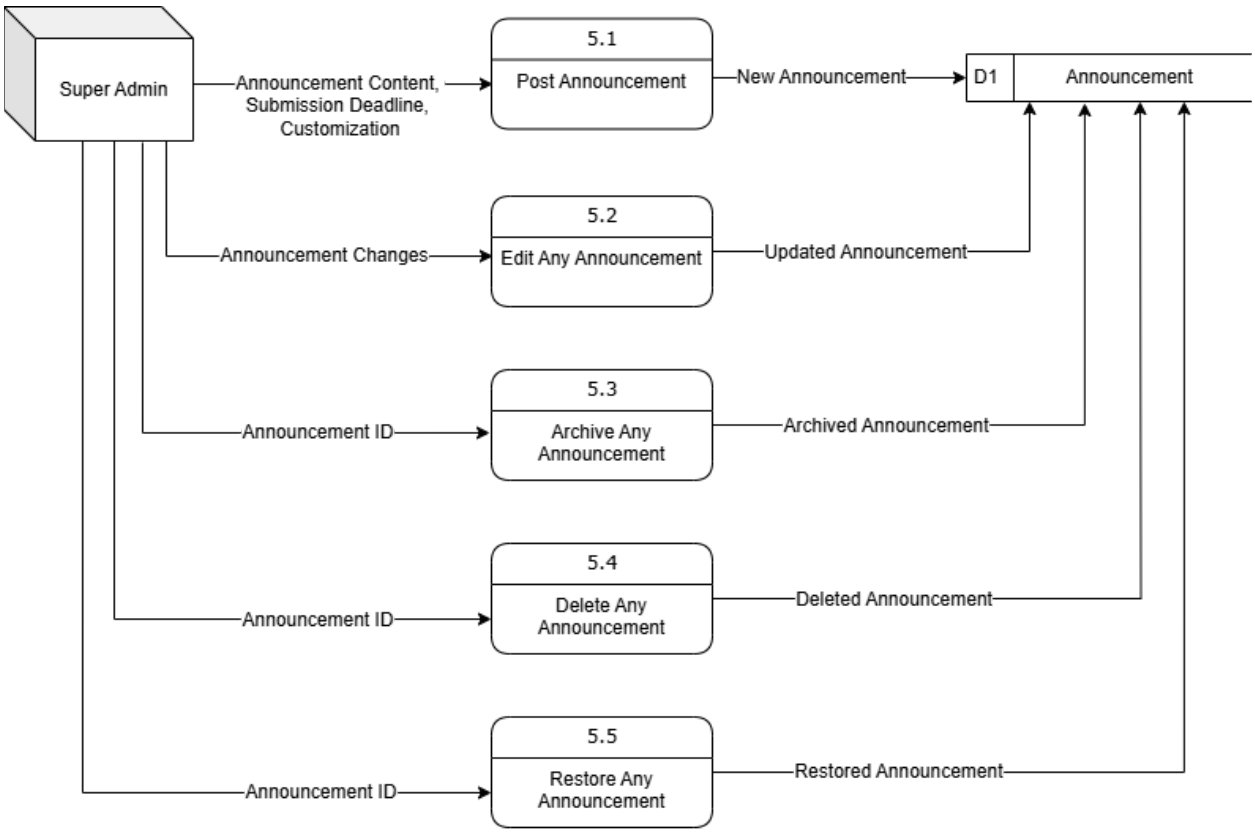


Figure 5. Data Flow Diagram Proposed – Campus Administration Manage Announcement Level 1

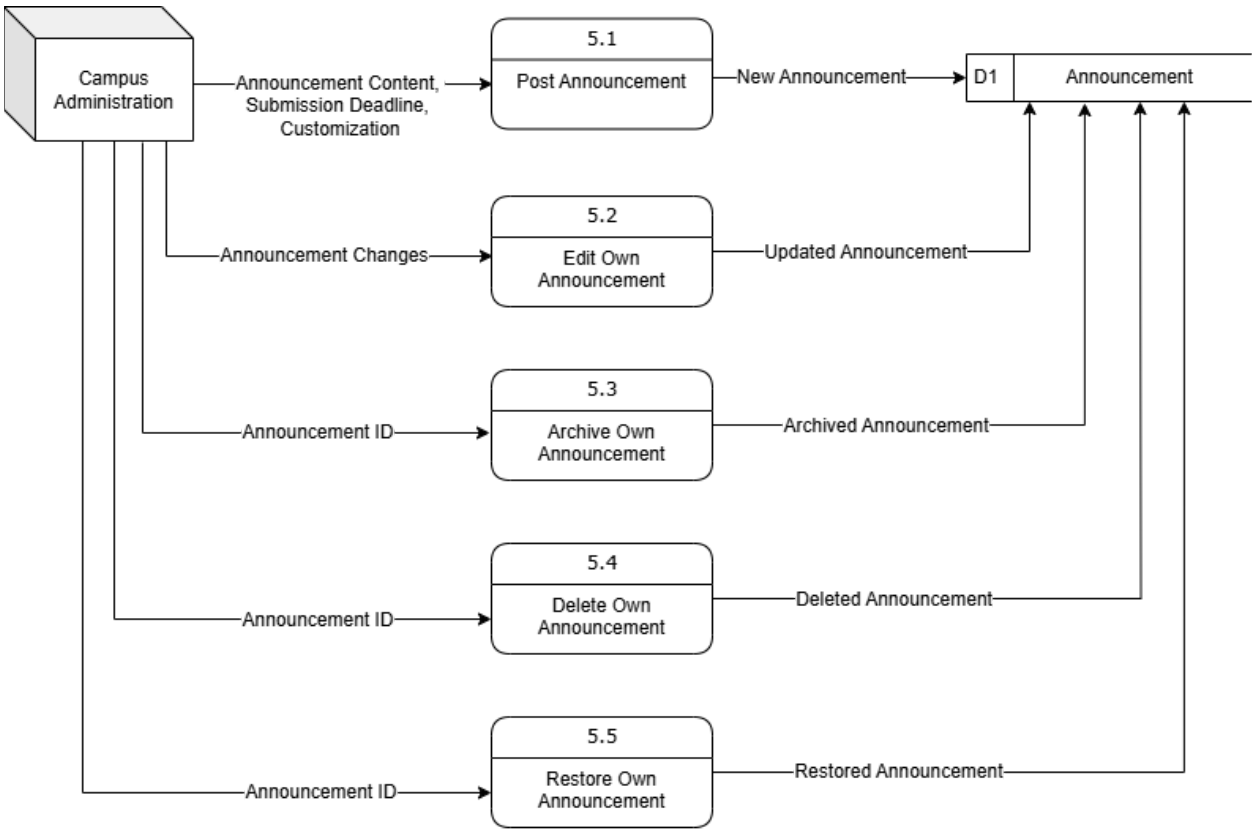


Figure 6. Data Flow Diagram Proposed – Super Admin Settings Level 0

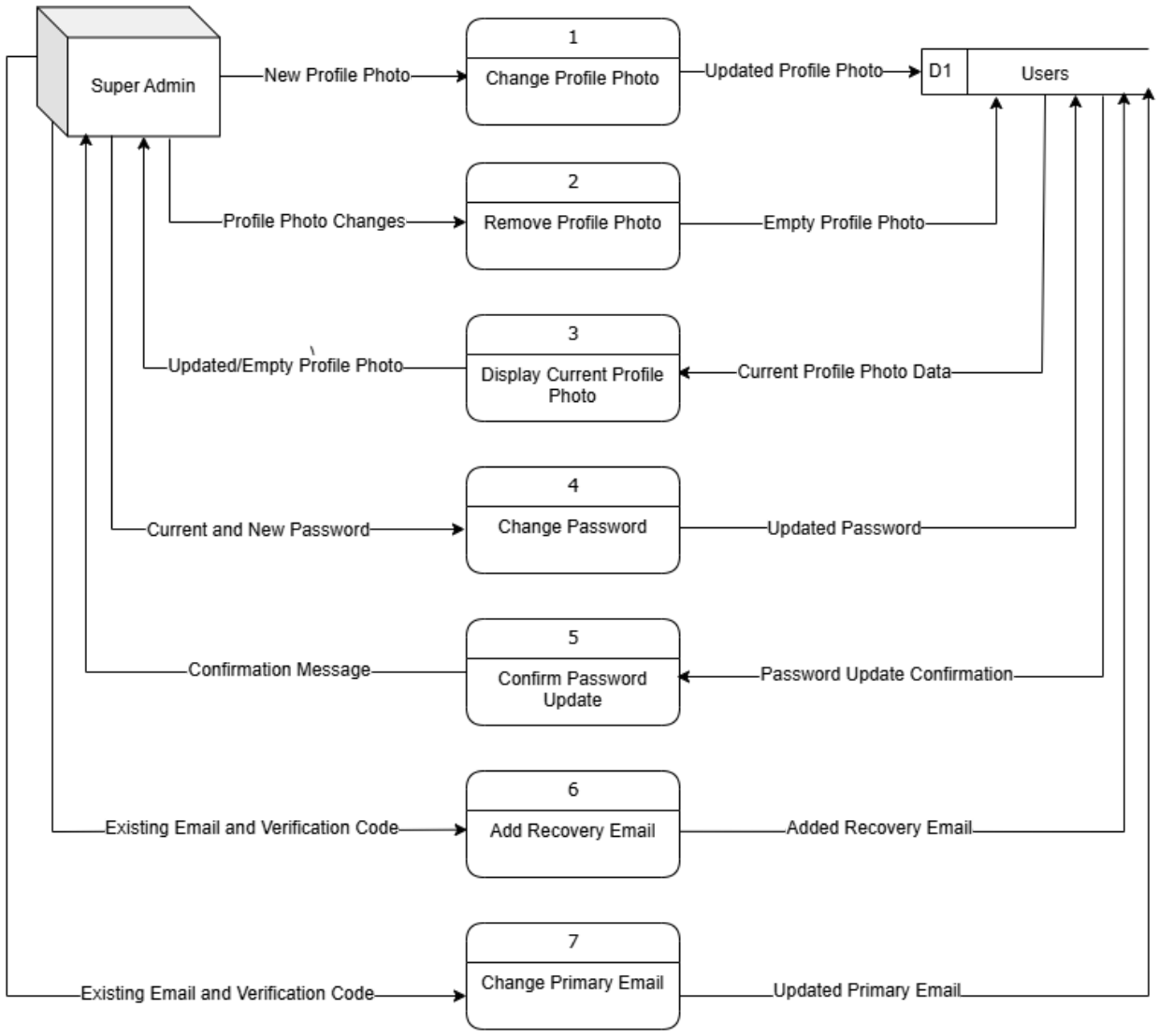


Figure 7. Data Flow Diagram Proposed – Super Admin Change Password Level 1

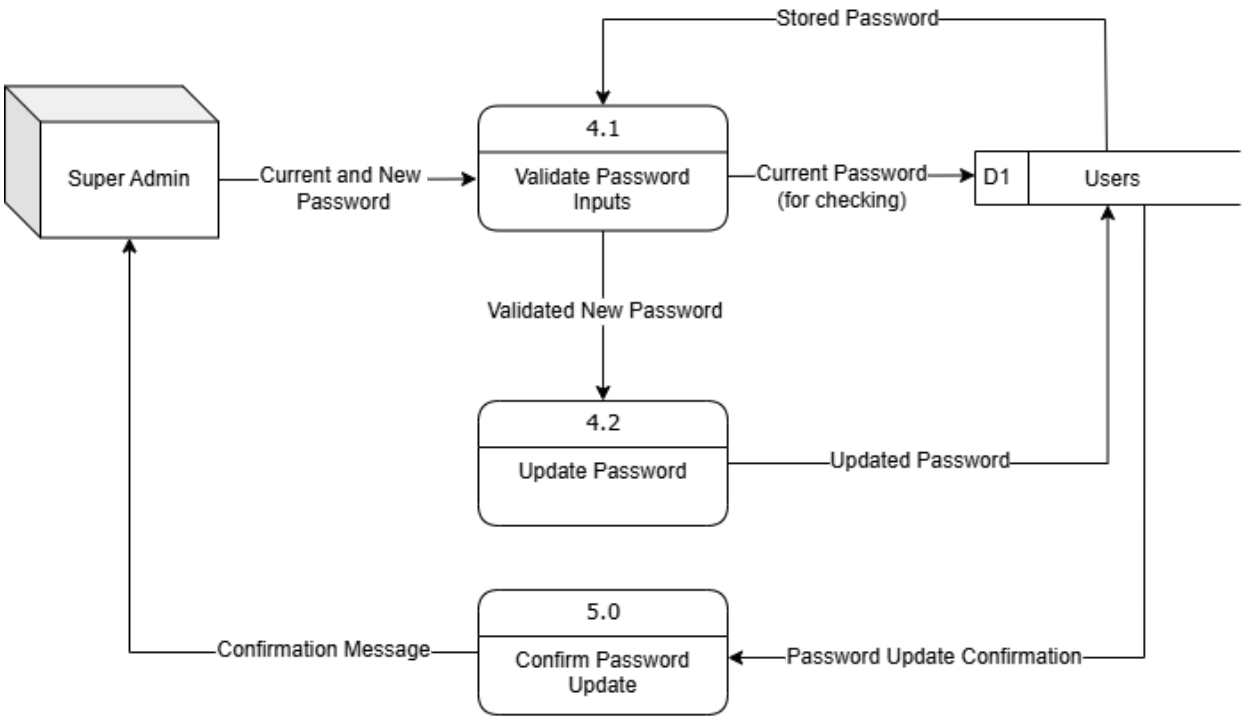


Figure 8. Data Flow Diagram Proposed – Campus Administration Settings Level 0

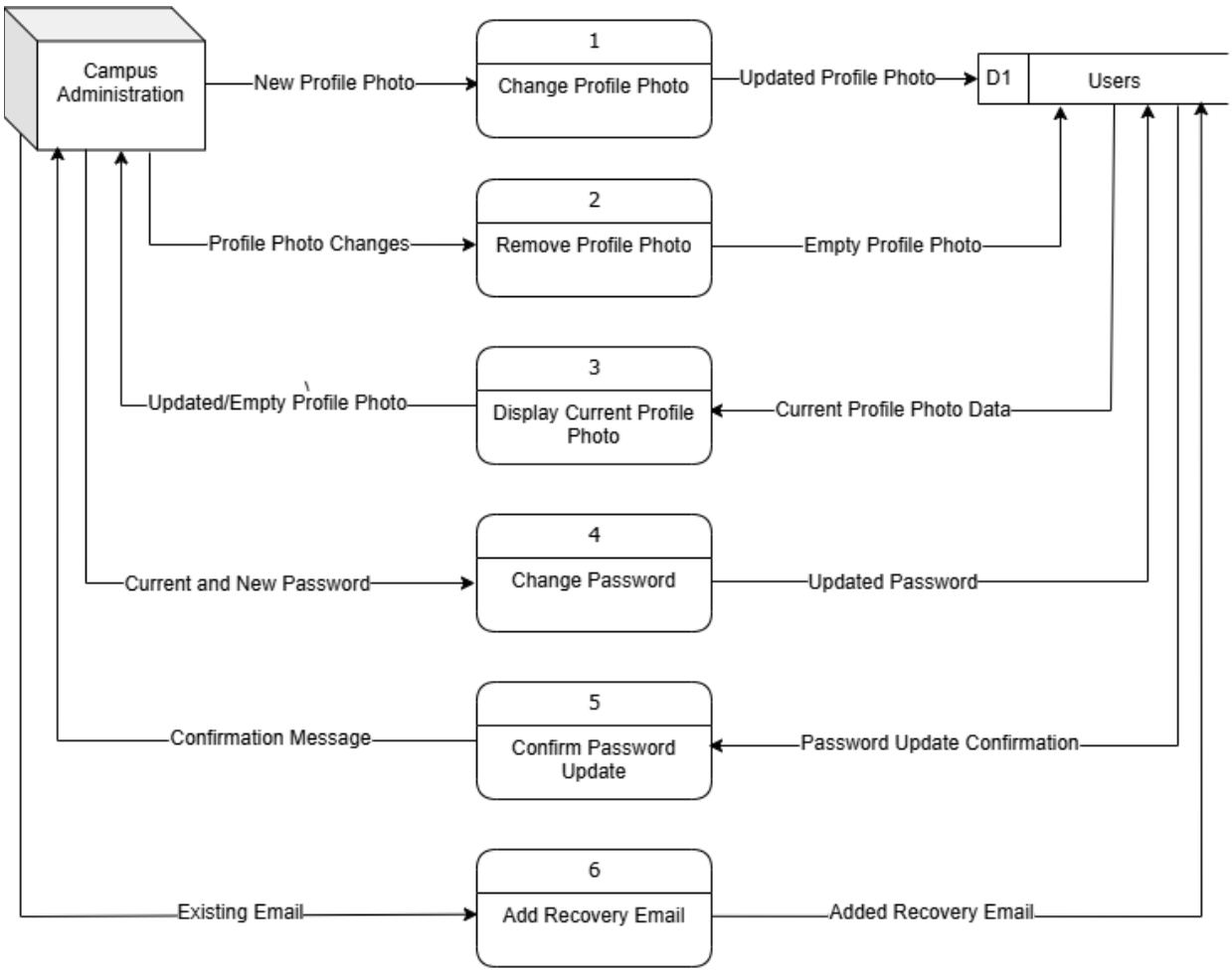


Figure 9. Data Flow Diagram Proposed – Campus Administration Change Password Level 1

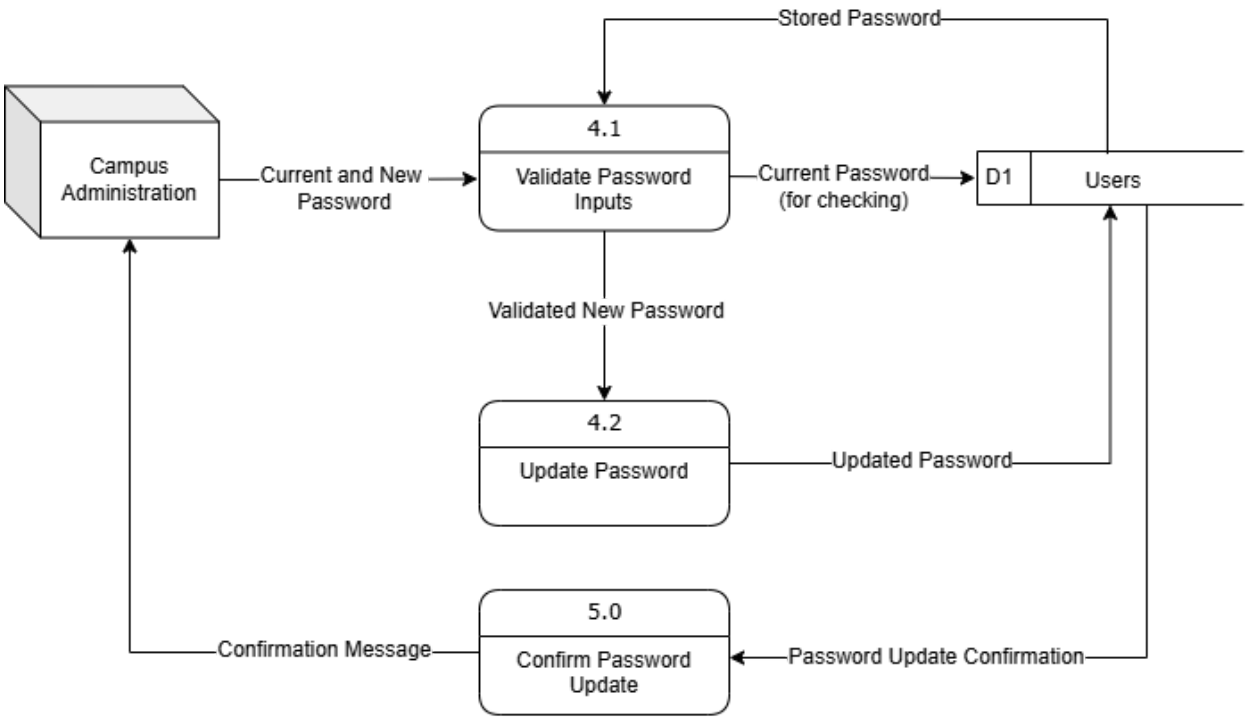


Figure 10. Data Flow Diagram Proposed – Student Organization Settings Level 0

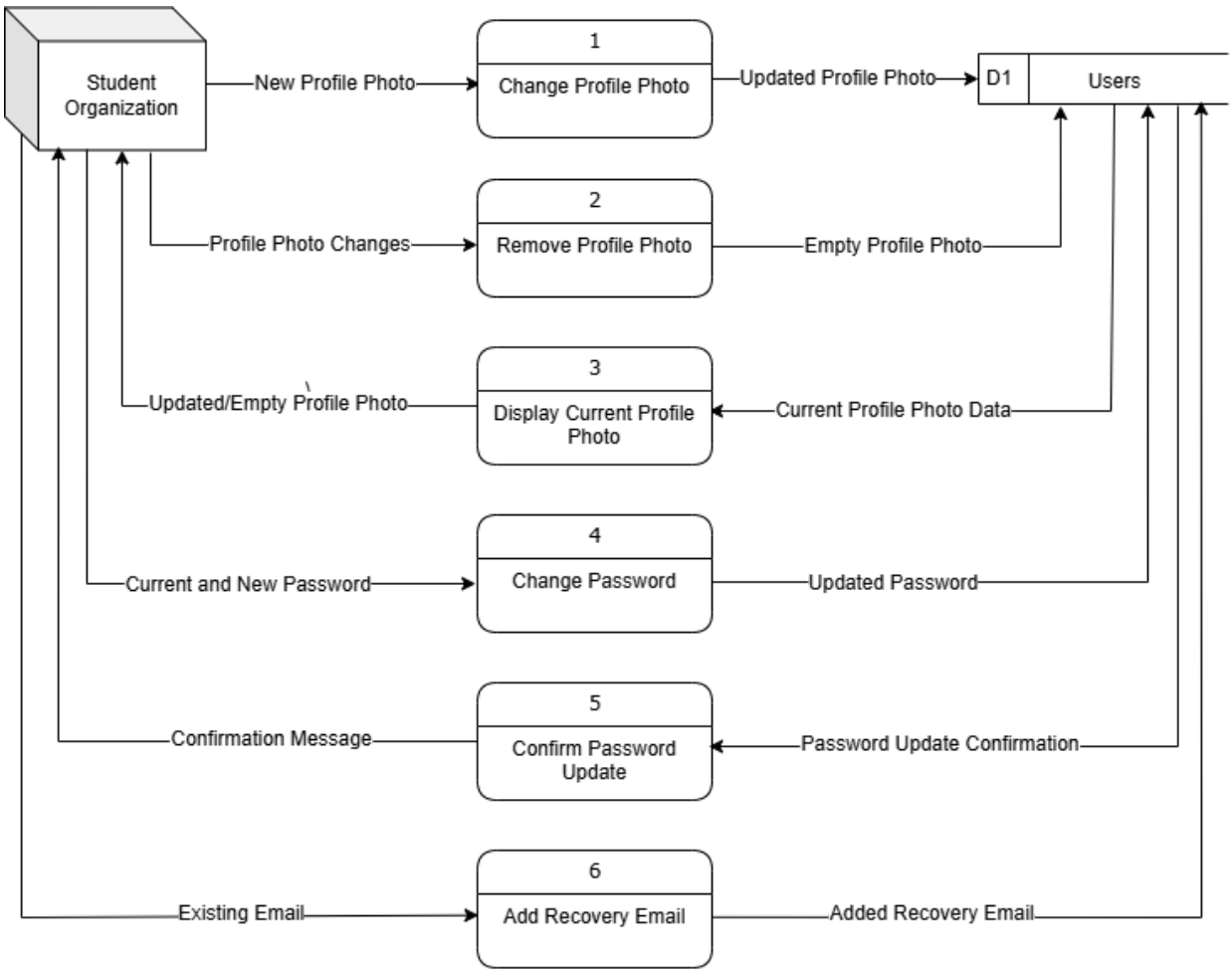


Figure 11. Data Flow Diagram Proposed – Student Organization Change Password Level 1

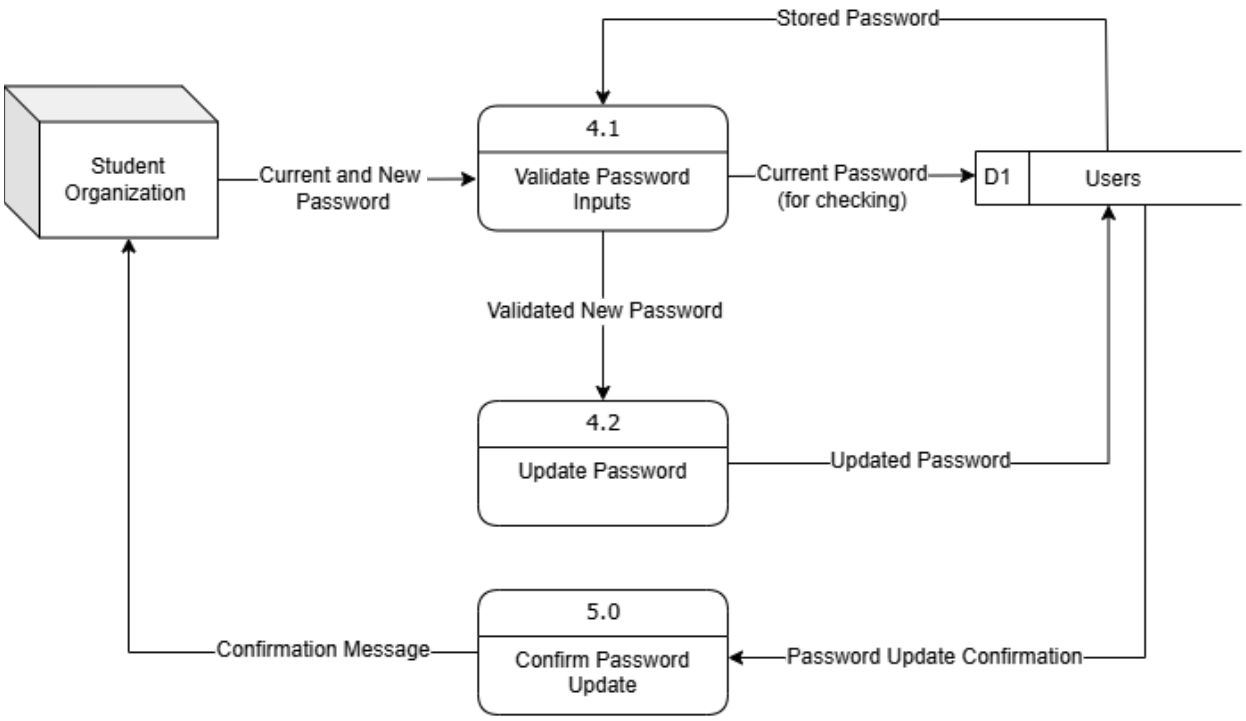


Figure 12. Entity Relationship Diagram

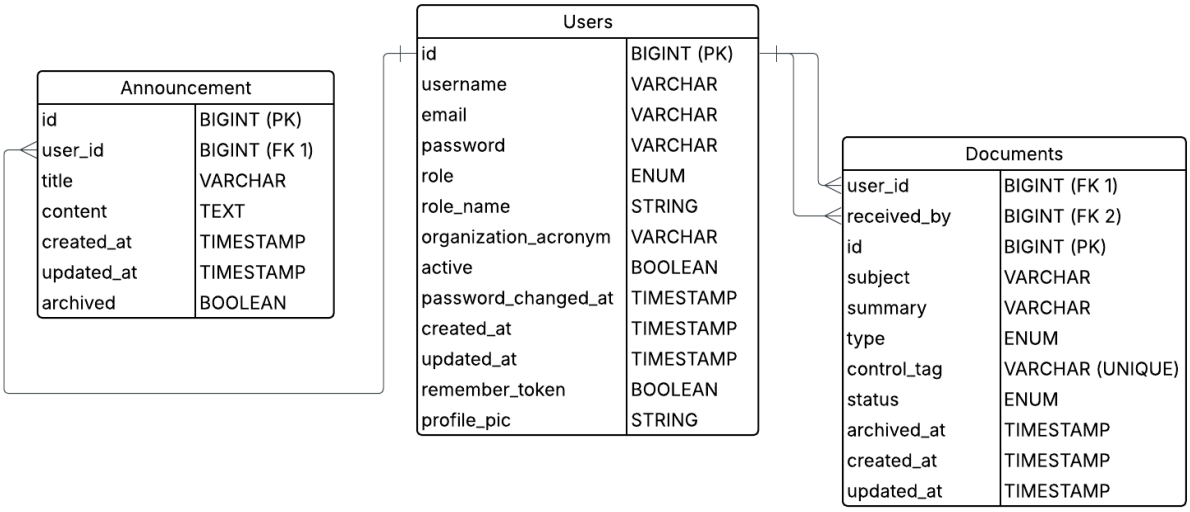


Figure 13. Business Process Flow Diagram Proposed – Super Admin Dashboard and Settings

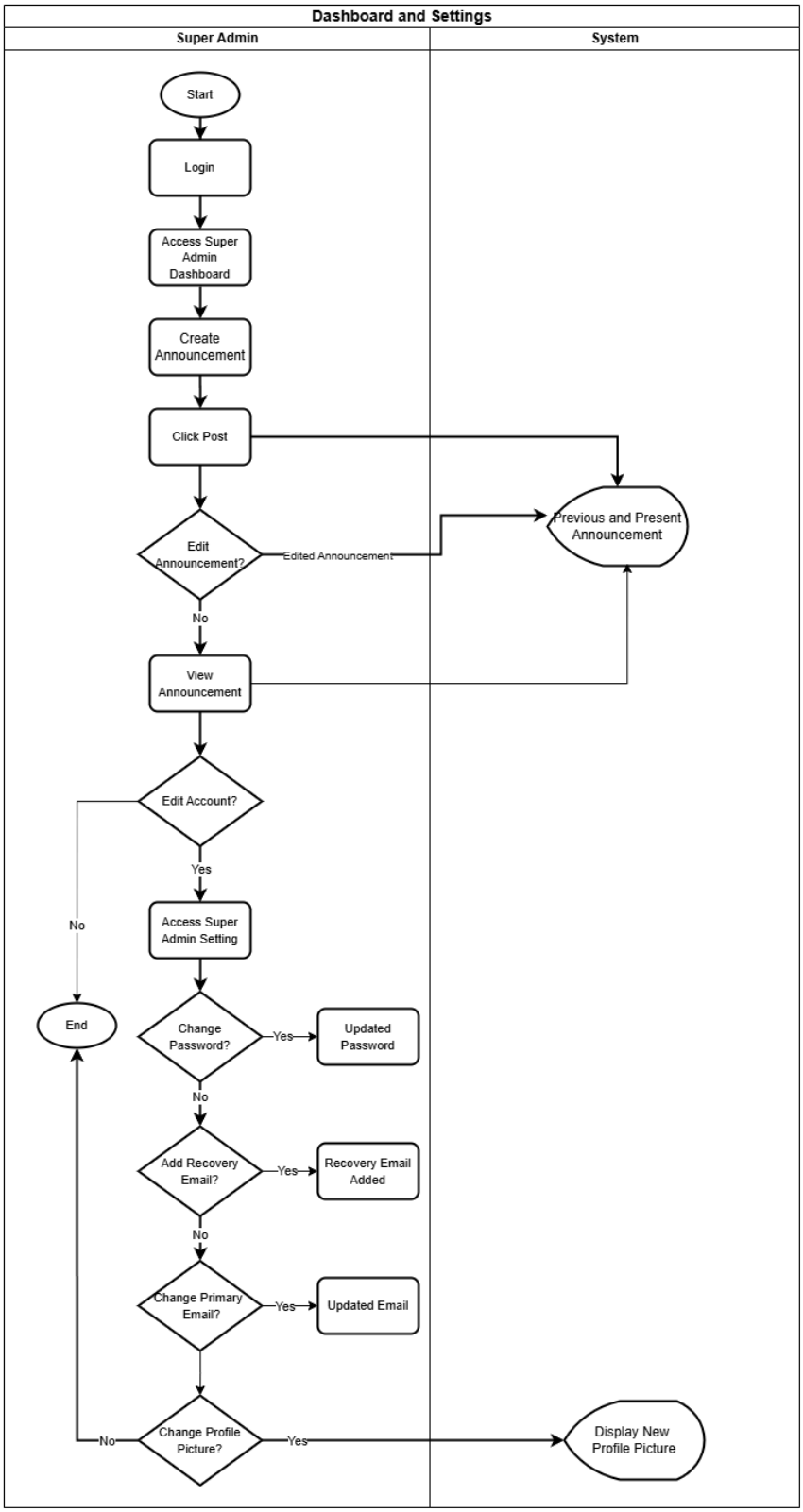


Figure 14. Business Process Flow Diagram Proposed – Campus Administration Dashboard and Settings

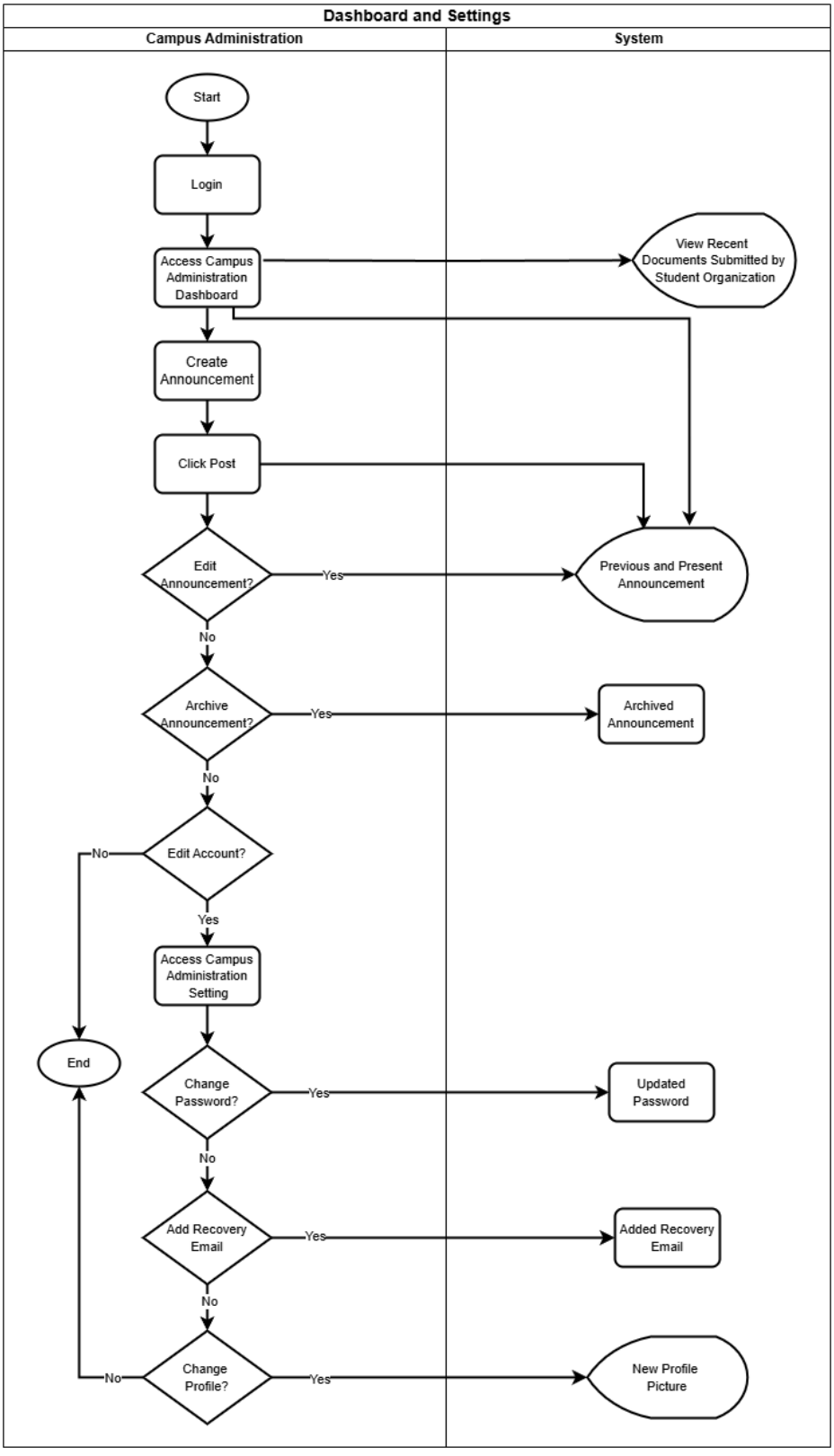


Figure 15. Business Process Flow Diagram Proposed – Student Organization Dashboard and Settings

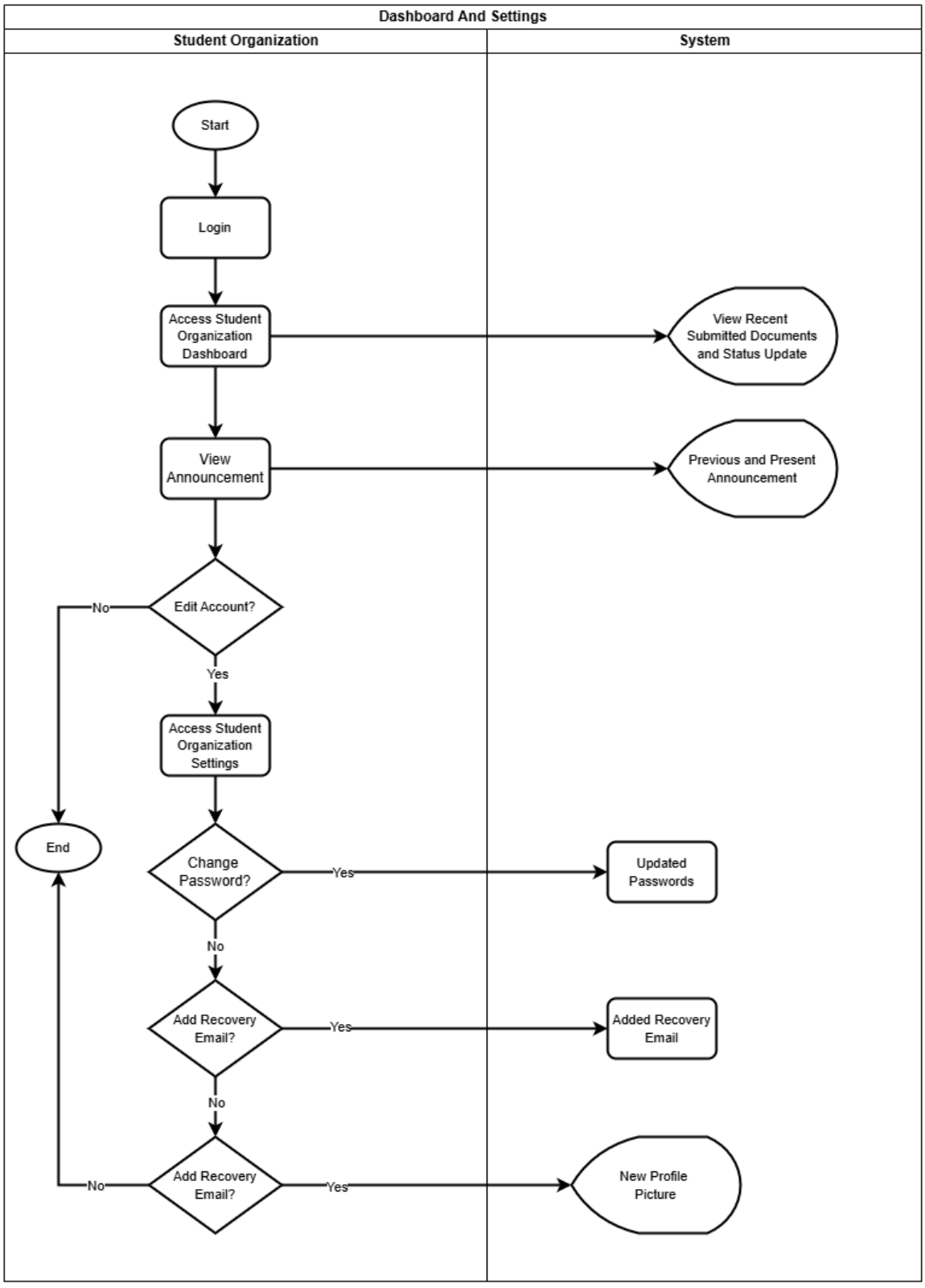


Figure 16. System Flow Diagram Proposed – Access Super Admin Dashboard

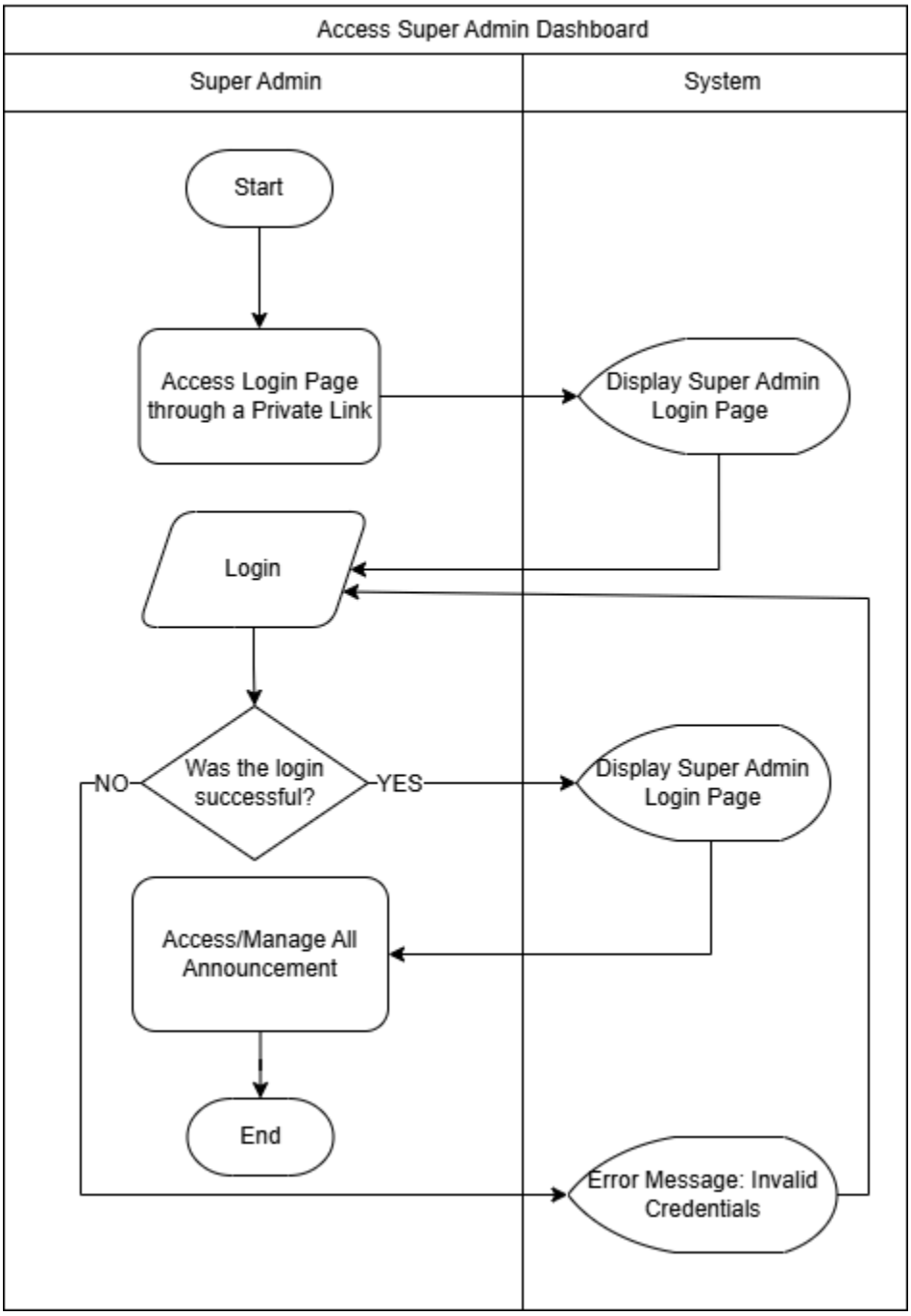


Figure 17. System Flow Diagram Proposed – Access Campus Administration Dashboard

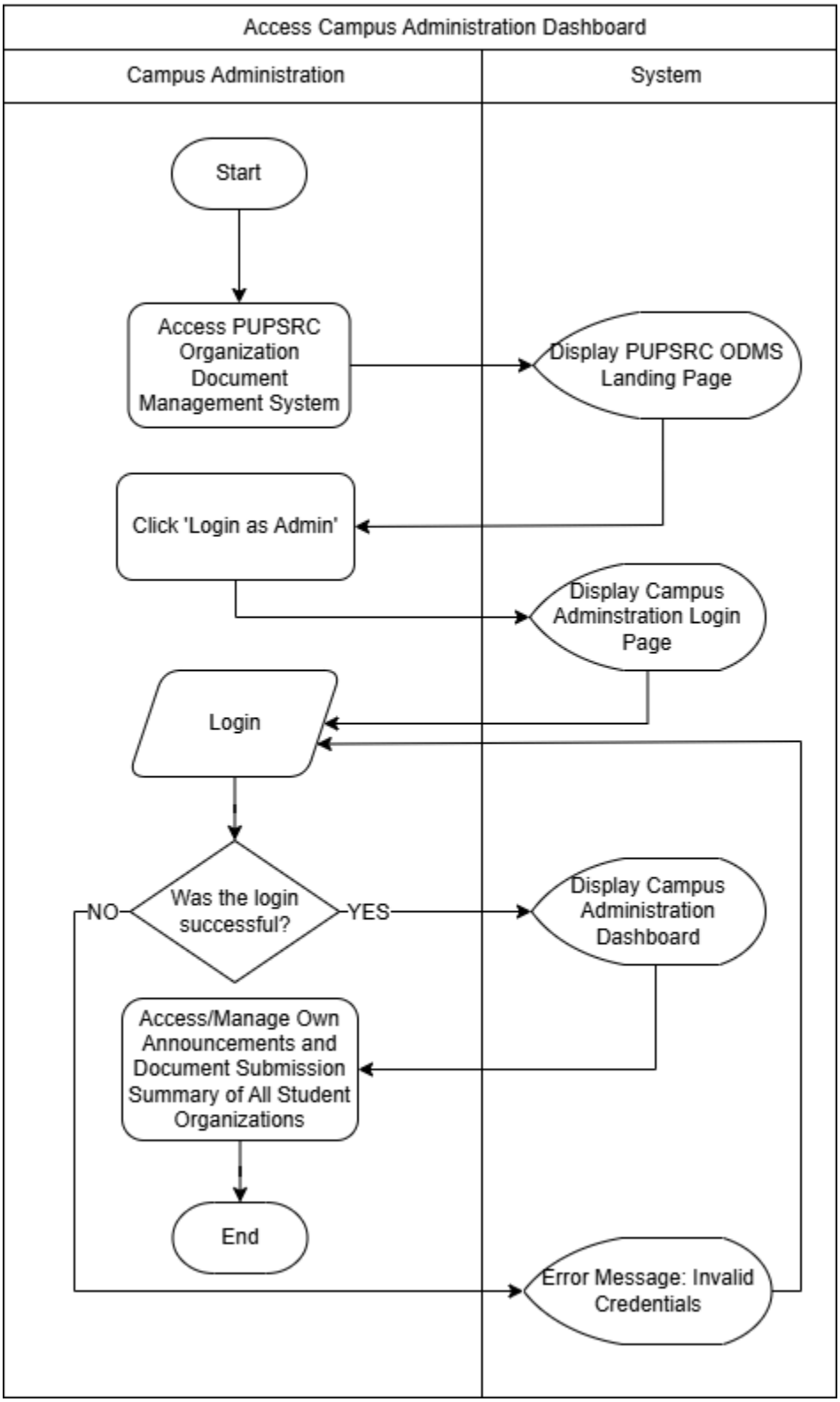


Figure 18. System Flow Diagram Proposed – Access Student Organization Dashboard

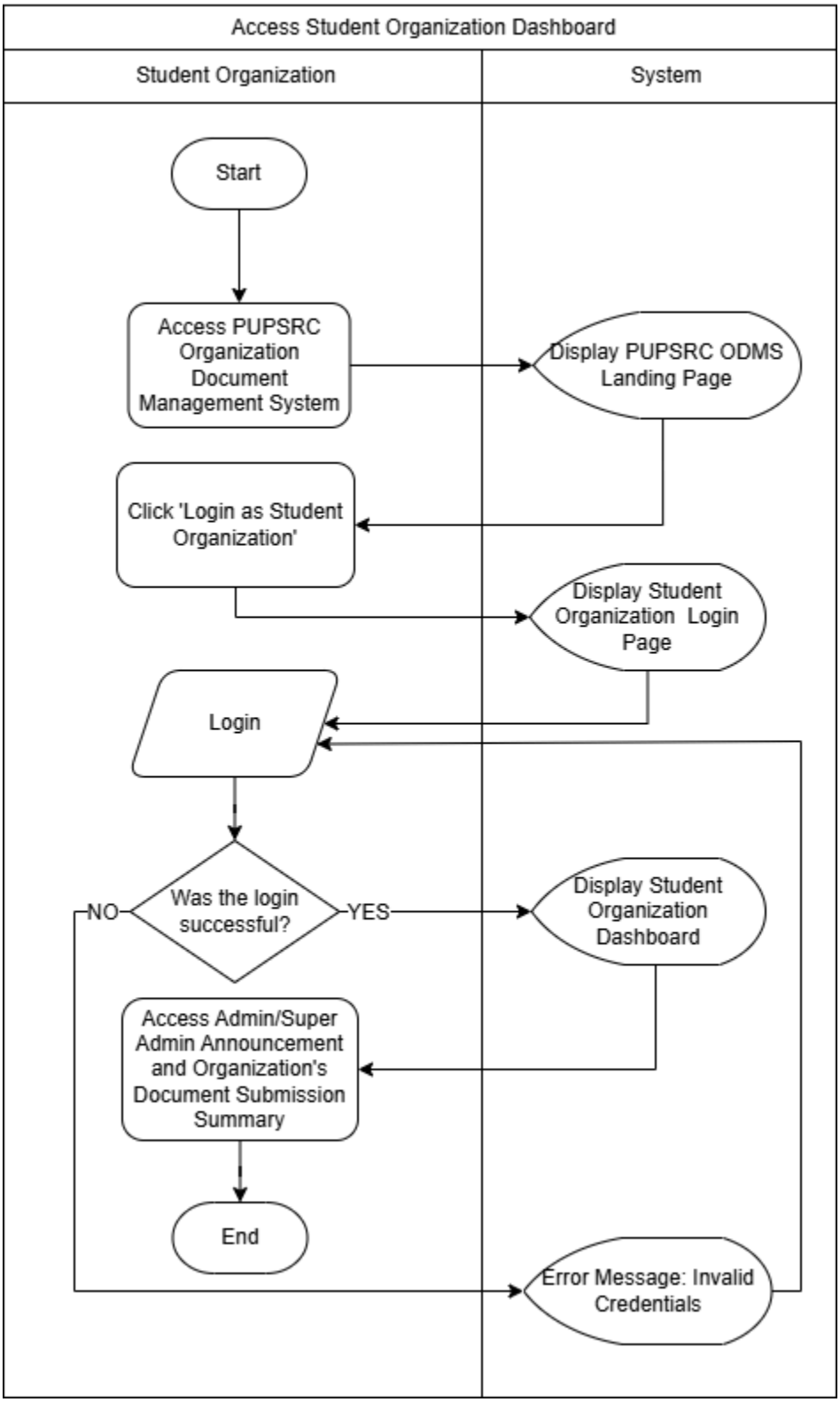


Figure 19. System Flow Diagram Proposed – Super Admin Post Announcement

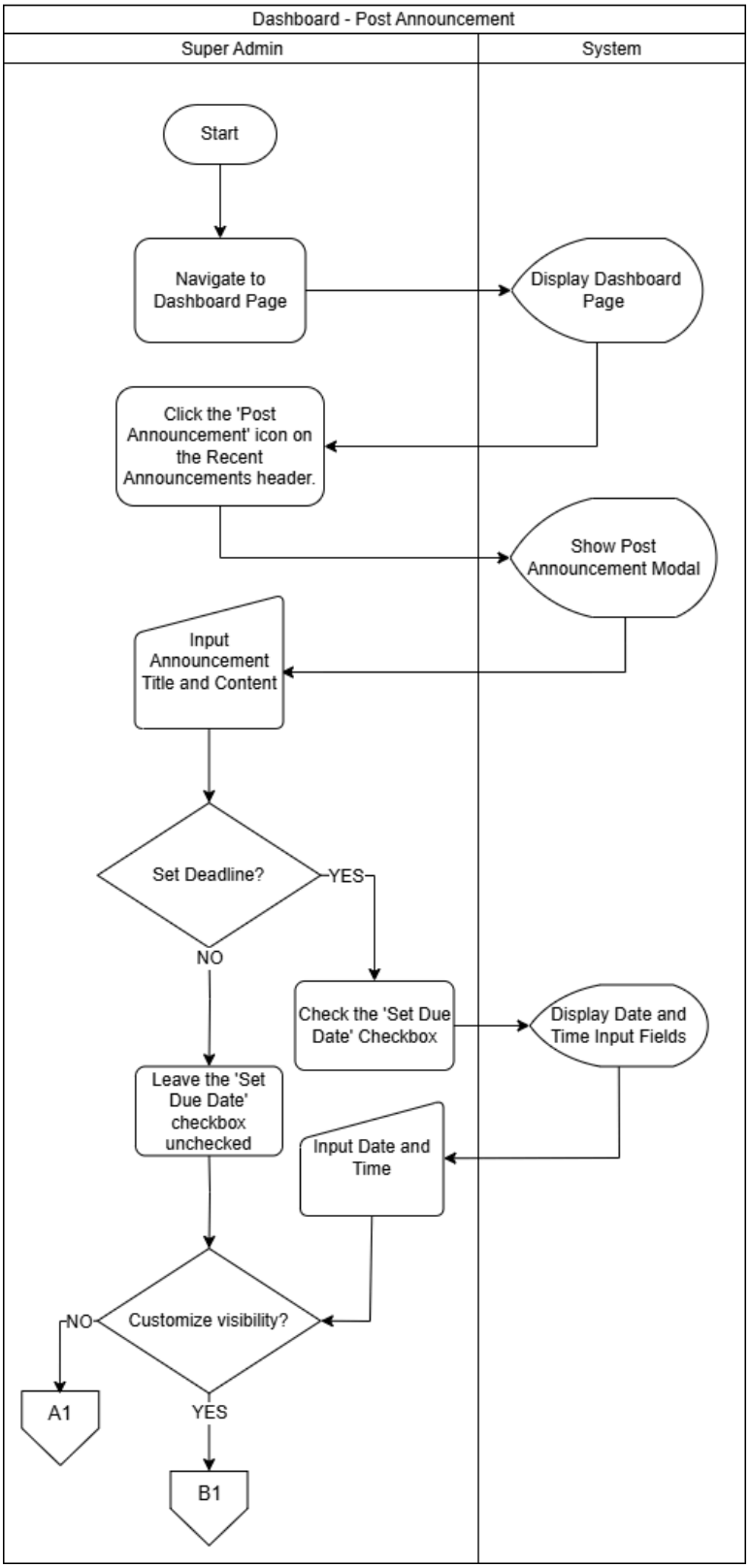


Figure 20. System Flow Diagram Proposed – Super Admin Post Announcement (Continued)

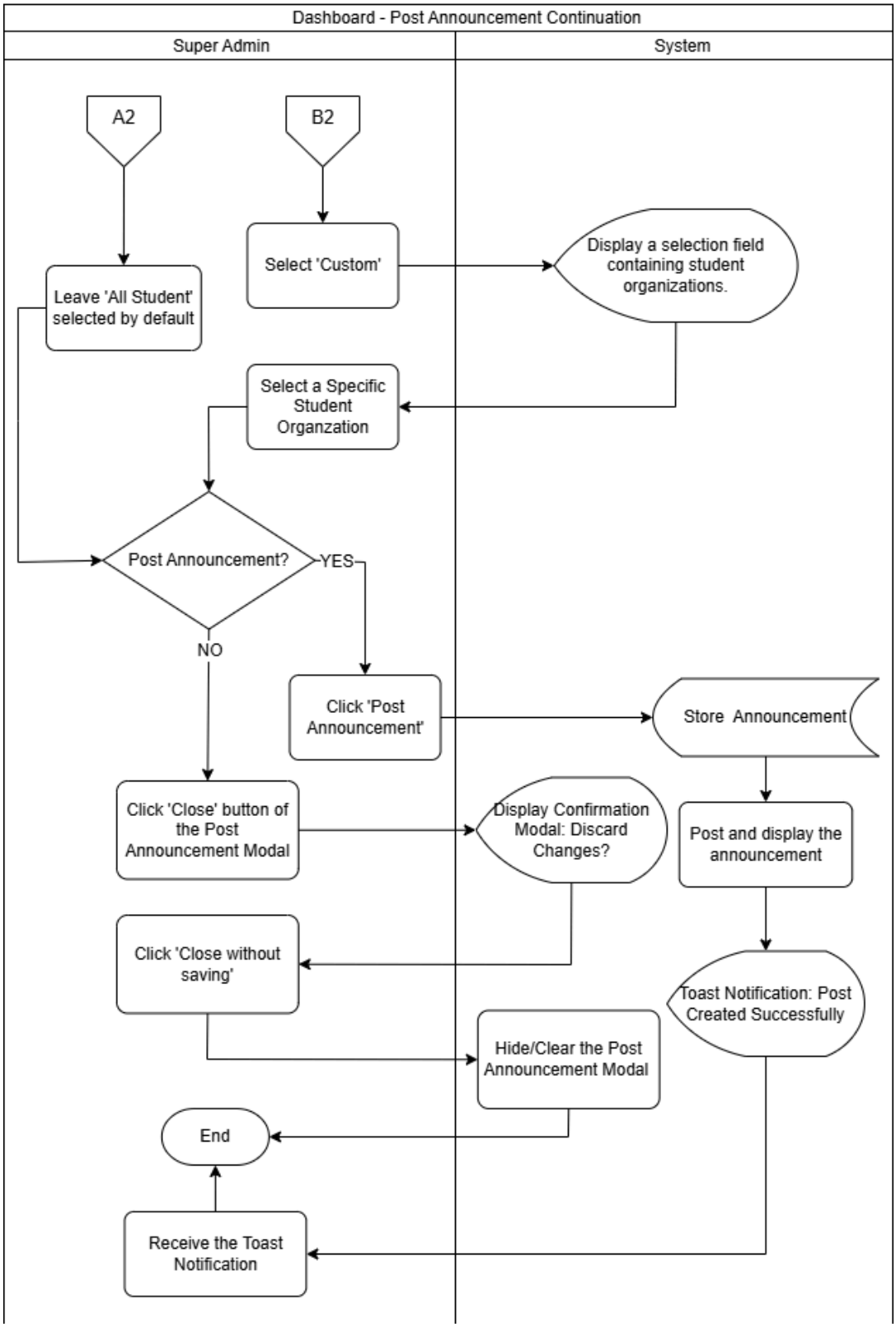


Figure 21. System Flow Diagram Proposed – Campus Administration Post Announcement

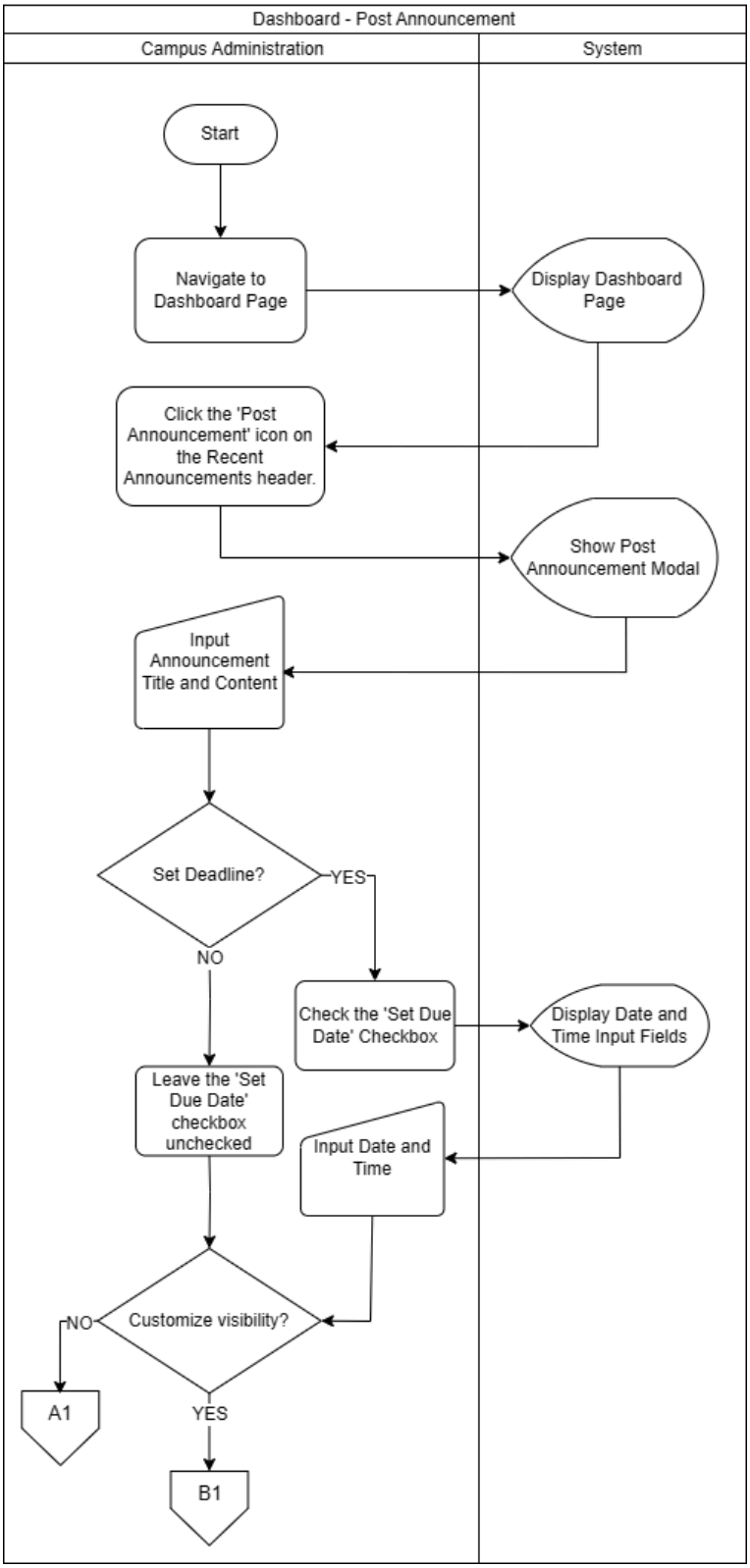


Figure 22. System Flow Diagram Proposed – Campus Administration Post Announcement (Continued)

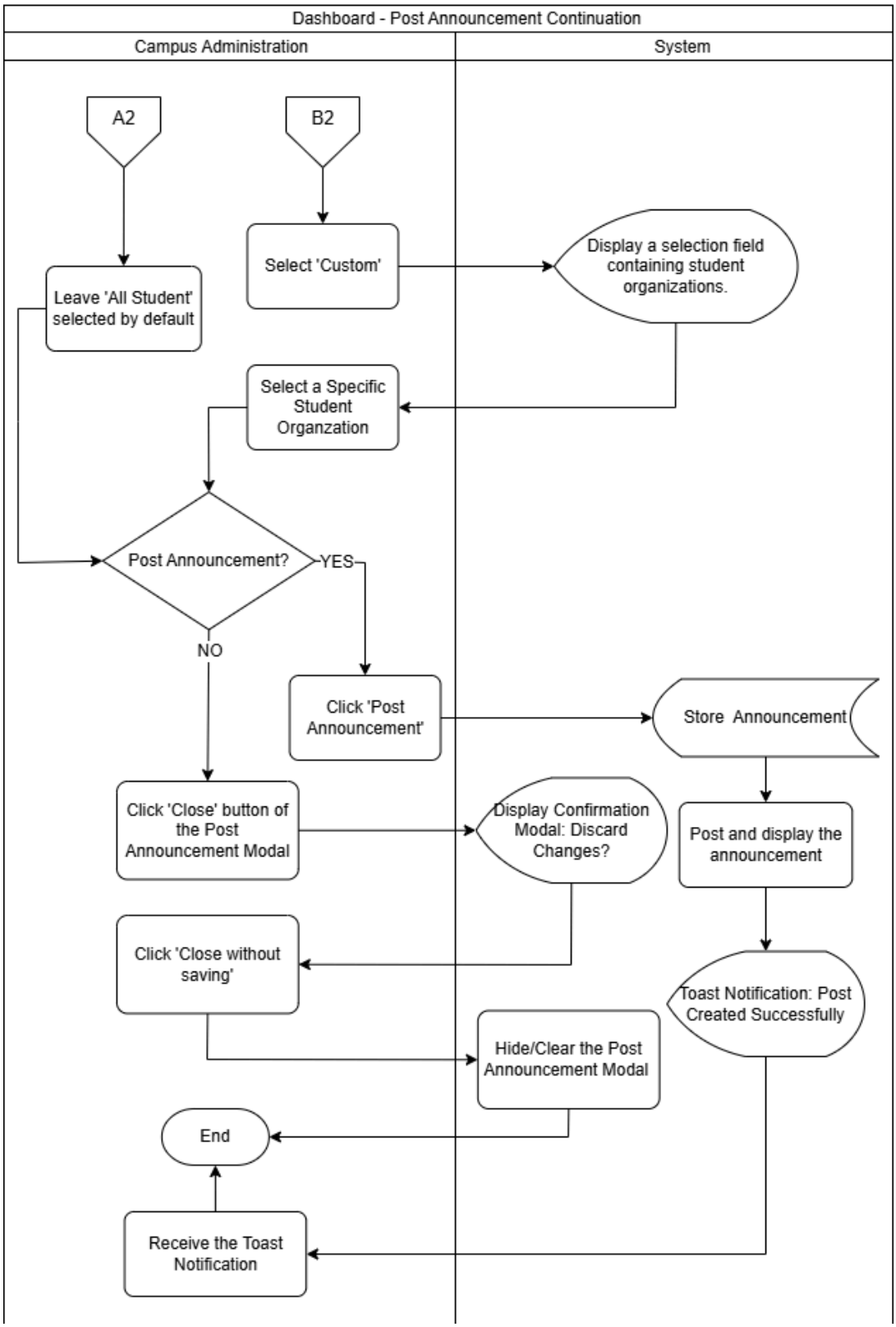


Figure 23. System Flow Diagram Proposed – Super Admin Edit Announcement

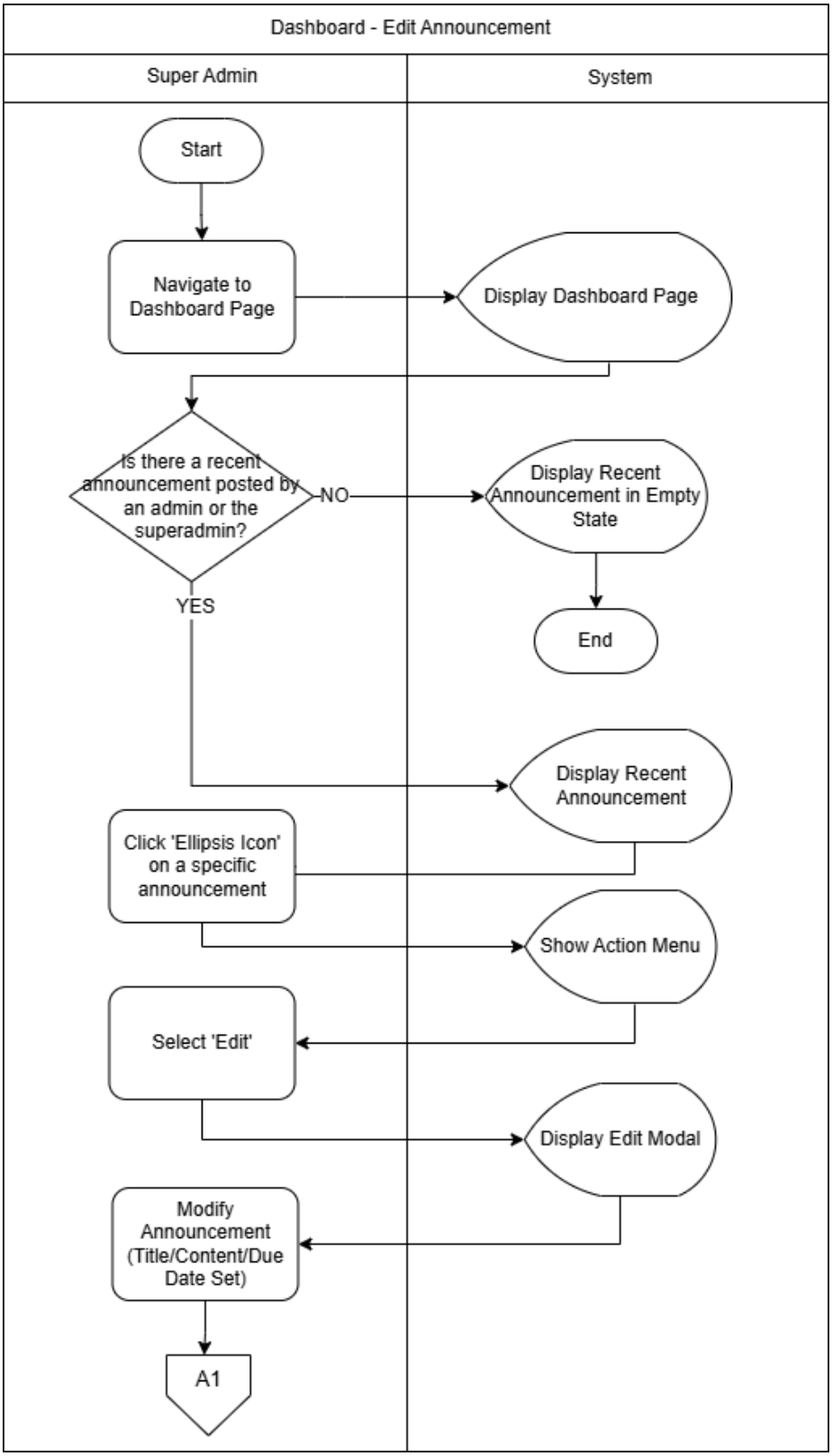


Figure 24. System Flow Diagram Proposed – Super Admin Edit Announcement (Continued)

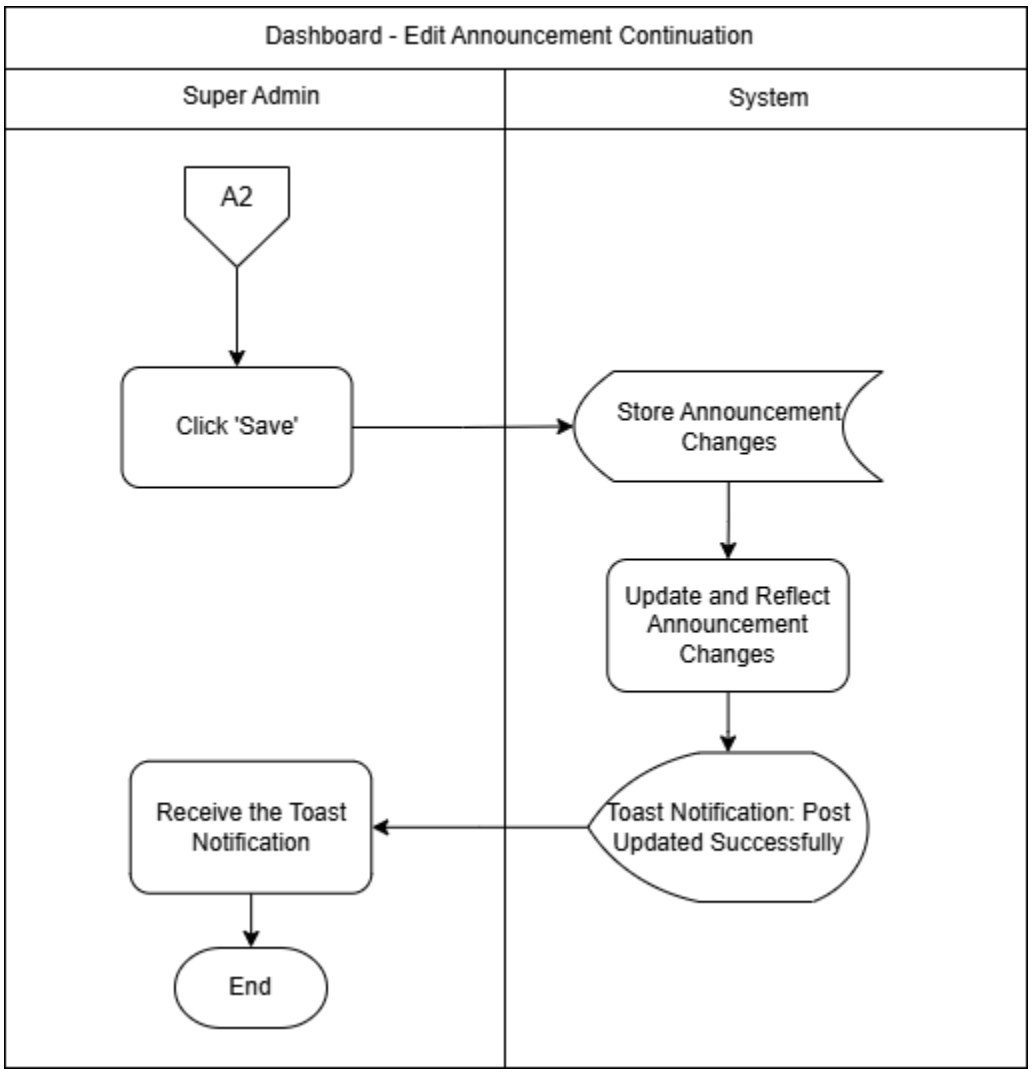


Figure 25. System Flow Diagram Proposed – Campus Administration Edit Announcement

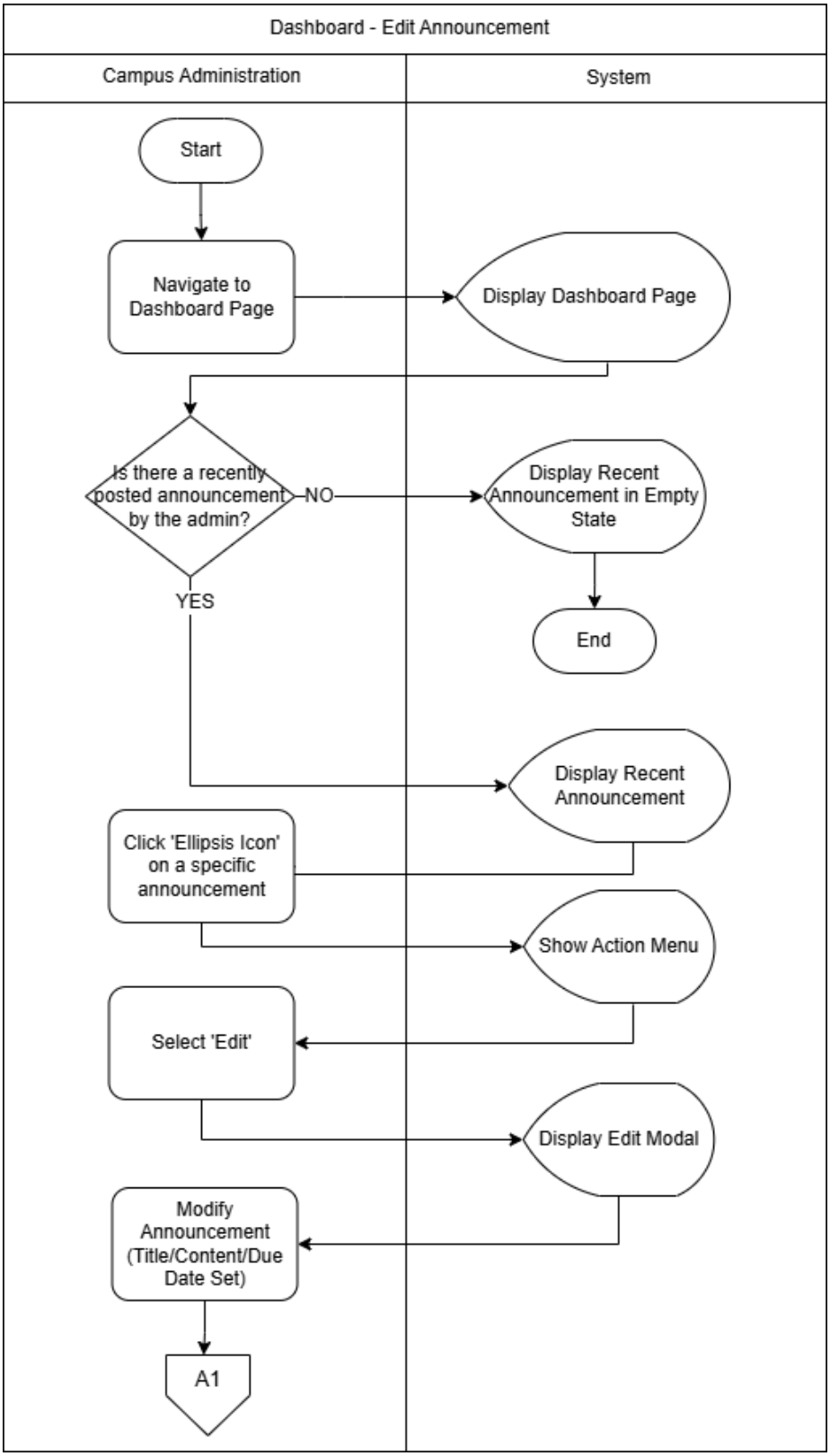


Figure 26. System Flow Diagram Proposed – Campus Administration Edit Announcement (Continued)

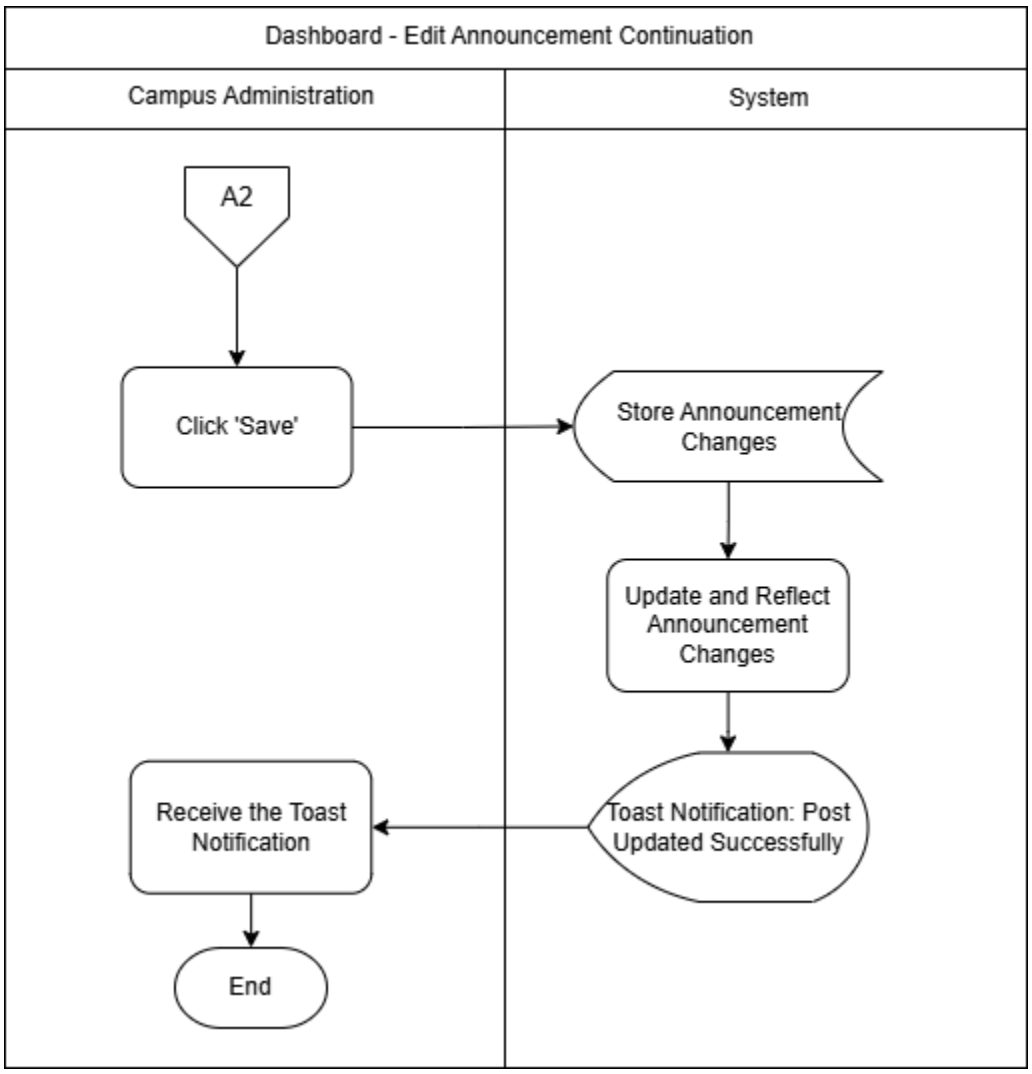


Figure 27. System Flow Diagram Proposed – Super Admin Archive Announcement

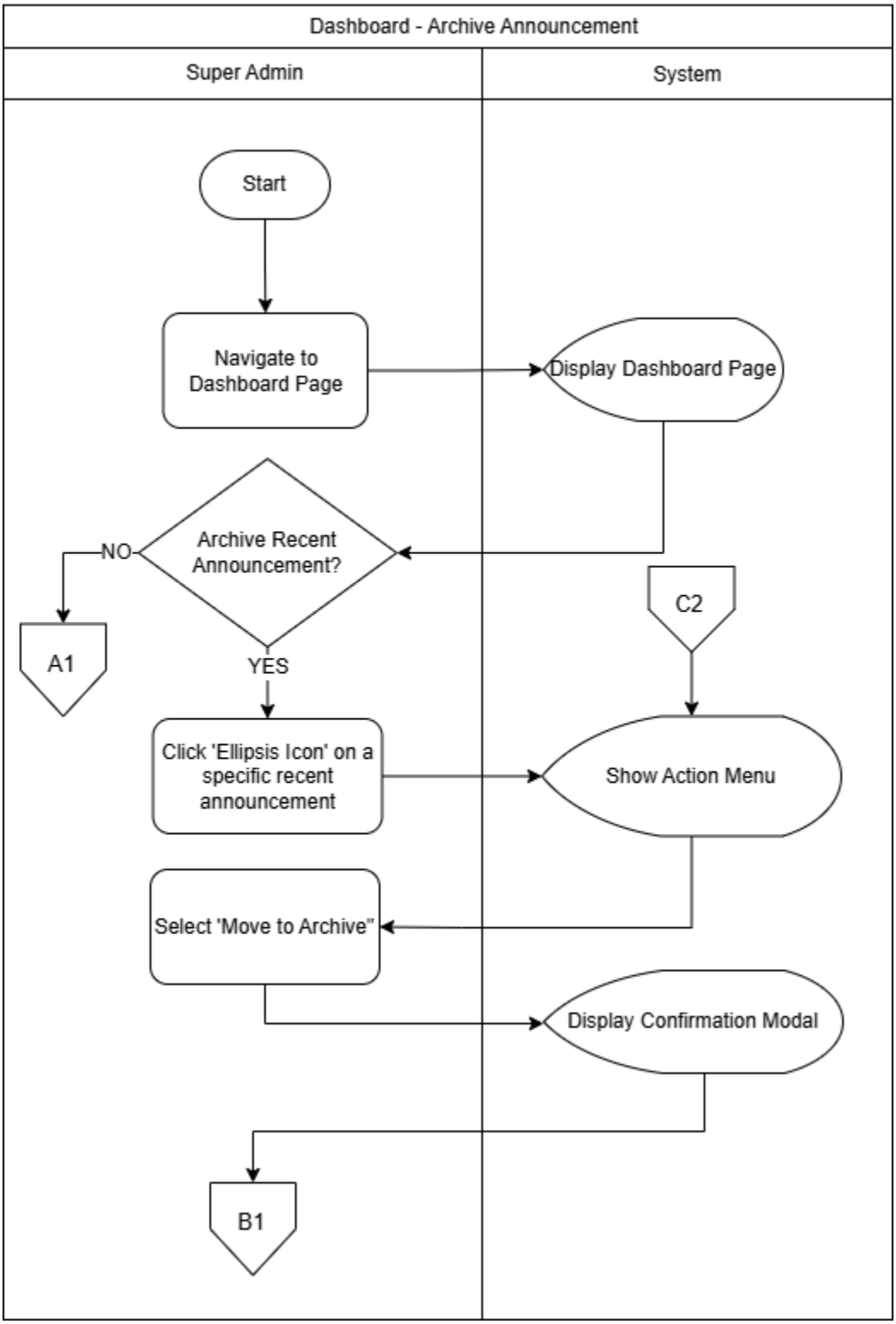


Figure 28 System Flow Diagram Proposed – Super Admin Archive Announcement (Continued)

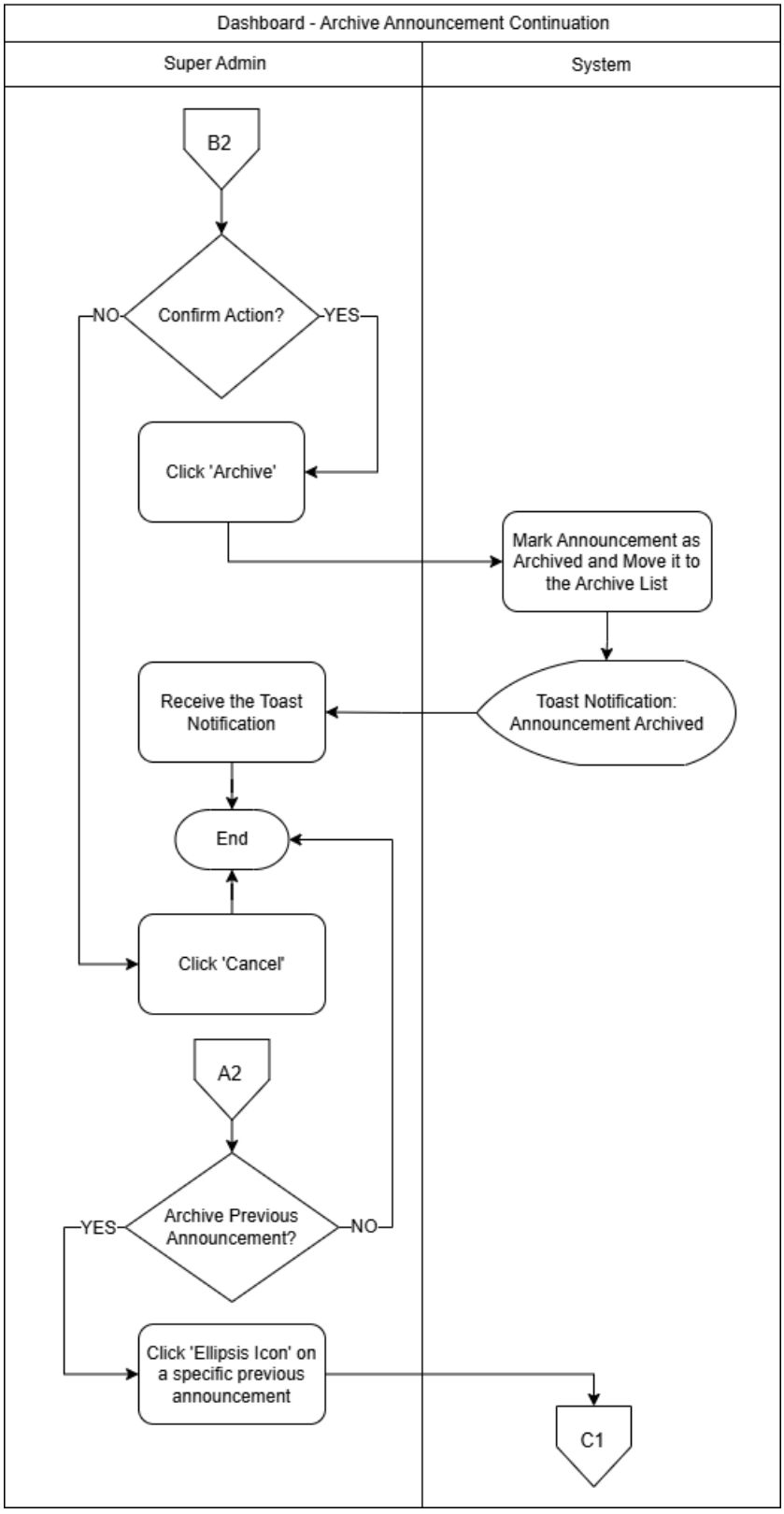


Figure 29. System Flow Diagram Proposed – Campus Administration Archive Announcement

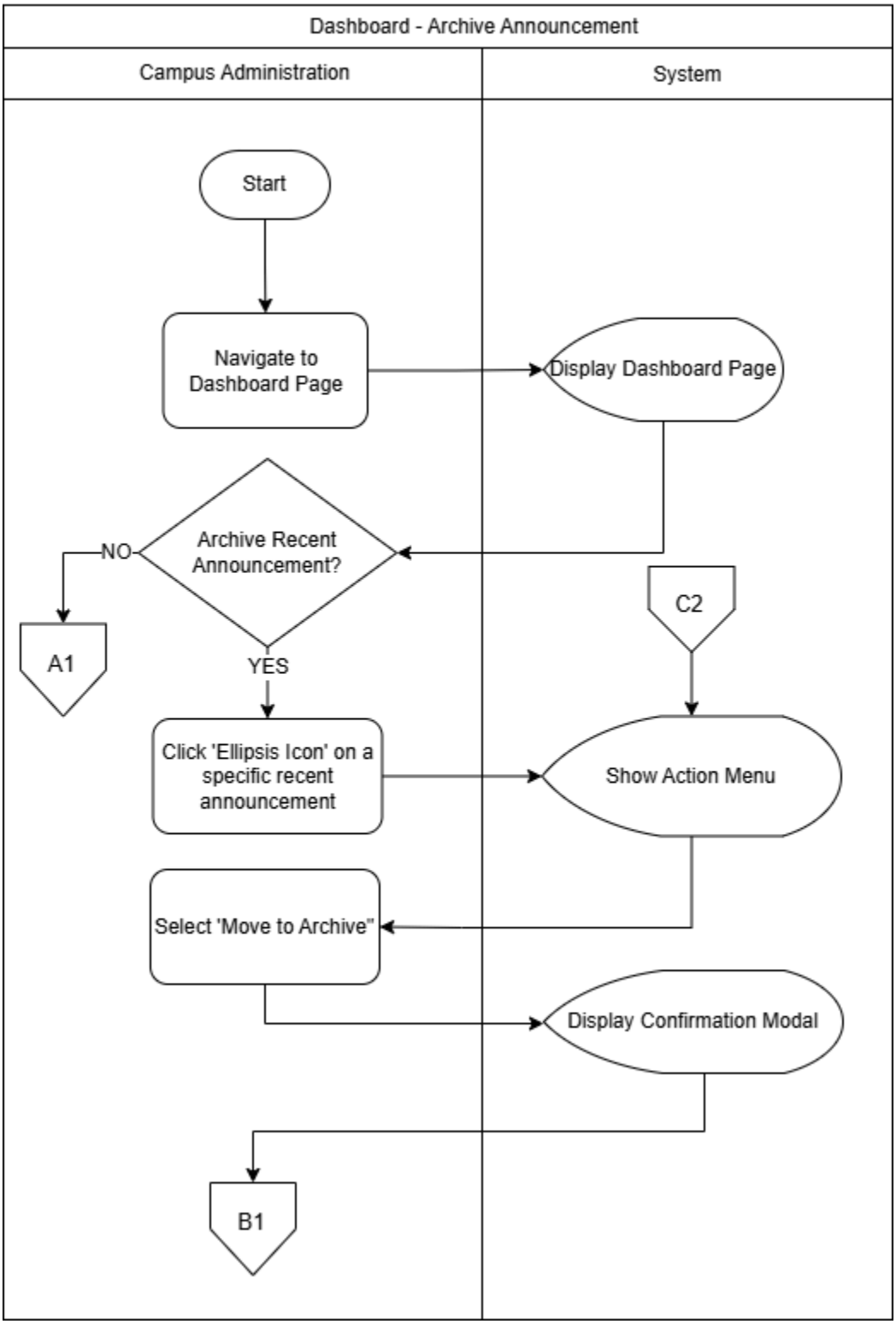


Figure 30. System Flow Diagram Proposed – Campus Administration Archive Announcement (Continued)

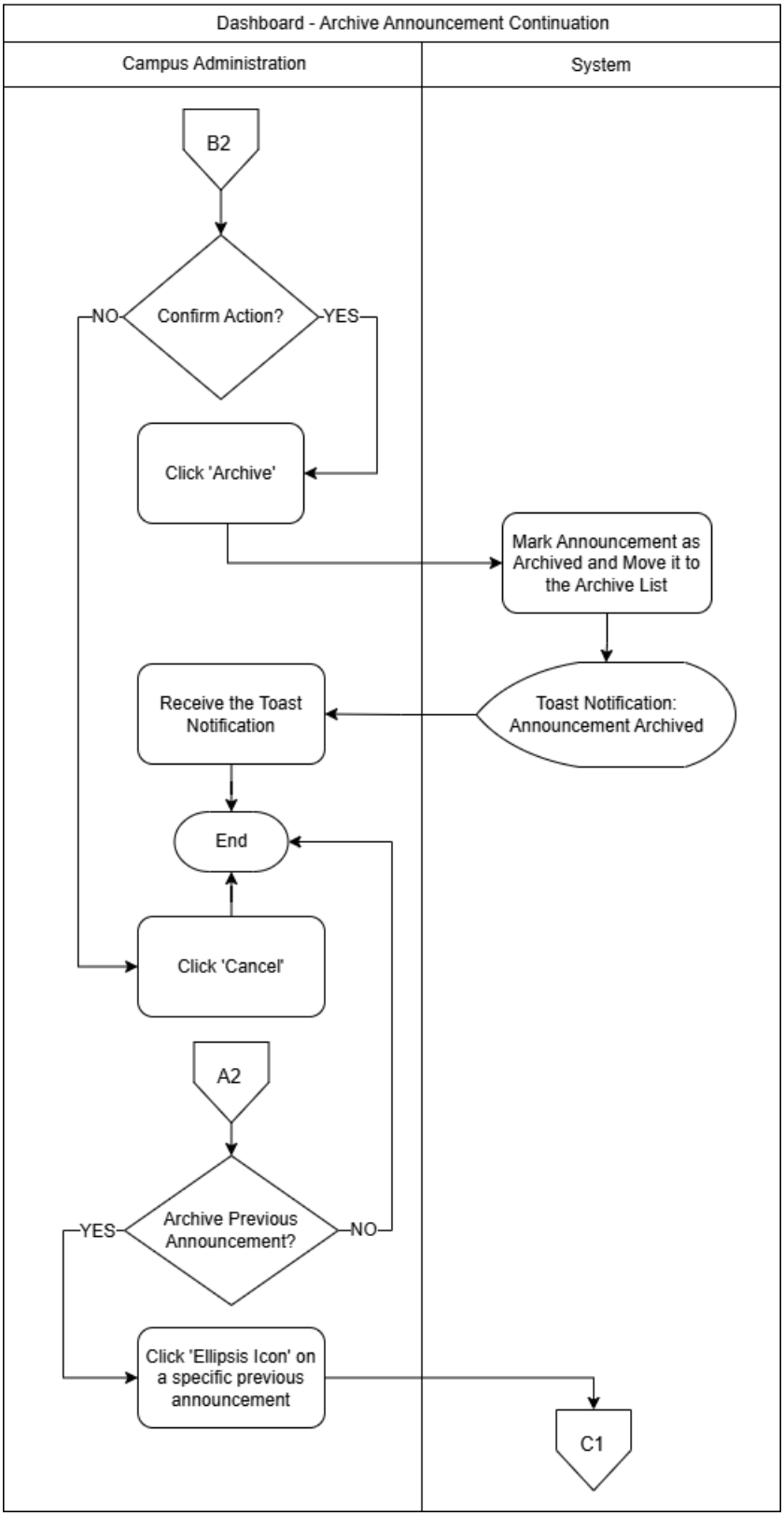


Figure 31. System Flow Diagram Proposed – Super Admin Delete Announcement

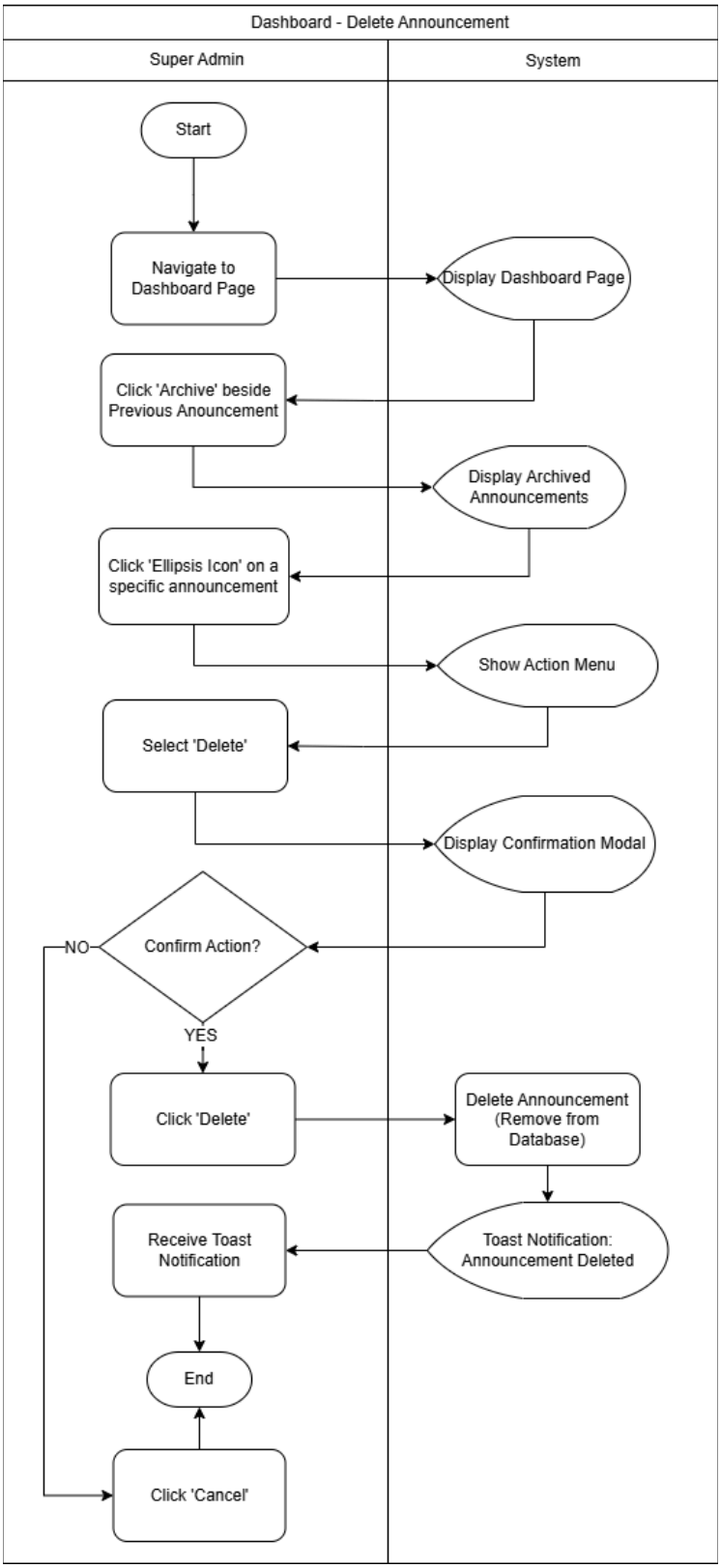


Figure 32. System Flow Diagram Proposed – Campus Administration Delete Announcement

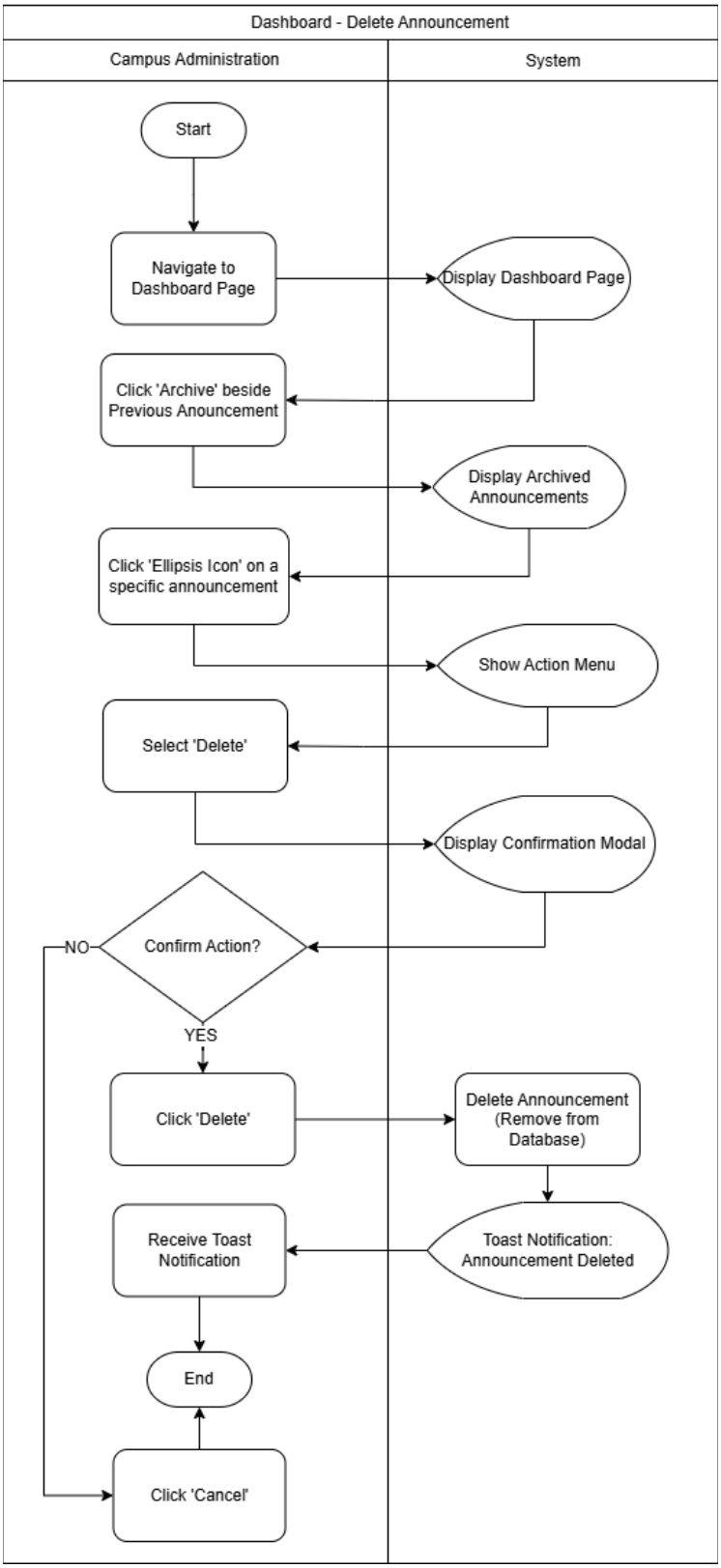


Figure 33. System Flow Diagram Proposed – Super Admin Restore Announcement

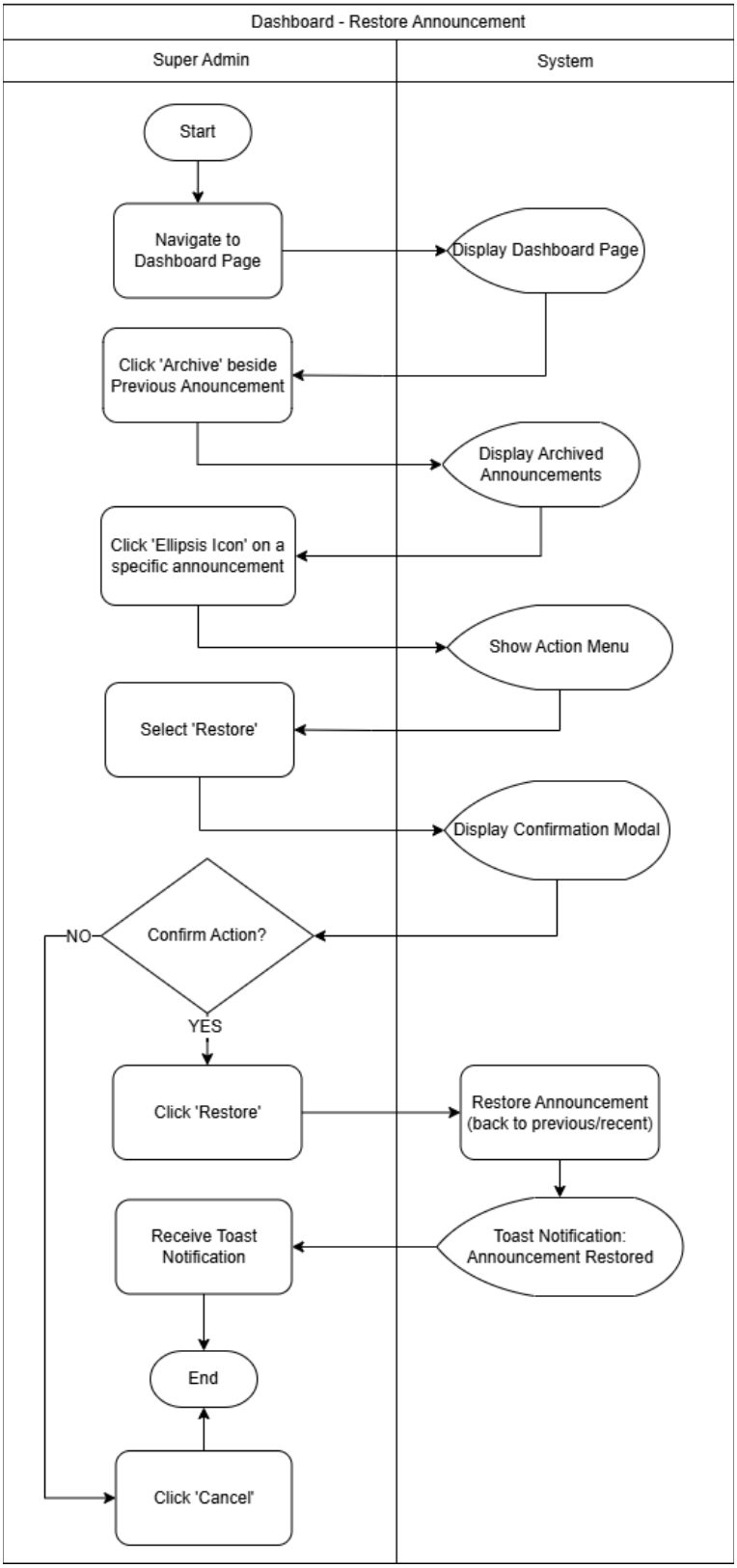


Figure 34. System Flow Diagram Proposed – Campus Administration Restore Announcement

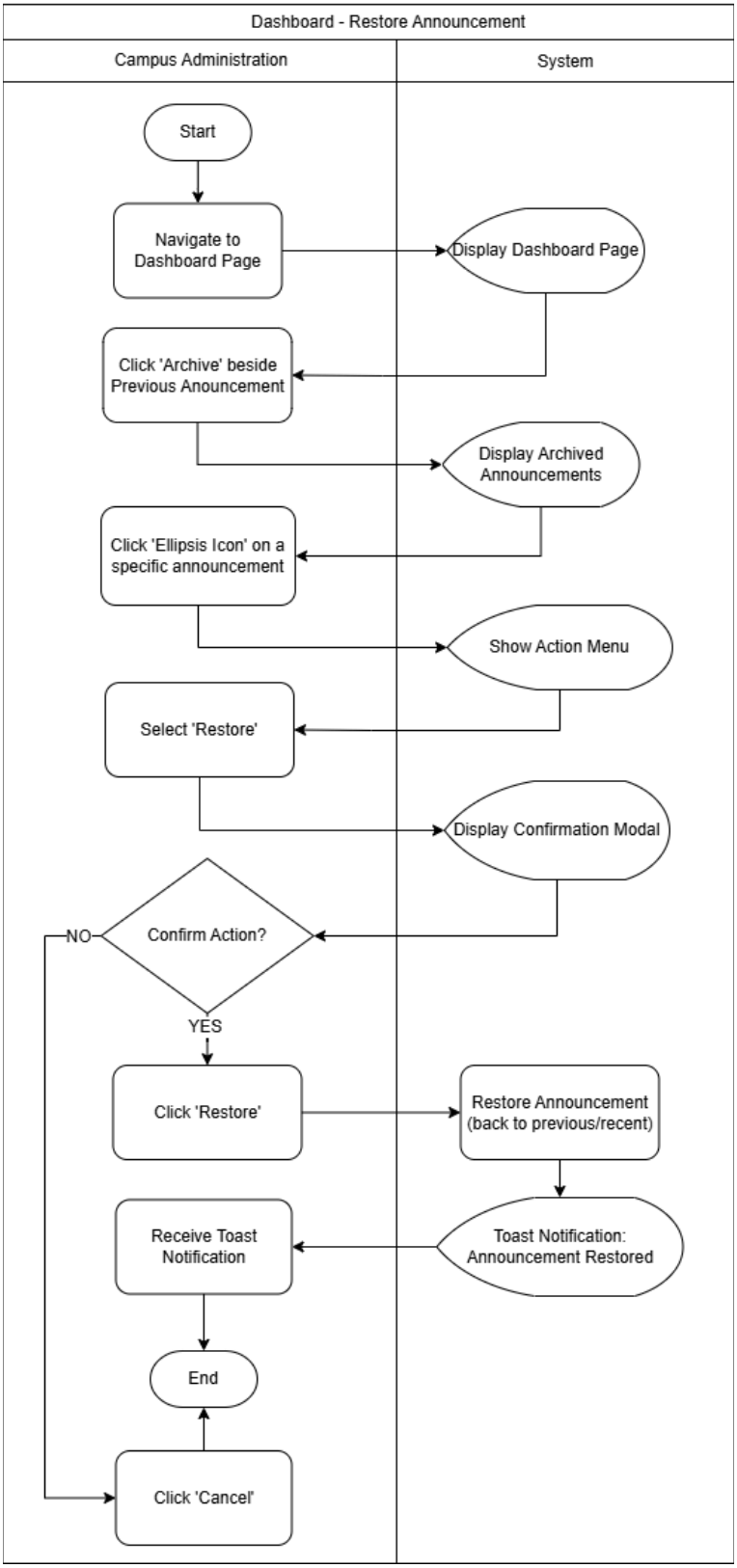


Figure 35. System Flow Diagram Proposed – User Settings Change/Upload Profile Photo

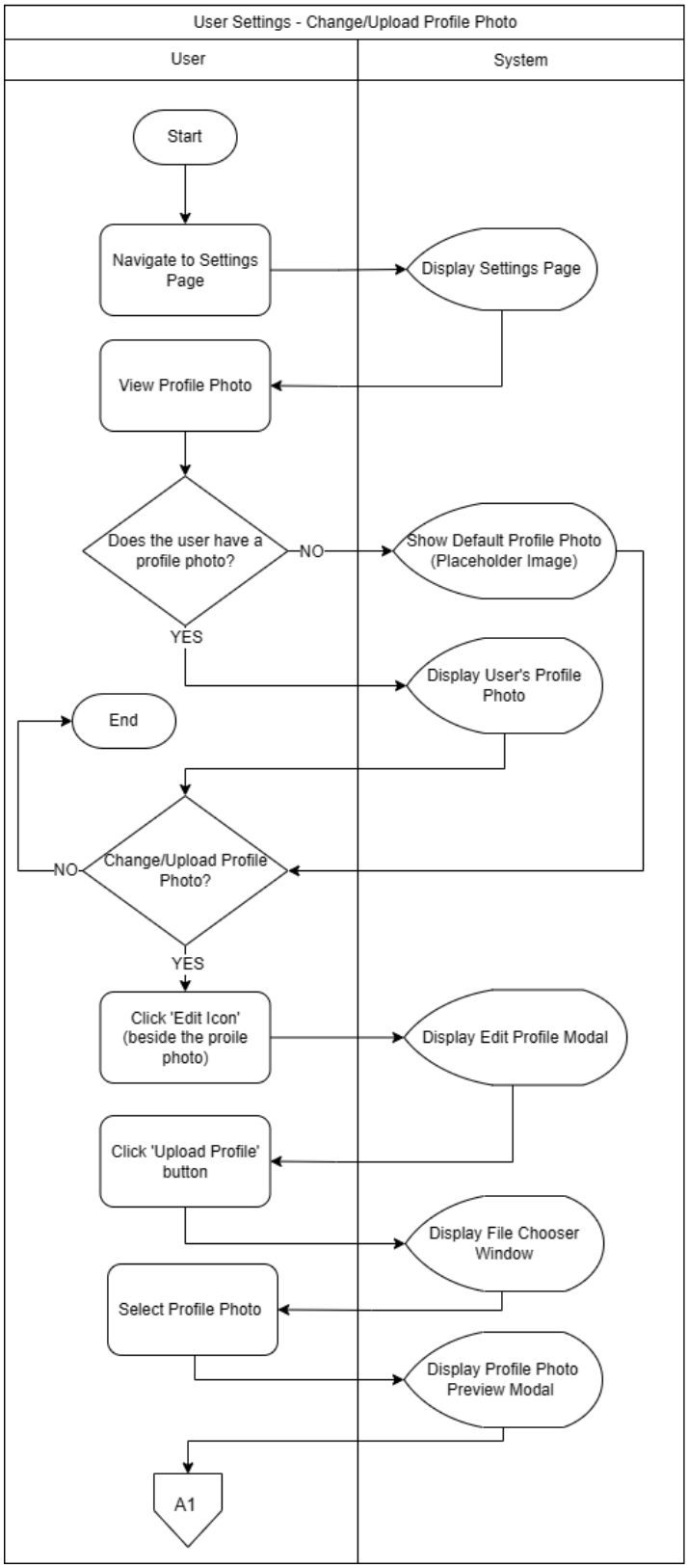


Figure 36. System Flow Diagram Proposed – User Settings Change/Upload Profile Photo (Continued)

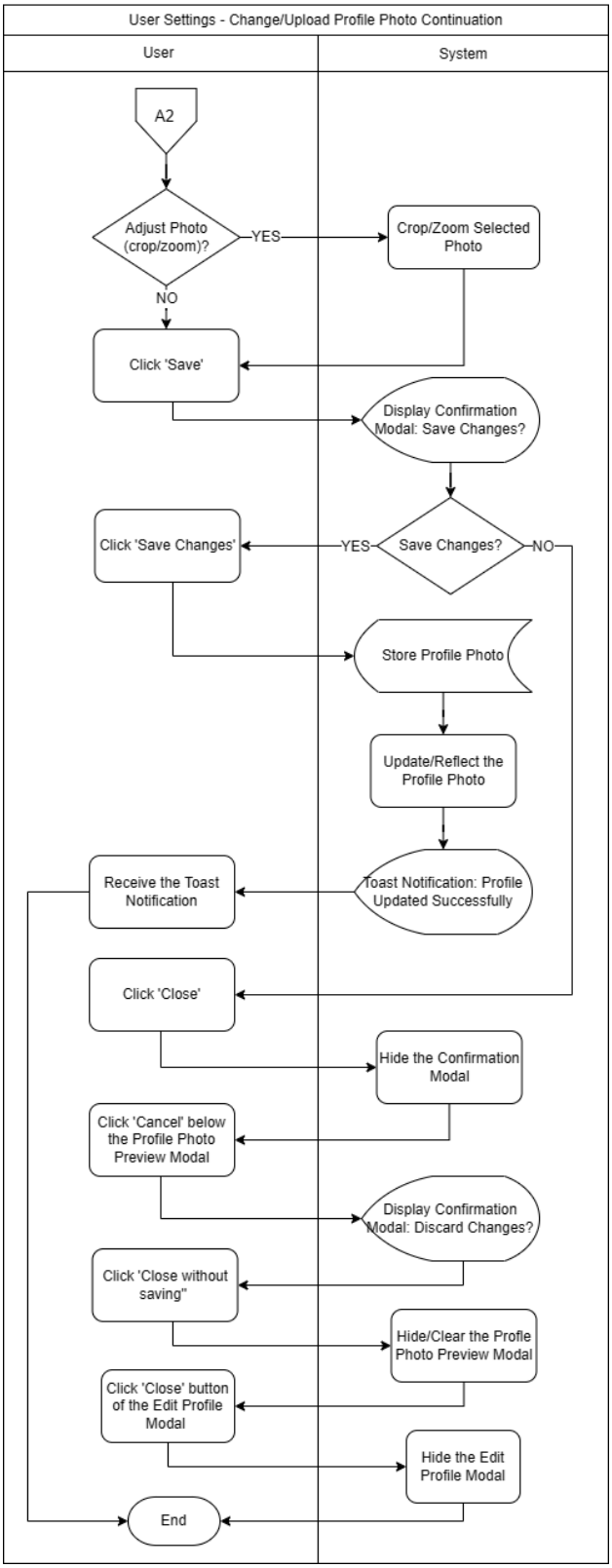


Figure 37. System Flow Diagram Proposed – User Settings Remove Profile Photo

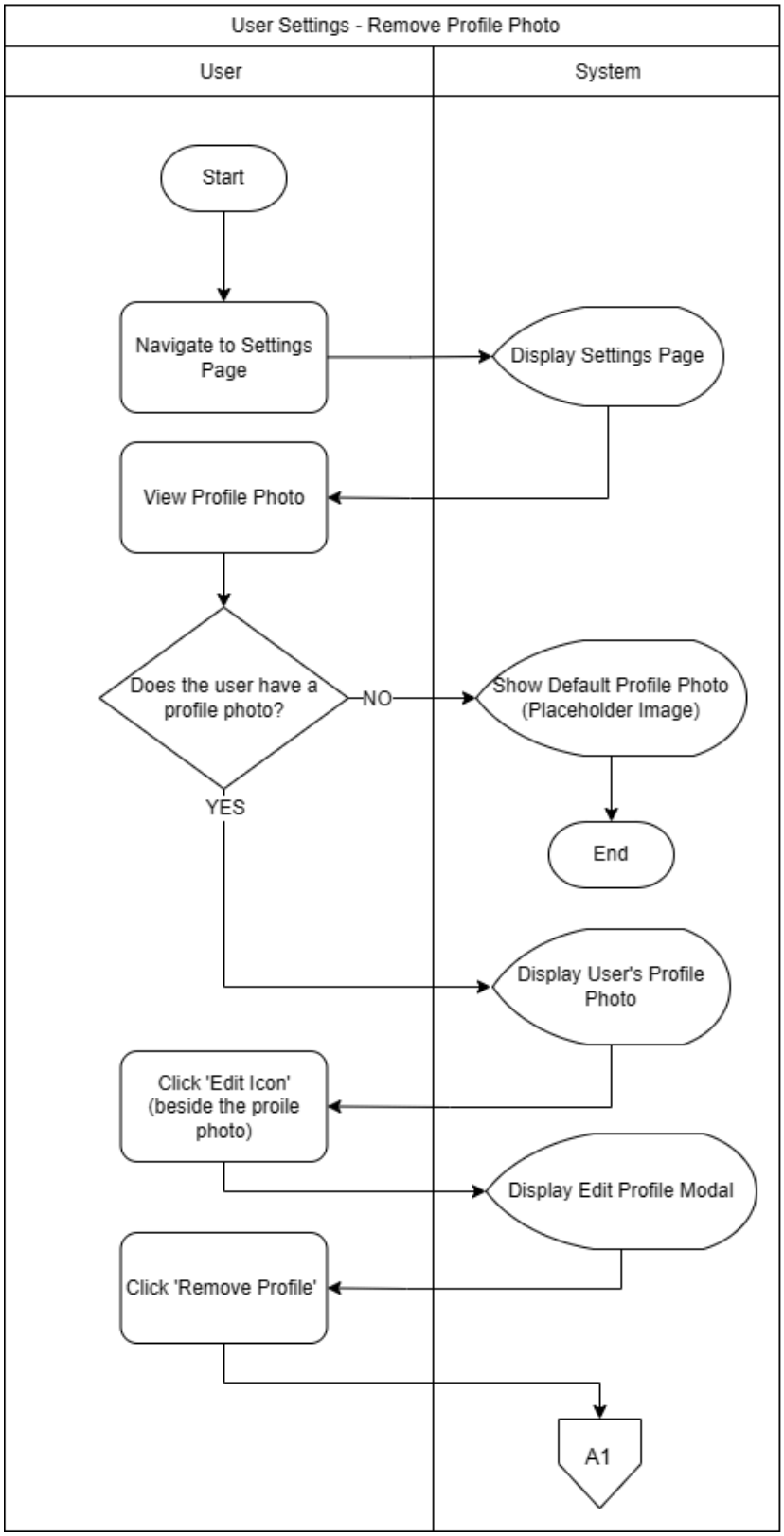


Figure 38. System Flow Diagram Proposed – User Settings Remove Profile Photo (Continued)

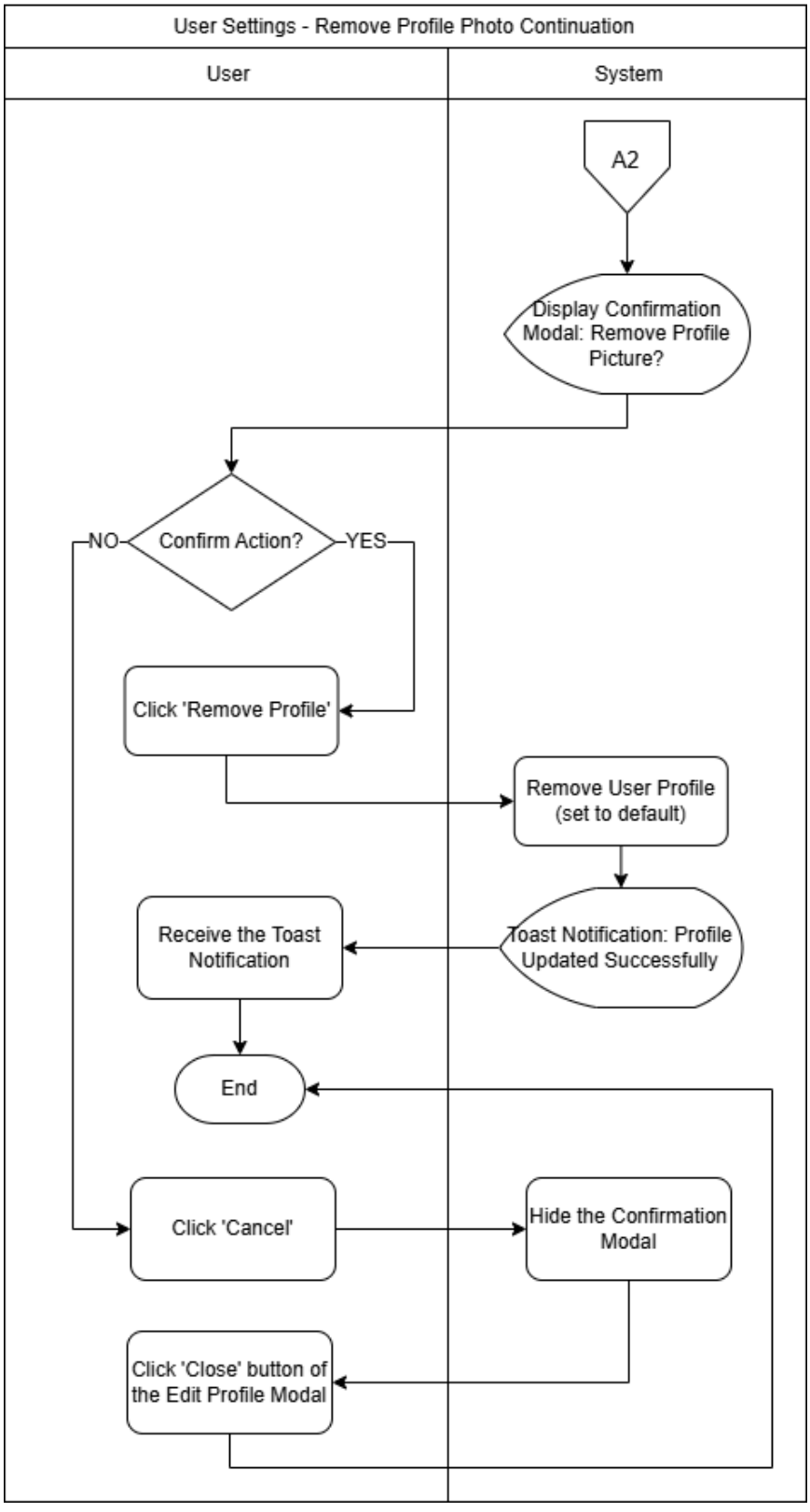


Figure 39. System Flow Diagram Proposed – User Settings Change Password

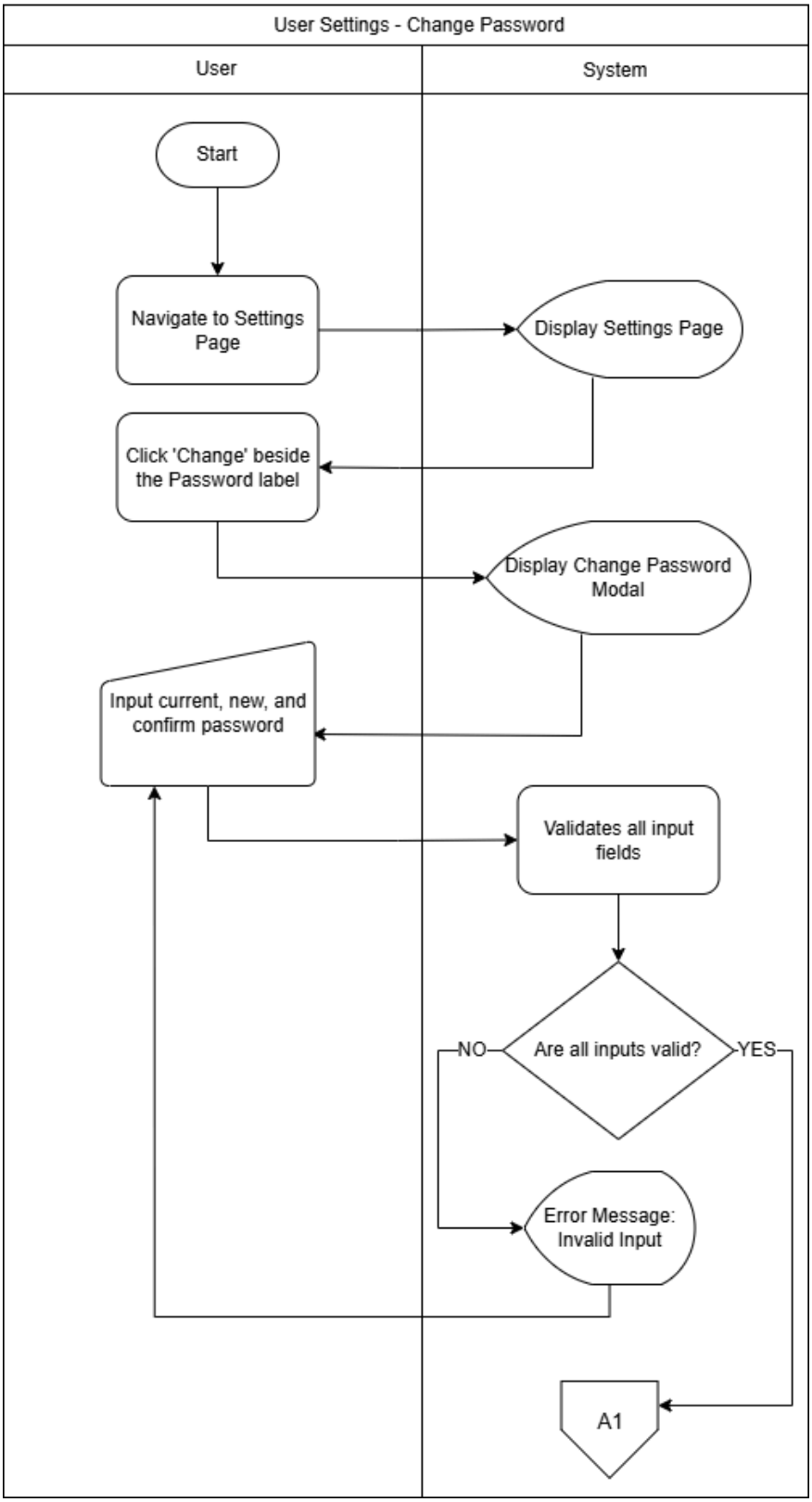


Figure 40. System Flow Diagram Proposed – User Settings Change Password (Continued)

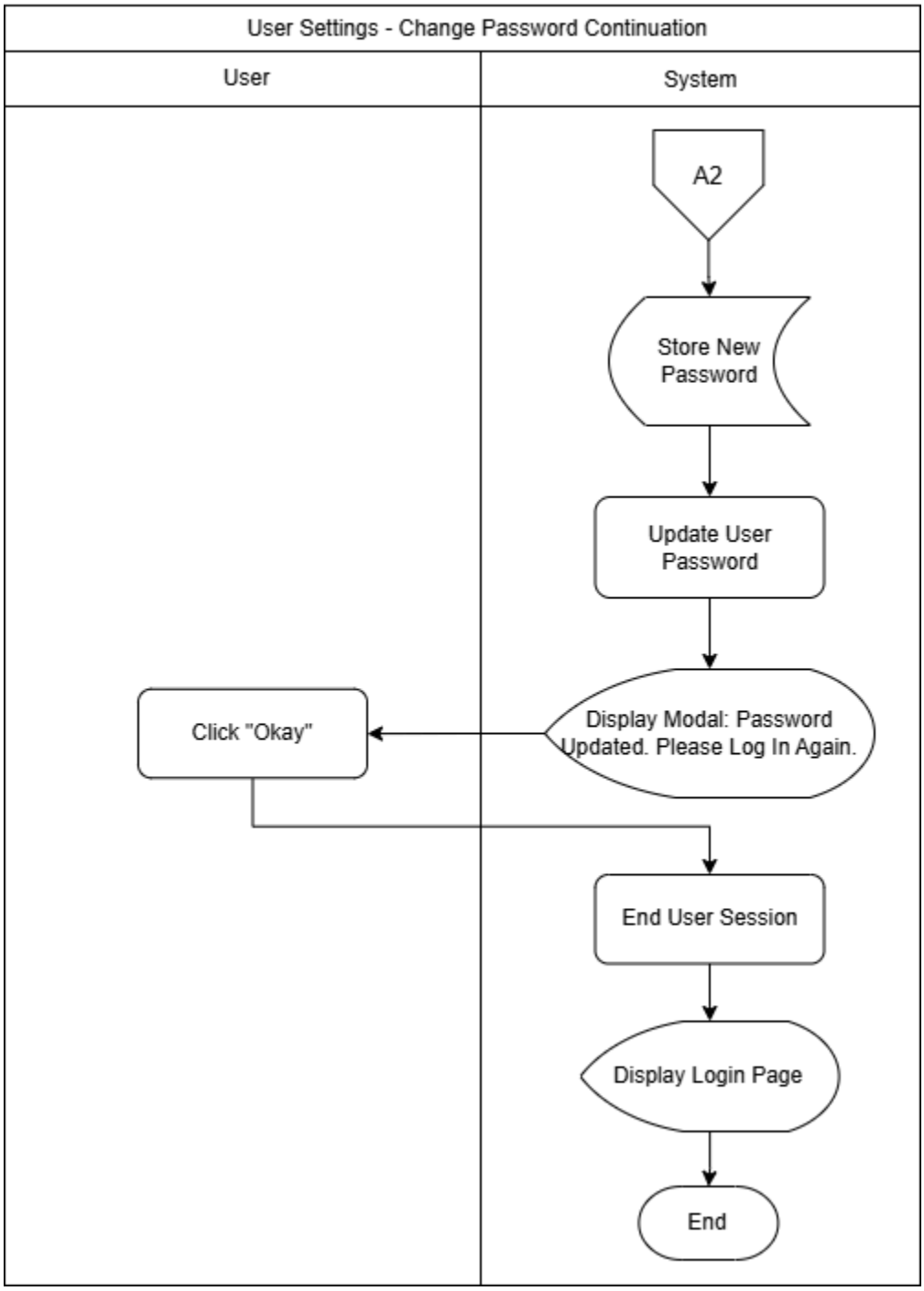


Figure 41. System Flow Diagram Proposed – User Settings Add Recovery Email

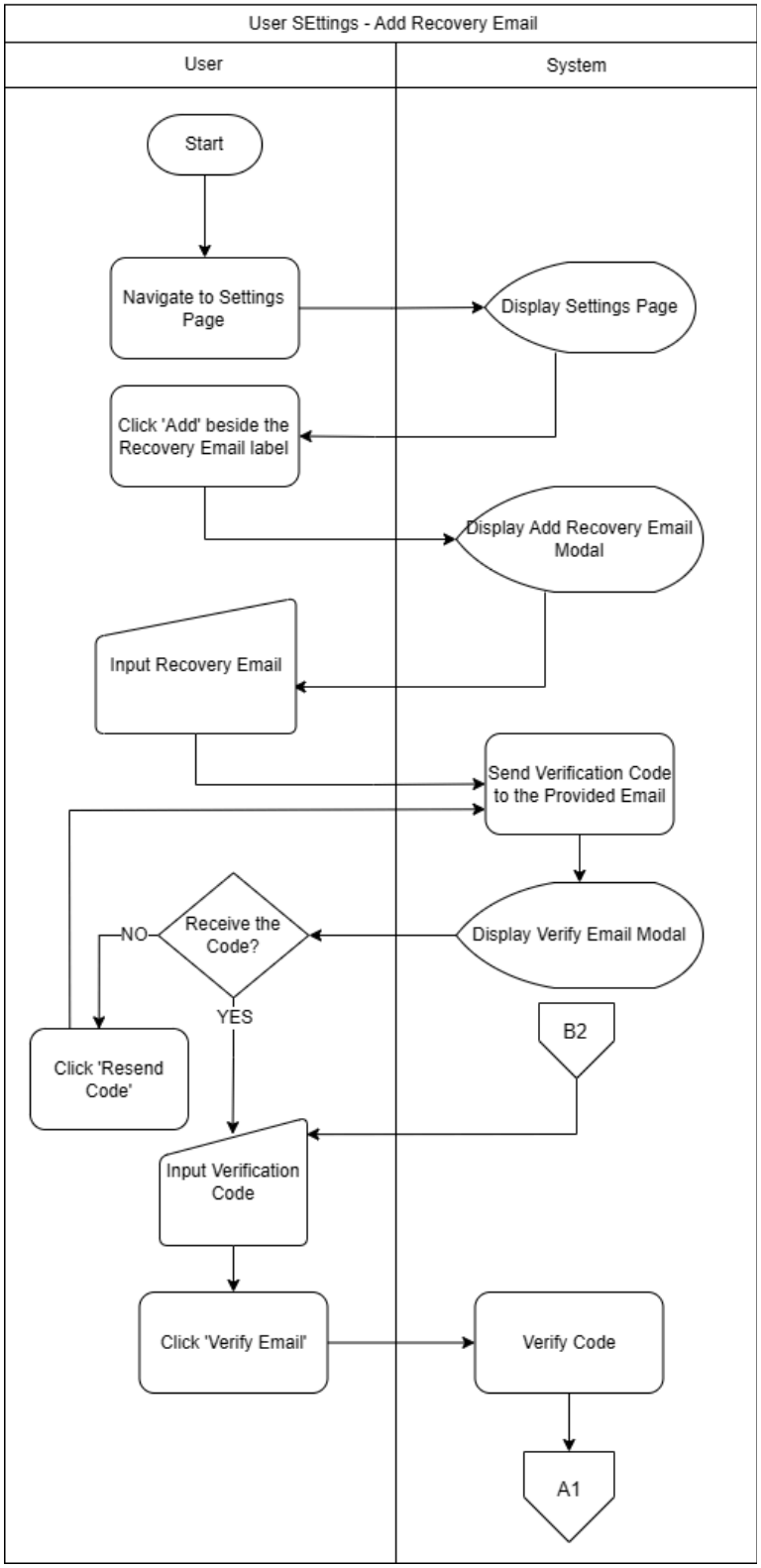


Figure 42. System Flow Diagram Proposed – User Settings Add Recovery Email (Continued)

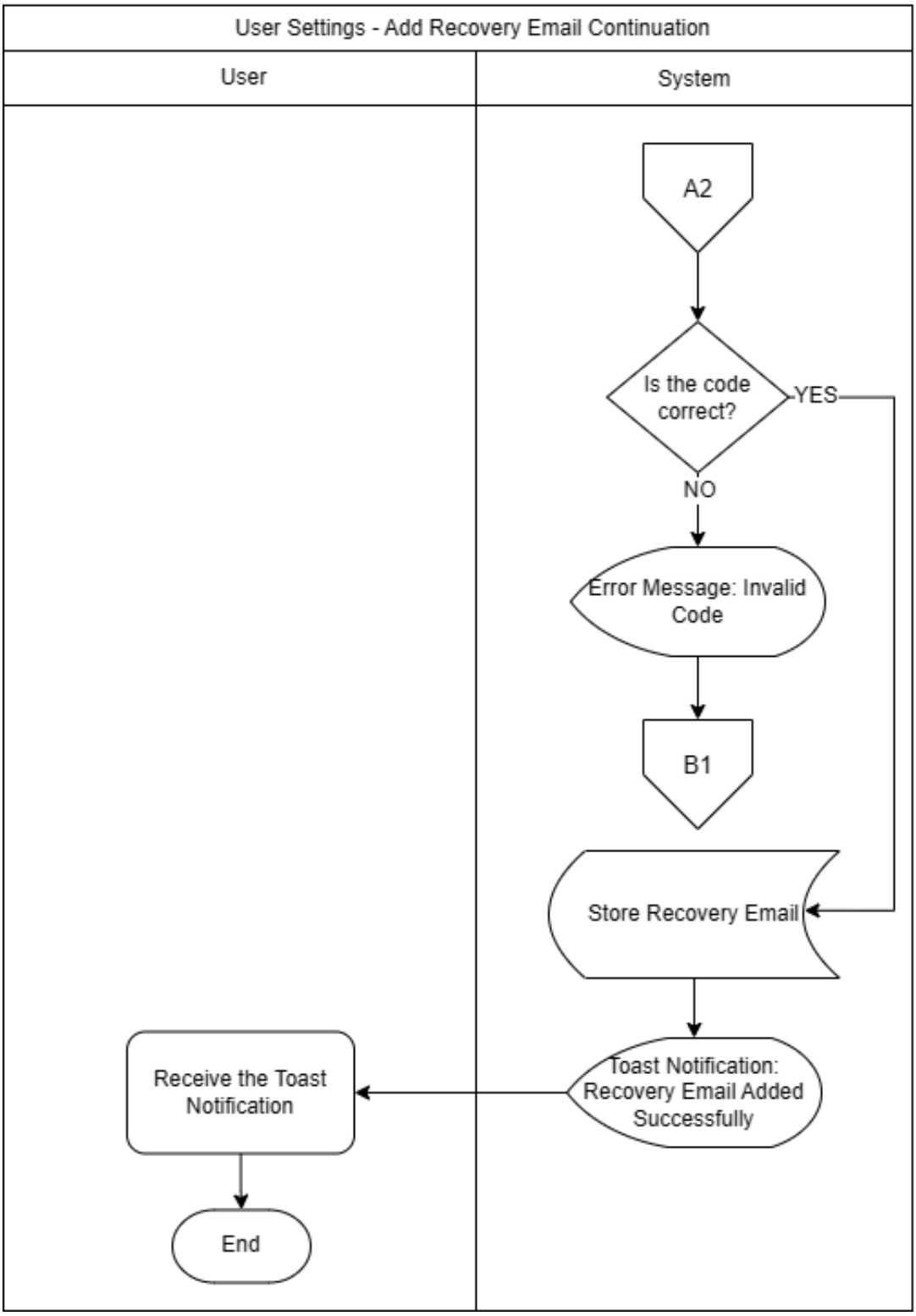


Figure 43. System Flow Diagram Proposed – User Settings Remove Recovery Email

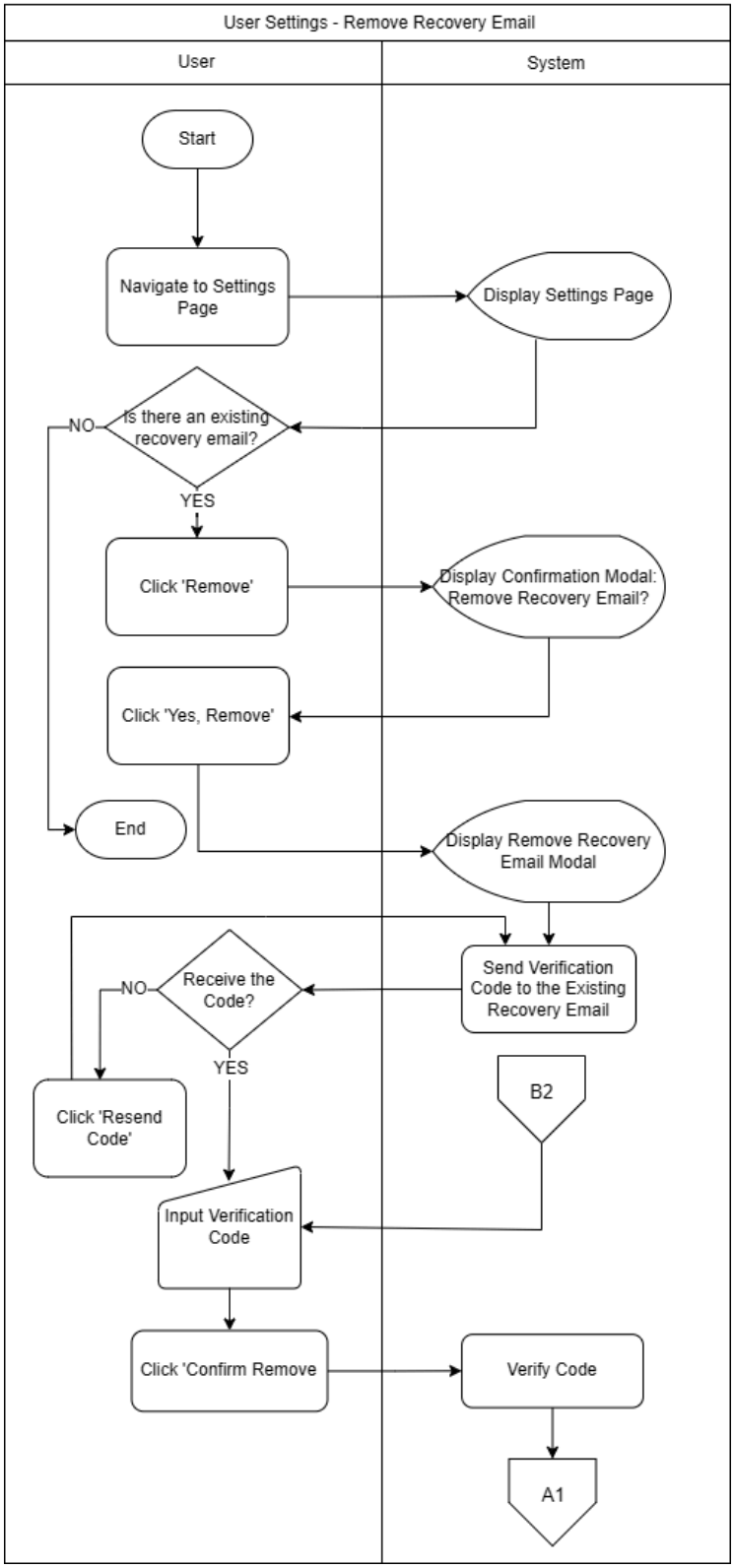


Figure 44. System Flow Diagram Proposed – User Settings Remove Recovery Email (Continued)

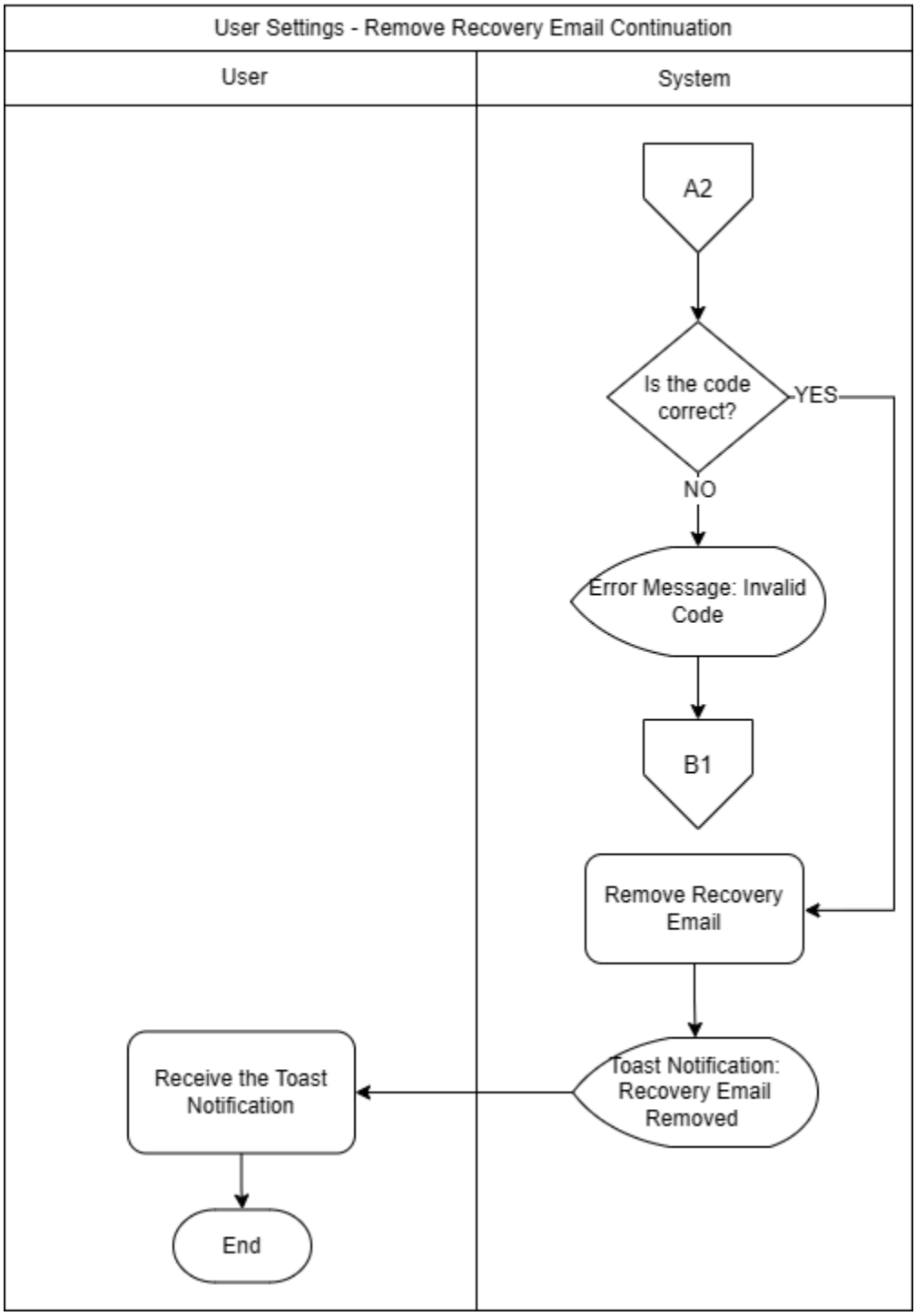

User Settings - Remove Recovery Email Continuation
User
System
A2
Is the code correct?
YES
NO
Error Message: Invalid Code
B1
Remove Recovery Email
Toast Notification: Recovery Email Removed
Receive the Toast Notification
End

Figure 44. System Flow Diagram Proposed – Super Admin Settings Change Primary Email

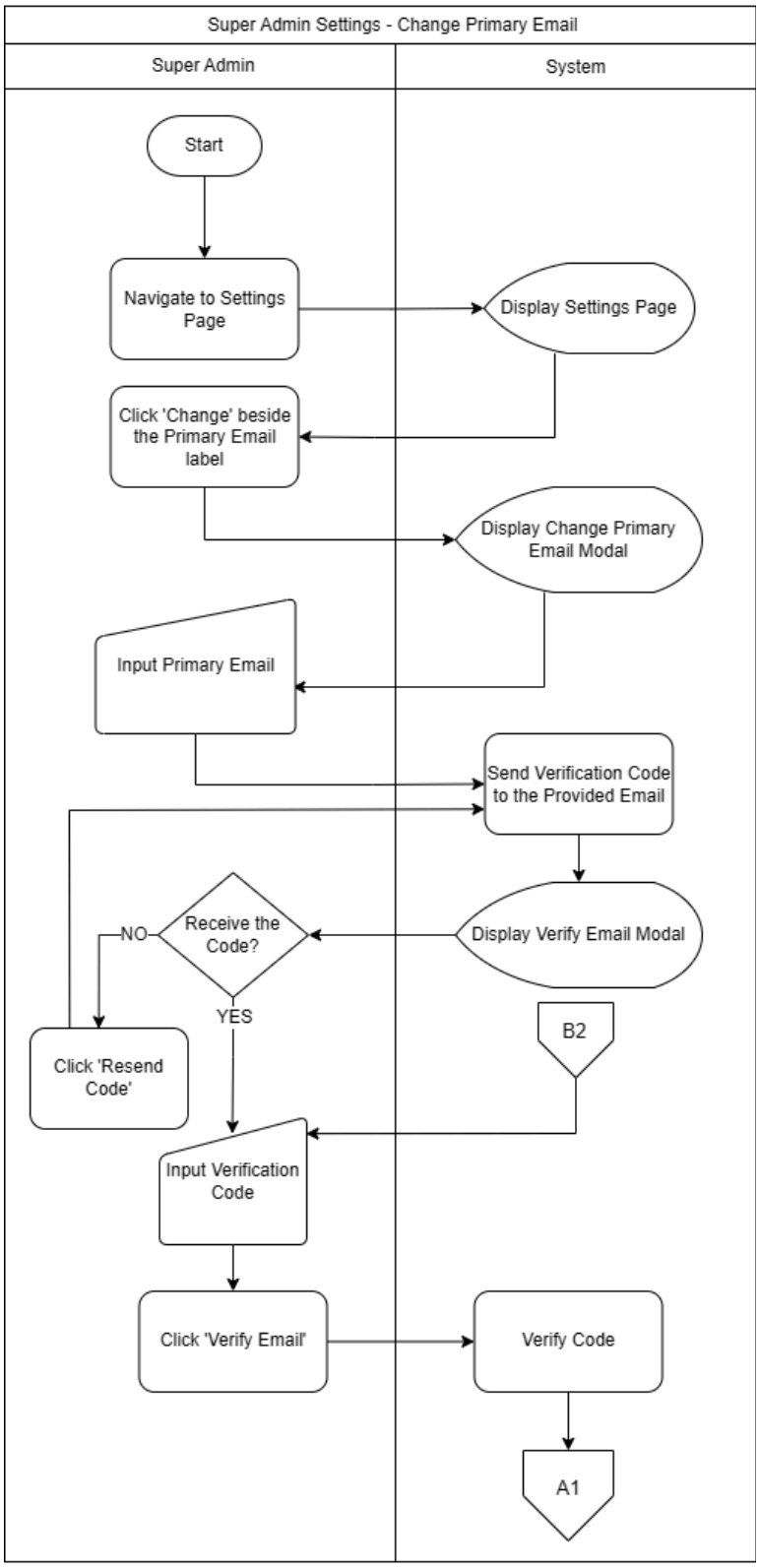


Figure 44. System Flow Diagram Proposed – Super Admin Settings Change Primary Email (Continued)

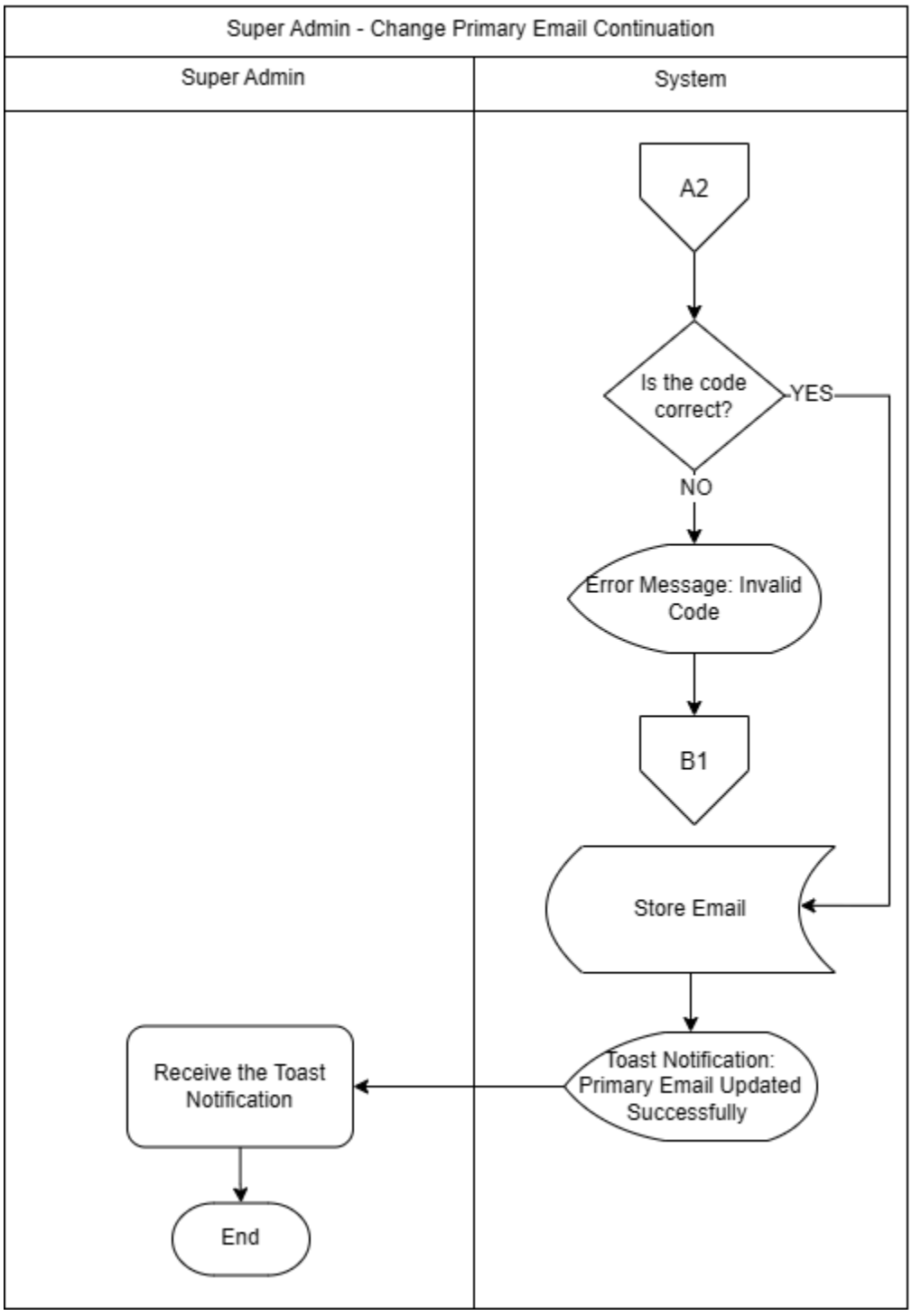


Figure 45. Application Screen Flow Diagram – Admin Dashboard

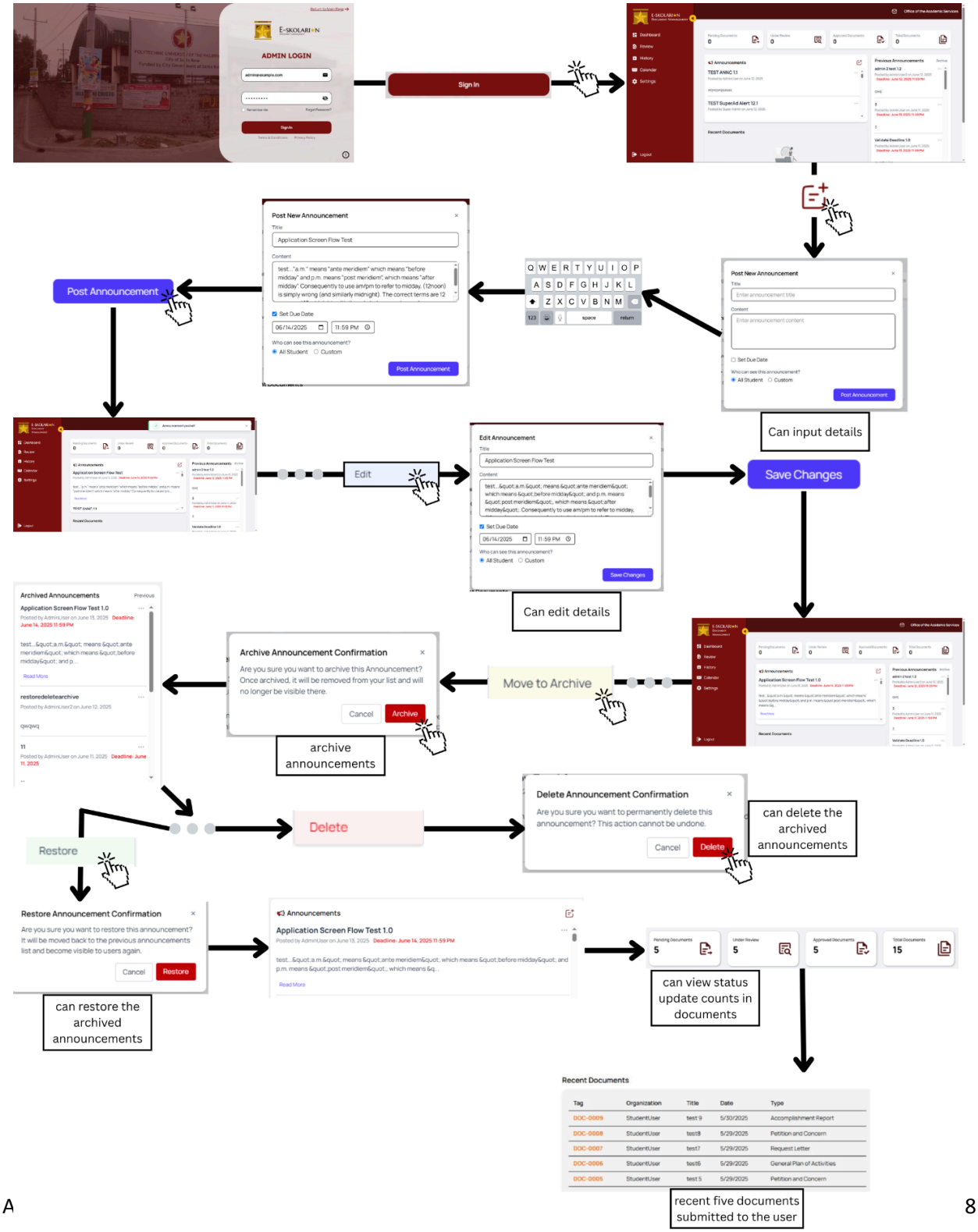


Figure 46. Application Screen Flow Diagram – Admin Settings (Update Profile)

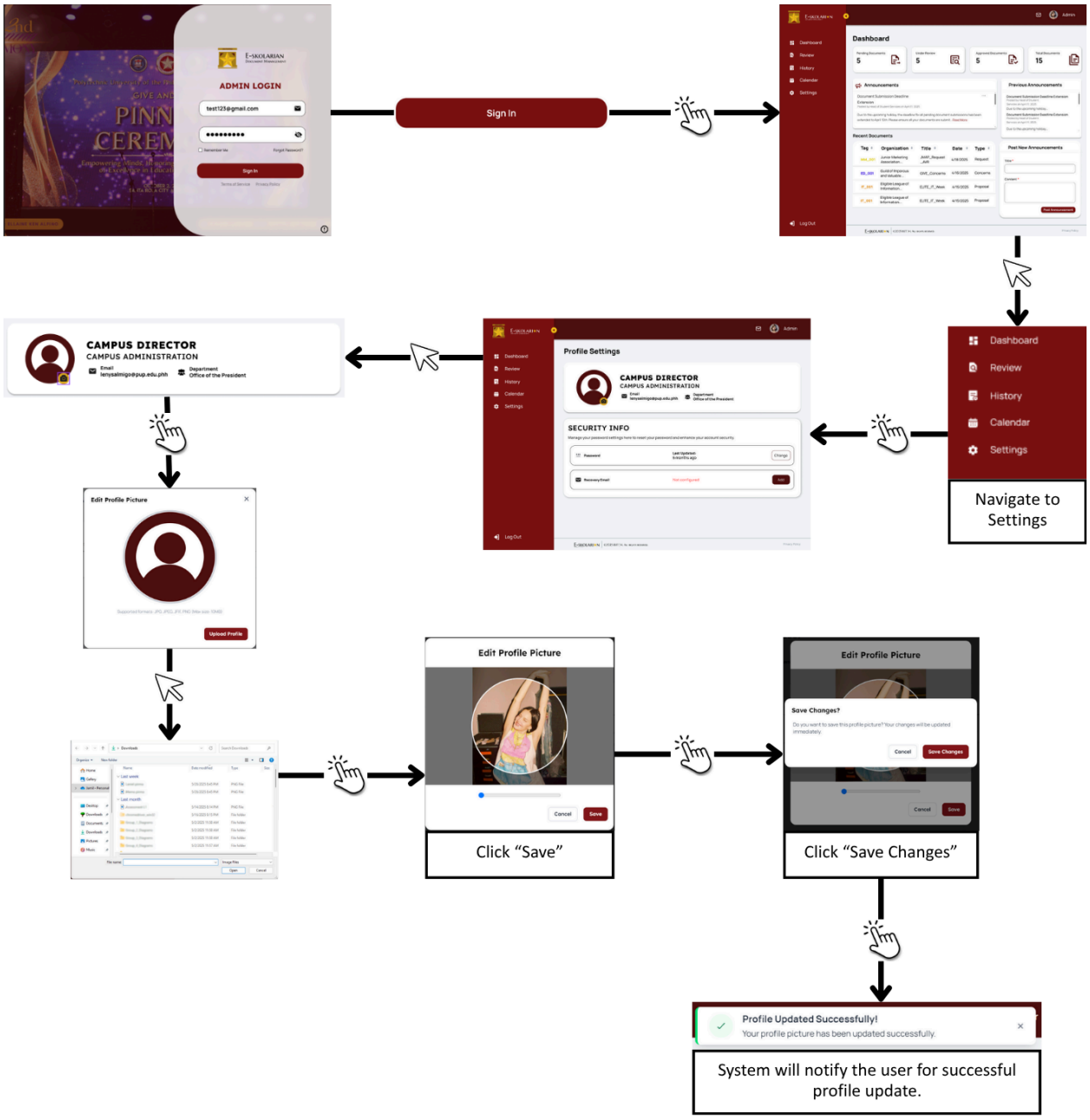


Figure 47. Application Screen Flow Diagram – Admin Settings (Change Password)

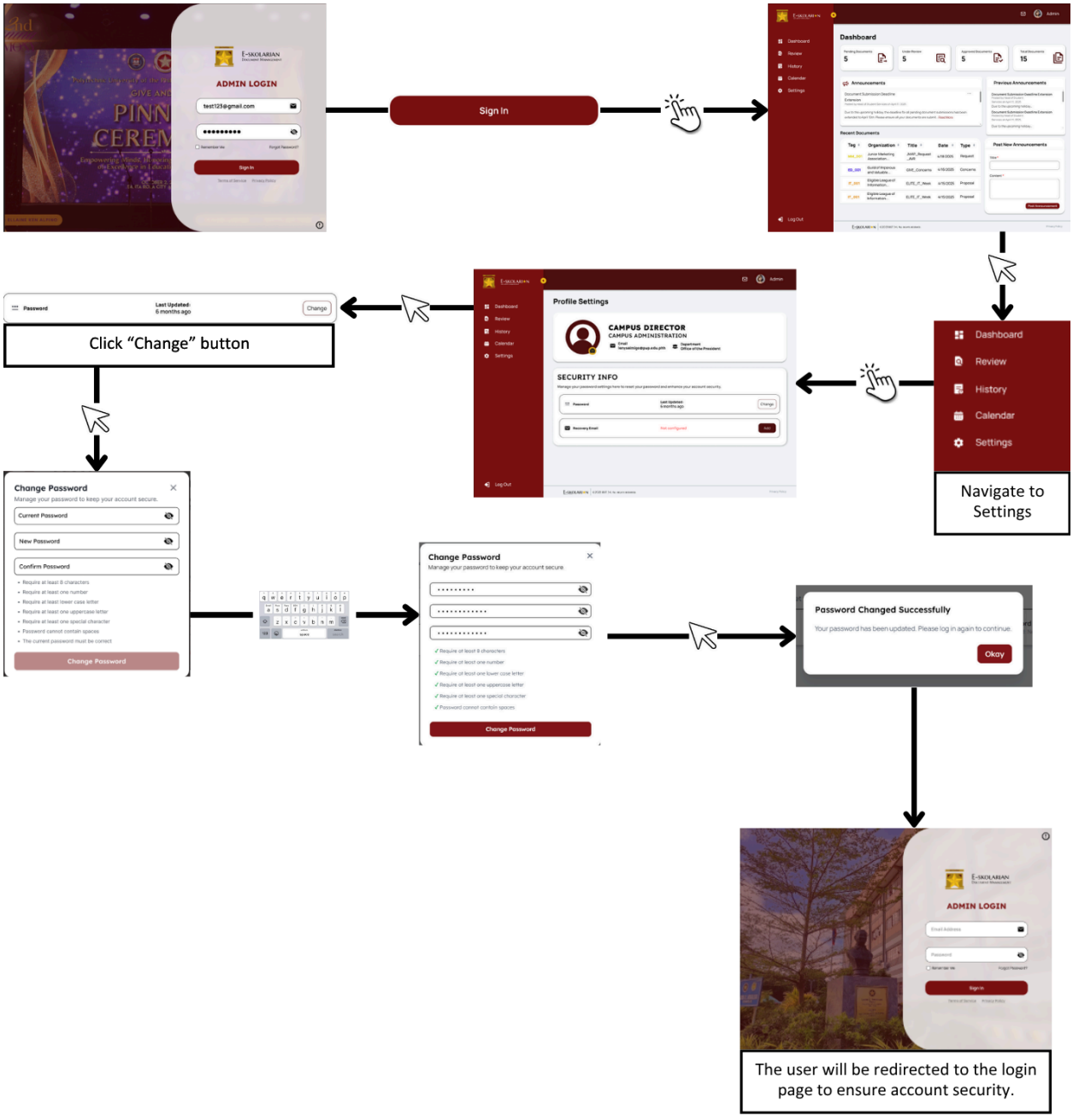


Figure 48. Application Screen Flow Diagram – Admin Settings (Add Recovery Email)

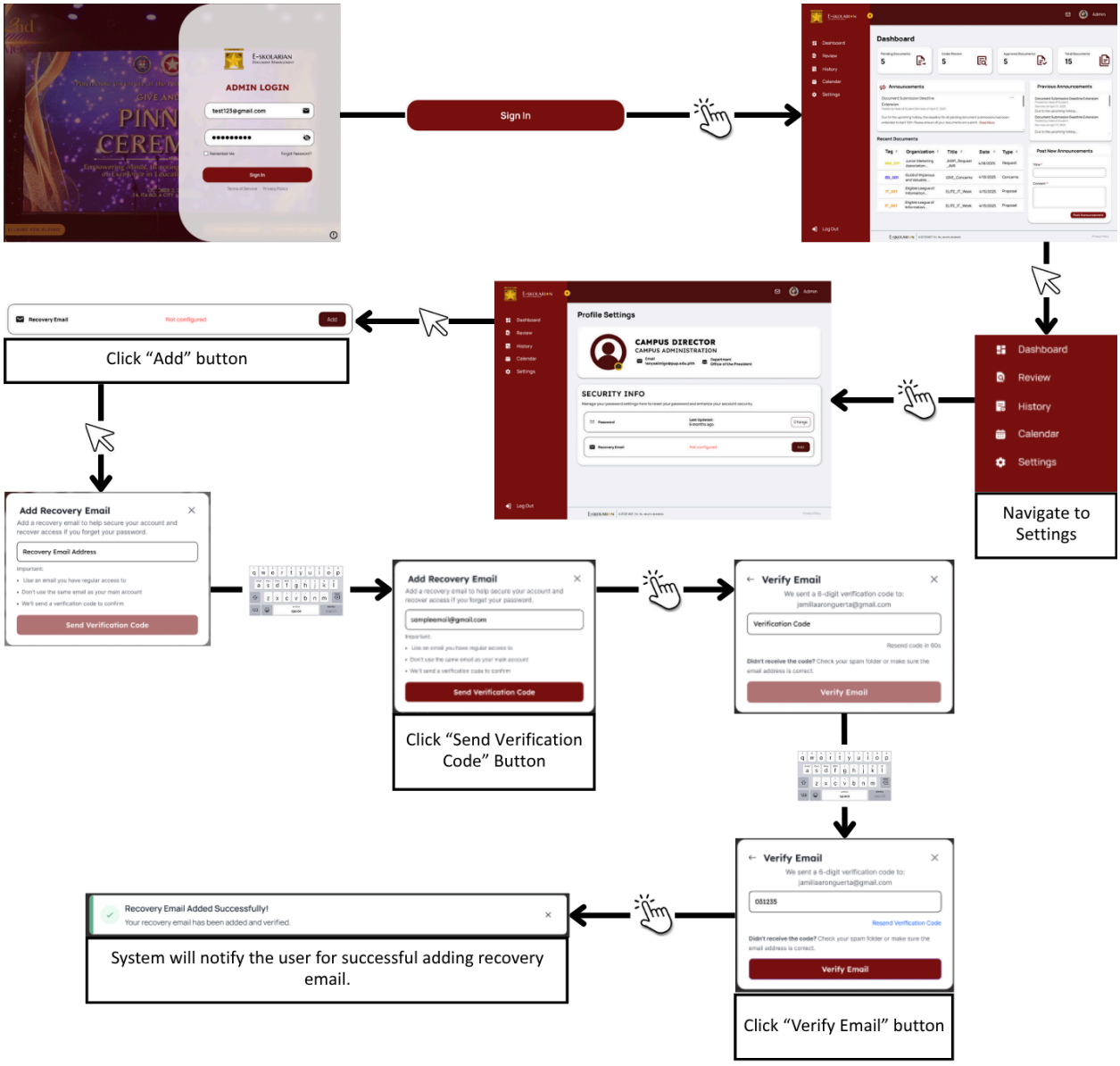


Figure 49. Application Screen Flow Diagram – Student Organization Dashboard



Figure 50. Application Screen Flow Diagram – Admin Settings (Update Profile)

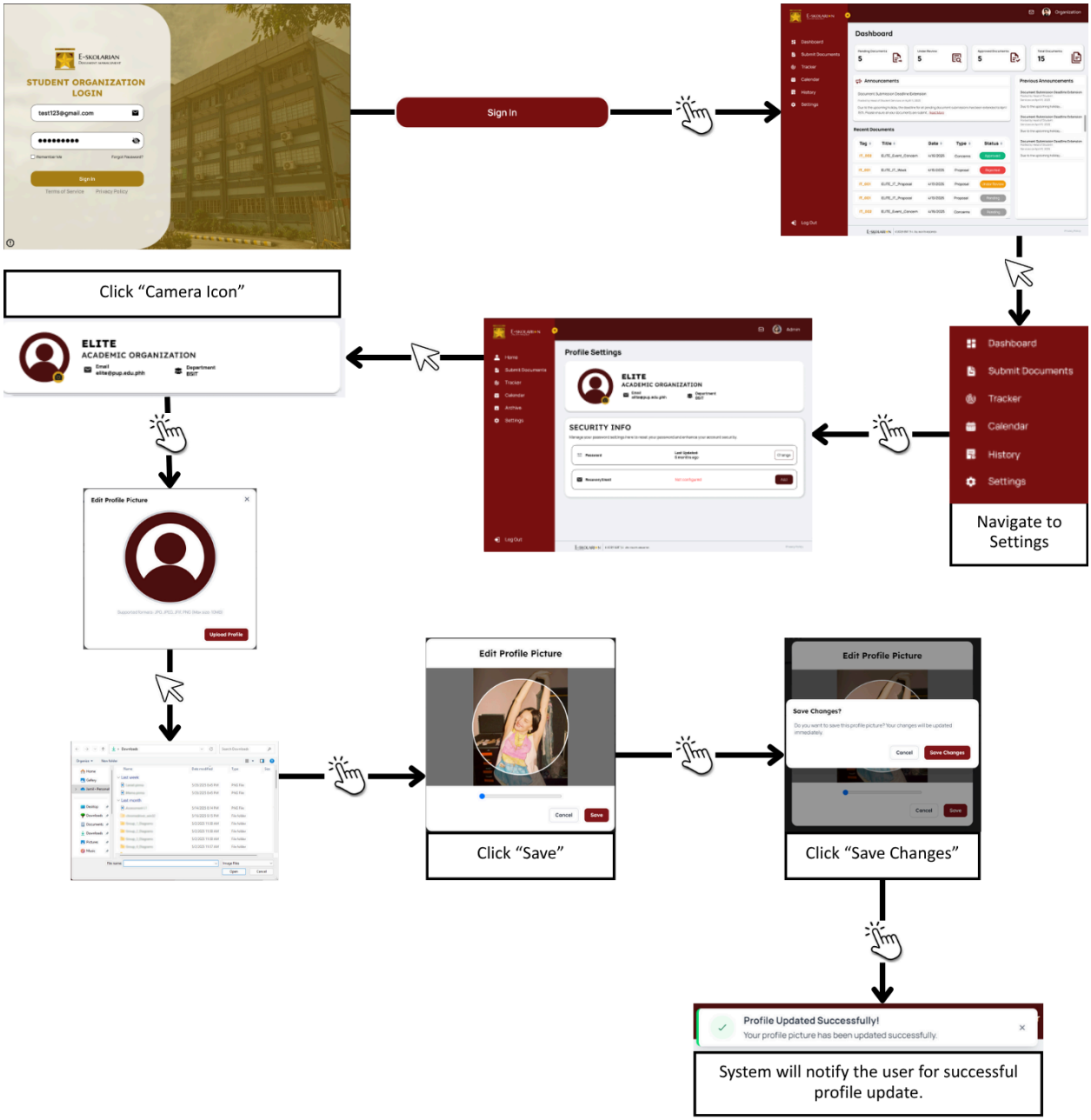


Figure 51. Application Screen Flow Diagram – Admin Settings (Change Password)

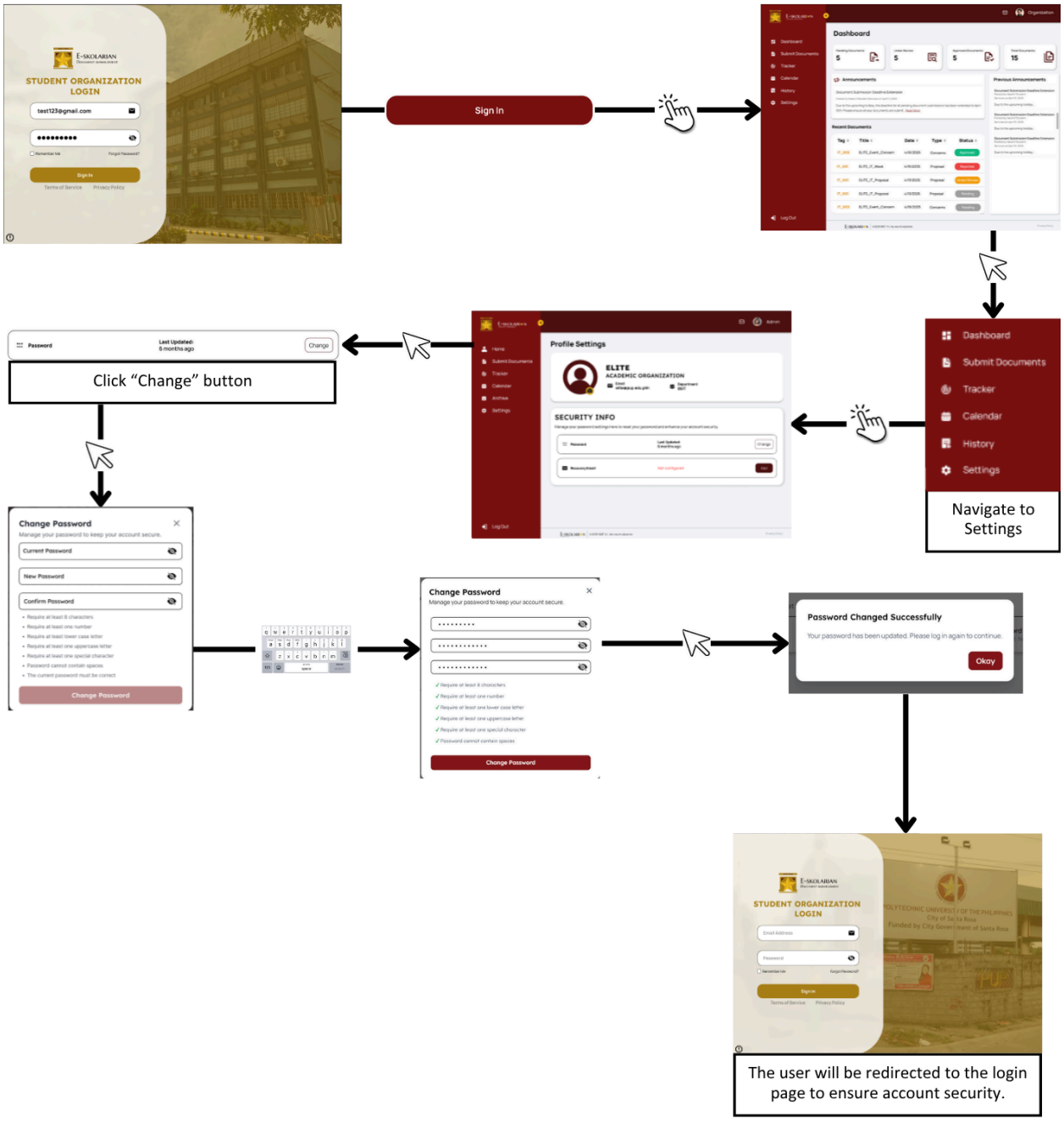


Figure 52. Application Screen Flow Diagram – Admin Settings (Add Recovery Email)

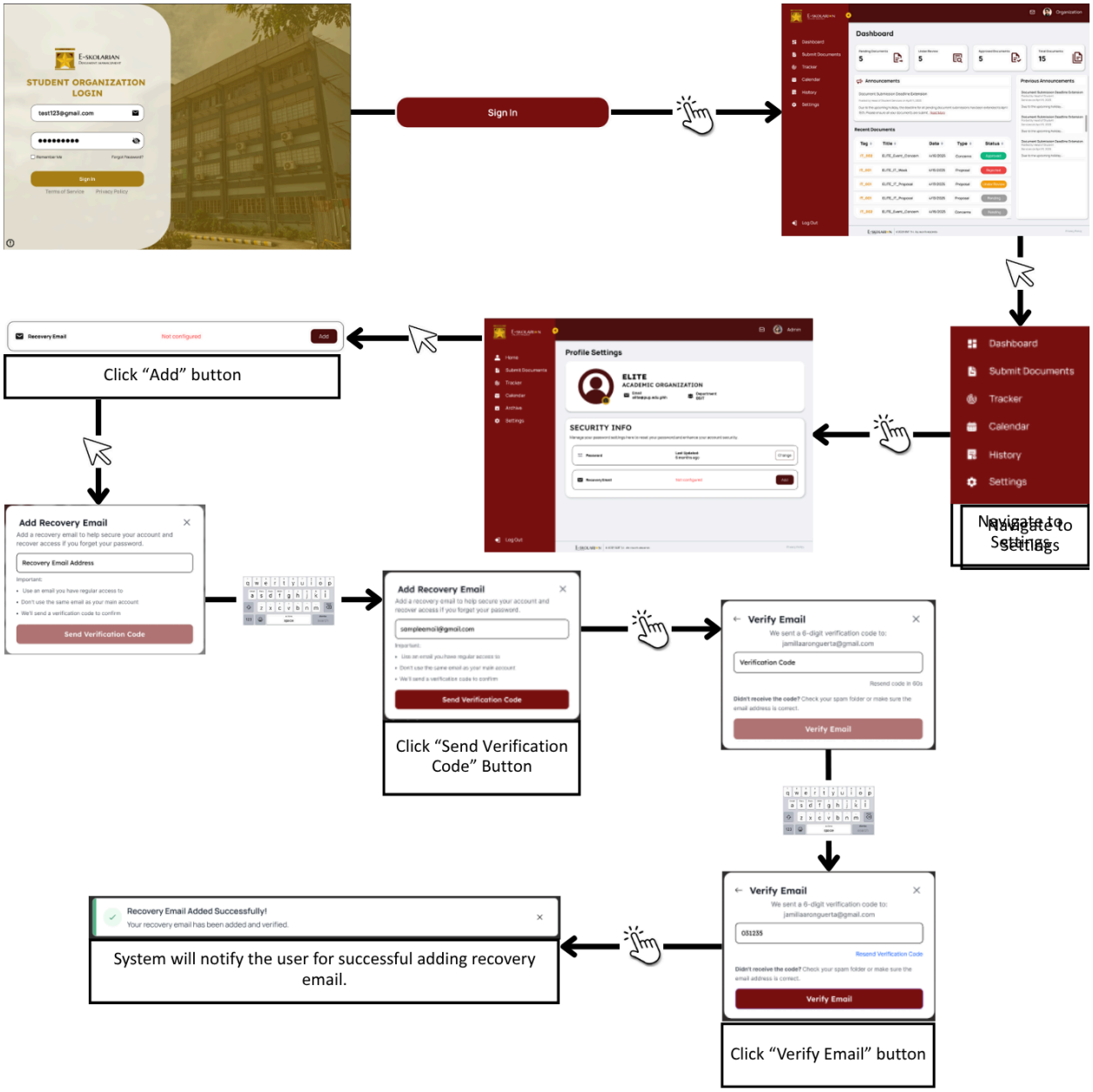


Figure 53. Application Screen Flow Diagram – Super Admin Dashboard (Announcement Functionality)

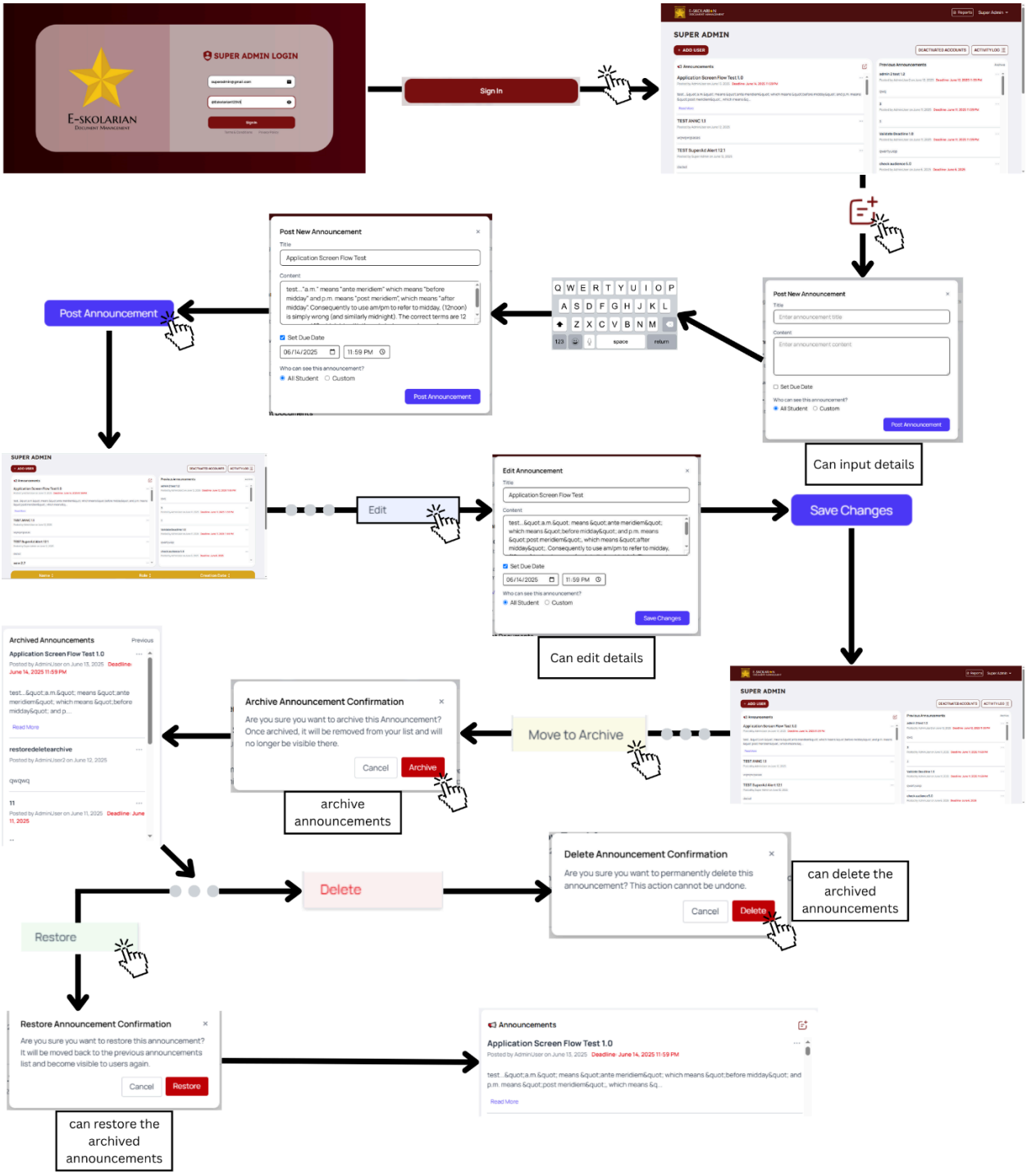


Figure 54. Sitemap



2.2 System Actors

2.2.1 User Roles and Responsibilities / Authority Requirements

User/Role	Example	Frequency of Use	Security/Access, Features Used	Additional Notes
Student Organization	PUP Santa Rosa Campus – Recognized Student Organization	Frequent	Access is limited to their own organization's dashboard. The module displays document statistics (Pending, Under Review, Approved, Total), announcements from the admin, and recent document submissions.	Dashboard updates automatically as statuses change.
Campus Administration	PUP Santa Rosa Campus – Designees from the Office of Student Services, Administrative Services, and Academic Services	Frequent	Granted full access to view submission data across all organizations. Responsibilities include managing announcements (posting and editing) and monitoring recent document submissions per organization.	Responsible for monitoring all org submissions and communicating through announcements.
Super Admin	PUP Santa Rosa Campus – ICT Head/System Admin	Occasional	Ability to add new users to the system and assign them roles (academic org, non-academic org, student services, administrative services, etc.). Oversees user role assignments and access permissions.	Not involved in daily document processing tasks (mainly responsible for user access control).

2.3 Dependencies and Change Impacts

2.3.1 System Dependencies

- **Web Server –** The system requires a web server such as Apache to host the application like XAMPP.
- **Browser Requirements** – The system requires a modern browser for proper functionality such as Google Chrome, Mozilla Firefox and Microsoft Edge.
- **Device Compatibility** – Employees and admins need laptops, smartphones, or tablets to access the system.
- **Database Platform** – MySQL handles data storage for administrative and student account information.
- **Development Technology Stack**– The application is built using the Laravel PHP framework for backend functionality, with Tailwind CSS implementing the responsive user interface design.

2.3.2 Change Impacts

The Dashboard and Analytics Module of the E-skolarian Organization Document Management System can go through the following changes, which could impact system operations and user behavior for both administrators and students:

- Operational process - Through this dashboard, it makes document monitoring and communication easy. Student representatives and admin facilitators will shift from manual tracking or fragmented systems to a unified, real-time overview. This may require adjustments in daily routines and coordination processes.
- User Familiarization - The interface and dashboard features, such as live document stats and announcement cards, will require users to adapt to a new way of accessing and using information. Introductory instructions and step-throughs might be required to enable smooth transition.
- Document Transparency and Monitoring - Greater transparency in document processing will benefit both user groups through the use of visual indicators (pending, under review, approved). This enhancement promotes better accountability with reduced requirements for unnecessary follow-ups.
- Announcement Management and Delivery - For students, access and centralization of announcements are greatly enhanced. For administrators, this module places a responsibility to manage and update announcements within the system itself, changing the way communications are handled across the organization.

3. Functional Specifications

The Dashboard and Analytics Module of the E-skolarian: PUPSRC Organization Document Management System (ODMS) provides two types of user dashboards: Admin Dashboard and Student

Organization Dashboard. The Admin Dashboard provides administrators with the tools to monitor and manage document submissions, track their statuses, and oversee announcements across all student organizations. On the other hand, the Student Organization Dashboard is intended for student organization representatives, giving them the ability to track their organization's document submissions, view relevant announcements, and stay informed about important deadlines and updates. The two tables below provide an overview of the main features of each dashboard.

Admin Dashboard:

Functionalities	Description
Display Pending Document Count	Shows the number of documents awaiting processing. The count should be updated in real-time as new documents are submitted or statuses are updated.
Display Under Review Document Count	Displays the number of documents currently under review. Updates automatically as documents move into the review phase.
Display Approved Document Count	Displays the number of documents that have been approved. This count is updated in real-time as documents are approved by administrators.
Display Total Document Count	Shows the total number of documents across all organizations, regardless of status. This provides an overall view of document submissions.
Show Recent Document	Displays the most recent document submissions with essential information such as document title, organization name, date submitted, and document type.
Post Announcement	Allows the admin to post announcements to all student organizations. Admin can include a title and content for each announcement.
Edit Announcement	Enables the admin to modify existing announcements, ensuring that all information stays up to date.
View Past Announcement	Displays previously posted announcements along with details of who posted them and when, allowing for easy tracking of past communications.

Student Organization Dashboard:

Functionalities	Description
Display Organization's Pending Document Count	Shows the number of documents submitted by the student organization that are awaiting processing.
Display Organization's Under Review Document Count	Displays the number of documents submitted by the student organization that are currently under review.
Display Organization's Approved Document Count	Shows the number of documents from the student organization that have been approved.
Display Organization's Total Document Count	Displays the total number of documents submitted by the student organization, regardless of their status.
Display Latest Announcement	Displays the most recent announcement posted by the admin, including the title, content, and posting date.
View Previous Announcement	Displays past announcements, including the title, who posted it, and when it was posted. This allows the representative to track prior communications.
Show Recent Organization Submissions	Displays the most recent document submissions from the student organization, showing key details like document tag, title, type, date submitted, and status. Sorting is available based on column headers.

3.1 Dashboard and Analytics Module

3.1.1 Purpose/ Description

The Dashboard and Analytics Module is the main user interface of the E-skolarian: PUPSRC Organization Document Management System (ODMS), featuring role-based access. The module provides organization presidents with real-time information about their document submissions, including status overviews, recent activity, and official announcements, thus keeping them well-informed and responsible. Meanwhile, it provides administrators with enhanced management tools to track and manage document submissions from all organizations, send announcements, and enable timely communication. By displaying relevant data and controls in a well-organized and role-based format, this module enables transparency, efficiency, and coordination among student leaders and administrative offices.

3.1.2 Use case

UC-1	View Document Status Count and Total (Admin Dashboard)
Primary Actor(s)	Admin
Stakeholders and Interest	Campus Designees from the Office of Student Services, Administrative Services, and Academic Services aims to track document submissions across student organizations by viewing the real-time counts of documents in different statuses (Pending, Under Review, Approved).
Trigger	The admin accesses the Admin Dashboard to view the real-time document status counts.
Pre-conditions	Admin is logged into the system currently accessing the dashboard, and documents with various statuses (Pending, Under Review, Approved) exist in the system.
Post-conditions	The Admin Dashboard displays the correct real-time counts for Pending, Under Review, Approved, and Total documents across all student organizations.
Main Success Scenario	1. Admin logs into the system and accesses the dashboard. 2. Admin views the Document Status Overview widget showing: ● Pending Documents Count ● Under Review Documents Count ● Approved Documents Count ● Total Documents Count 3. Admin verifies that the counts match the actual status of documents.
Extensions	If document status counts do not update in real-time, the admin refreshes the page or contacts technical support.
Priority	High
Special Requirements	● The system should update document counts in real-time. ● Data privacy and security measures should protect sensitive student organization information.
Open Questions	How will the system handle discrepancies in document counts between statuses?

UC-2	View Document Status Count and Total (Student Dashboard)
Primary Actor(s)	Student Organization Representative
Stakeholders and Interest	Student Representatives of each organization **need** to track the status of their organization's document submissions (Pending, Under Review, Approved).
Trigger	The student representative logs into the system (redirected into the dashboard) to view their organization's document submission status.
Pre-conditions	The student representative is logged in, and the organization has documents in various statuses (Pending, Under Review, Approved).

Post-conditions	The Student Dashboard displays the correct real-time counts of Pending, Under Review, Approved, and Total documents for the student organization.
Main Success Scenario	1. The student representative logs into the system and accesses the dashboard. 2. The student representative views the Document Status Overview widget showing: ● Pending Documents Count ● Under Review Documents Count ● Approved Documents Count ● Total Documents Count 3. The student representative verifies that the counts are correct for their organization.
Extensions	If the data does not update in real-time, the student representative can refresh the page or contact the admin/super admin for clarification.
Priority	High
Special Requirements	● The system should provide real-time updates for document counts. ● Data privacy ensures the student representative can only access their own organization's documents.
Open Questions	How will the system handle discrepancies in document counts between statuses?

UC-3	Post Announcement
Primary Actor(s)	Admin, Super Admin
Stakeholders and Interest	System Admin and Campus Designees from the Office of Student Services, Administrative Services, and Academic Services need to post an announcement to inform all student organizations about important updates, events, or reminders. The posted announcement should appear immediately on the Student Dashboard.
Trigger	The admin clicks the Post Announcement button after entering the announcement details in the Admin Dashboard.
Pre-conditions	Admin is logged into the system currently accessing the dashboard.
Post-conditions	● The new announcement is displayed in real-time on the Student Dashboard. ● The newly posted announcement becomes the most recent announcement.
Main Success Scenario	1. Admin logs into the system and accesses the dashboard. 2. Admin navigates to the Post New Announcement section. 3. Admin enters the Title and Content of the announcement. 4. Admin reviews the details entered. 5. Admin clicks the Post Announcement button.

	6. The system posts the announcement in real-time, and the new announcement is displayed on the Student Dashboard for all student organizations.
Extensions	If the admin decides to cancel, he/she can clear the fields or navigate away without posting.
Priority	High
Special Requirements	● Only authorized users (Admin) can post announcements. ● The system must post the announcement in real-time and immediately reflect it on the Student Dashboard. ● The Super Admin can manage and access all announcements, while the Admin can only manage and access their own announcements.
Open Questions	What improvements could be made to the announcement posting process to ensure admins can quickly create clear, engaging, and accurate announcements for all student organizations?

UC-4	Edit Recent Announcement
Primary Actor(s)	Admin, Super Admin
Stakeholders and Interest	Campus Designees from the Office of Student Services, Administrative Services, and Academic Services need the ability to edit recent announcements to correct any errors, update details, or remove outdated information.
Trigger	The admin clicks the ellipsis icon on a specific recent announcement and selects the Edit option to open the Edit Modal for that announcement.
Pre-conditions	● Admin is logged into the system currently accessing the dashboard. ● There should be at least one posted announcement that can be edited and must be recent.
Post-conditions	The most recent announcement is updated with the new changes, and the updates are reflected on the Student Dashboard in real-time.
Main Success Scenario	1. Admin logs into the system and accesses the dashboard. 2. Admin clicks the ellipsis icon on the recent announcement they want to modify. 3. Admin selects Edit from the dropdown menu. 4. The Edit Modal opens, allowing the admin to modify the Title and Content of the announcement. 5. Admin makes necessary changes to the announcement. 6. Admin clicks Save to confirm the changes. 7. The system updates the announcement in real-time, and the updated announcement is displayed on the Student Dashboard for all student representatives to view.
Extensions	● Admin can click Cancel in the modal, and no changes will be saved.

	● Only recent announcements are editable; the Edit option does not appear for previous or archived announcements.
Priority	High
Special Requirements	● Only the recent announcement can be edited. ● Real-time updates must be reflected on the Student Dashboard.
Open Questions	What additional features or safeguards could help admins confidently and efficiently edit announcements without risking mistakes or confusion for students?

UC-5	Archive Announcement
Primary Actor(s)	Admin, Super Admin
Stakeholders and Interest	System Admin and Campus Designees from the Office of Student Services, Administrative Services, and Academic Services need the ability to archive recent and previous announcements to keep the system focused on current and relevant information, while still allowing access to older posts for reference.
Trigger	The admin/super admin clicks the ellipsis icon on a specific recent/previous announcement and selects the Move to Archive option.
Pre-conditions	● Admin/Super Admin is logged into the system currently accessing the dashboard. ● At least one recent or previous announcement is available to be archived.
Post-conditions	● The selected announcement is removed from the visible list (Recent or Previous). ● The announcement is stored in the Archive section, accessible only to the admin/super admin for future reference or retrieval.
Main Success Scenario	1. Super Admin/Admin logs into the system and accesses the dashboard. 2. Super Admin/Admin identifies an announcement (recent or previous) to archive. 3. Super Admin/Admin clicks the ellipsis icon on a specific recent/previous announcement and selects the Move to Archive option. 4. A confirmation modal appears. 5. Admin/Super Admin clicks Confirm. 6. The system archives the announcement and removes it from the active list in real-time. 7. A success message confirms the action, and the announcement is added to the Archive section.
Extensions	● Admin/Super Admin can click Cancel in the confirmation modal that will appear, and no changes will be saved.

	● Archived announcements are accessible within the same container as Previous Announcements, where clicking the Archive link (located in the Previous Announcements header) transitions the view to display archived announcements.
Priority	High
Special Requirements	● Recent announcements and previous announcements must both be archivable. ● Real-time updates are required upon archiving. ● Archived announcements must be accessible in an organized Archive section, available only to admins/super admin. ● The Super Admin can manage and access all announcements, while the Admin can only manage and access their own announcements.
Open Questions	Should the Archive section support filters or categories for easier management?

UC-6	Delete Announcement
Primary Actor(s)	Admin, Super Admin
Stakeholders and Interest	System Admin and Campus Designees from the Office of Student Services, Administrative Services, and Academic Services need the ability to permanently delete outdated or irrelevant archived announcements to keep the archive section organized and manageable.
Trigger	The admin/super admin clicks the ellipsis icon on a specific archived announcement and selects the Delete option.
Pre-conditions	● Admin/Super Admin is logged into the system currently accessing the dashboard. ● At least one archived announcement is available for deletion.
Post-conditions	● The selected archived announcement is permanently deleted from the system. ● The deleted announcement can no longer be restored or referenced.
Main Success Scenario	1. Admin/Super Admin logs into the system and accesses the dashboard. 2. Admin/Super Admin clicks the "Archive" link in the Previous Announcements header. 3. The Archive view is displayed with a list of archived announcements. 4. Admin/Super Admin clicks the ellipsis icon on the specific announcement to delete. 5. Admin/Super Admin selects Delete from the dropdown menu. 6. A confirmation modal appears, warning that the action is permanent. 7. Admin/Super Admin clicks Confirm.

	8. The system permanently deletes the announcement and displays a success message.
Extensions	● Admin/Super Admin clicks Cancel in the confirmation modal, and no changes occur. ● Only archived announcements can be deleted; delete options are hidden for recent and previous announcements.
Priority	Medium
Special Requirements	● Only archived announcements can be deleted. ● A confirmation modal must appear before deletion. ● Deleted announcements must not be recoverable.
Open Questions	Should there be a log of deleted announcements for auditing purposes?

UC-7	Restore Announcement
Primary Actor(s)	Admin, Super Admin
Stakeholders and Interest	System Admin and Campus Designees from the Office of Student Services, Administrative Services, and Academic Services need the ability to restore archived announcements in case they need to be referenced or made active again.
Trigger	The admin/super admin clicks the ellipsis icon on a specific archived announcement and selects the Restore option.
Pre-conditions	● Admin/Super Admin is logged into the system currently accessing the dashboard. ● At least one archived announcement is available for restoration.
Post-conditions	● The selected archived announcement is moved back to the Recent or Previous announcement list, depending on where it was archived from. ● The announcement becomes visible again in the appropriate section of the dashboard.
Main Success Scenario	1. Admin/Super Admin logs into the system and accesses the dashboard. 2. Admin/Super Admin clicks the "Archive" link located in the Previous Announcements header. 3. The Archive view is displayed, showing all archived announcements. 4. Admin/Super Admin clicks the ellipsis icon on the specific announcement to be restored. 5. Admin/Super Admin selects Restore from the dropdown menu. 6. A confirmation modal appears. 7. Admin/Super Admin clicks Confirm.

	8. The system restores the announcement and places it back into either the Recent or Previous section, depending on its original category. 9. A success message is displayed to confirm the action.
Extensions	● Admin/Super Admin clicks Cancel in the confirmation modal, and the announcement remains archived. ● Only archived announcements can be restored; deleted announcements cannot be restored.
Priority	Medium
Special Requirements	● Only archived announcements can be restored. ● The system must track the original category of the announcement (Recent or Previous) and restore it accordingly. ● Real-time updates are required upon restoration. ● The Super Admin can manage and access all announcements, while the Admin can only manage and access their own announcements.
Open Questions	Should a timestamp be updated or retained when the announcement is restored?

UC-8	Change/Upload Profile Photo
Primary Actor(s)	Admin, Super Admin, Student Organization Representative
Stakeholders and Interest	Super Admin, Student Representatives of an Organization, and the Campus Designees need the ability to upload or change their profile photo to personalize their accounts and improve user identification within the system.
Trigger	The user navigates to Settings, clicks the Edit Icon beside their profile photo, and selects the Upload Profile option from the Profile Preview modal.
Pre-conditions	● The user (admin/student/super admin) is logged into the system. ● The user is on the Settings page.
Post-conditions	● The selected image is uploaded and set as the new profile photo. ● The updated profile photo is displayed across the system.
Main Success Scenario	1. The user logs into the system and navigates to the Settings page. 2. The user clicks the Edit Icon beside their profile photo. 3. A Profile Preview Modal appears, showing the current profile photo or an empty placeholder. 4. The user clicks Upload Profile. 5. A file chooser window opens, allowing the user to select a photo (JPEG, JPG, PNG, JFIF format, not exceeding 10MB). 6. After selection, the chosen photo is displayed in the Profile Preview Modal. 7. The user can crop or zoom the photo. 8. The user clicks Save.

	9. The system uploads the new photo, updates the profile, and displays a success message.
Extensions	● If the user clicks Remove Profile, the profile photo is cleared and replaced with a default placeholder (this option only appears if a profile photo already exists). ● If the selected image is in an unsupported format or exceeds 10MB, an error message is displayed, and the upload is blocked. ● If the user clicks Discard, changes are not saved and the modal closes with the original photo intact.
Priority	Medium
Special Requirements	● File formats allowed: .jpeg, .jpg, .png, .jfif. ● Maximum file size: 10MB. ● Basic image editing (crop and zoom) should be supported before saving. ● Changes must reflect in real time after saving.
Open Questions	Should there be support for animated images (e.g., GIF) in the future?

UC-9	Remove Profile Photo
Primary Actor(s)	Admin, Student Organization Representative, Super Admin
Stakeholders and Interest	Super Admin, Student Representatives of an Organization, and the Campus Designees need the ability to remove their profile photo to revert to a default avatar, ensuring control over how their profile is displayed within the system.
Trigger	The user navigates to Settings, clicks the Edit Icon beside their profile photo, and selects the Remove Profile option from the Profile Preview modal.
Pre-conditions	● The user is logged into the system. ● The user is on the Settings page. ● The user has an existing profile photo uploaded.
Post-conditions	● The existing profile photo is removed. ● A default placeholder/avatar is shown in place of the removed photo across the system.
Main Success Scenario	1. The user logs into the system and navigates to the Settings page. 2. The user clicks the Edit Icon beside their profile photo. 3. A Profile Preview Modal appears displaying the current profile photo. 4. The user clicks the Remove Profile button. 5. A confirmation modal appears asking the user to confirm the action. 6. The user clicks Confirm. 7. The profile photo is removed, replaced with a default avatar or placeholder.

	8. A success message confirms that the profile photo has been removed.
Extensions	● If the user clicks Cancel in the confirmation modal, the profile photo remains unchanged. ● The Remove Profile button only appears if there is an existing profile photo.
Priority	Low
Special Requirements	● The system should immediately reflect the change across all components that display the profile photo. ● A default placeholder or avatar must be in place after removal.
Open Questions	Should users be able to undo the removal if it was done by mistake?

UC-10	Add Recovery Email
Primary Actor(s)	Admin, Student Organization Representative, Super Admin
Stakeholders and Interest	All users (Super Admins, Admins, and Student Organization Representatives) need the ability to add a recovery email to enhance account security and support password recovery when needed. Verified recovery emails ensure that only authorized users can regain access to their accounts.
Trigger	The user navigates to the Settings page and clicks the Add button in the Recovery Email section.
Pre-conditions	● The user is logged into the system. ● The user is on the Settings page. ● No recovery email is currently added, or the user is updating it.
Post-conditions	● The recovery email is successfully added and verified. ● The verified email is stored and can be used for future account recovery actions.
Main Success Scenario	1. The user logs into the system and navigates to the Settings page. 2. The user clicks the Add button in the Recovery Email section. 3. A modal pop-up appears prompting the user to enter a valid recovery email address. 4. The user inputs the email and clicks Add Recovery Email. 5. A 6-digit verification code is sent to the entered email address. 6. A verification modal appears where the user inputs the code. 7. The user clicks Verify Email. 8. The recovery email is saved and verified. 9. A toast message confirms the successful addition of the recovery email.

Extensions	• If the email is invalid or already used by another account, an error message is shown. • If the entered verification code is incorrect, the system prompts the user to re-enter the correct code. • If the user cancels the modal, no email is added and no changes are made.
Priority	Low
Special Requirements	• The system must validate email format and uniqueness. • The verification code must expire after a defined time for security. • The system must notify the user once verification is complete.
Open Questions	Should users be allowed to change or add multiple recovery emails?

3.1.3 Mock-up

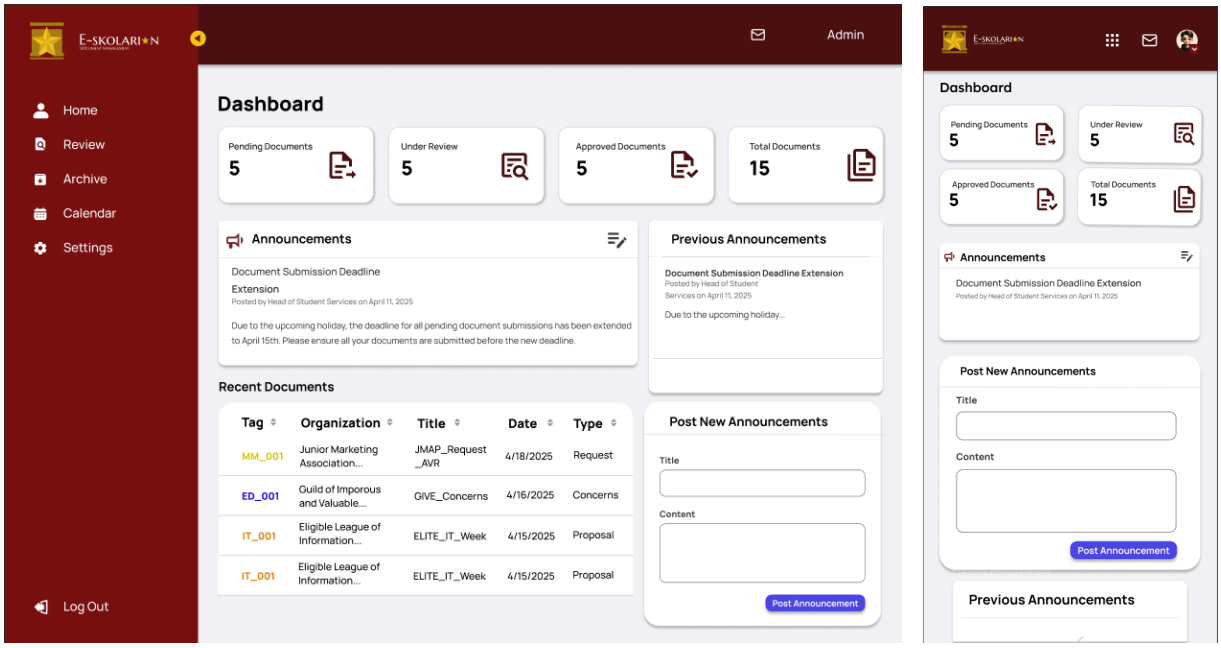


Figure 55: Admin Dashboard

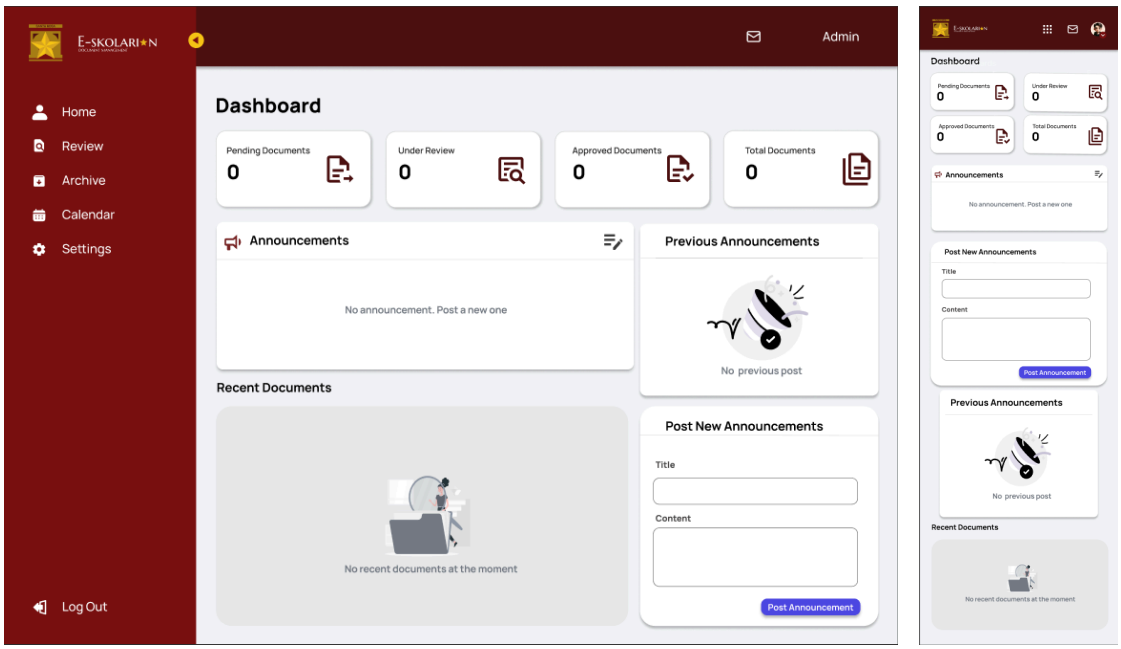


Figure 56: Admin Dashboard (Empty State)

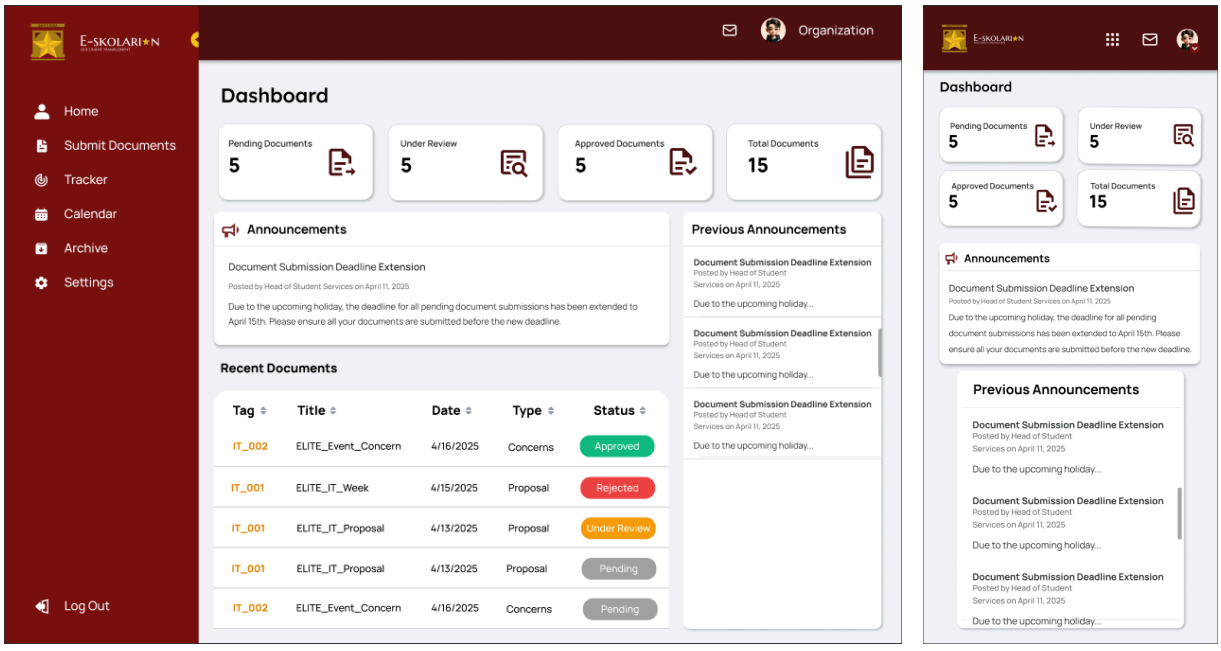


Figure 57: Student Dashboard

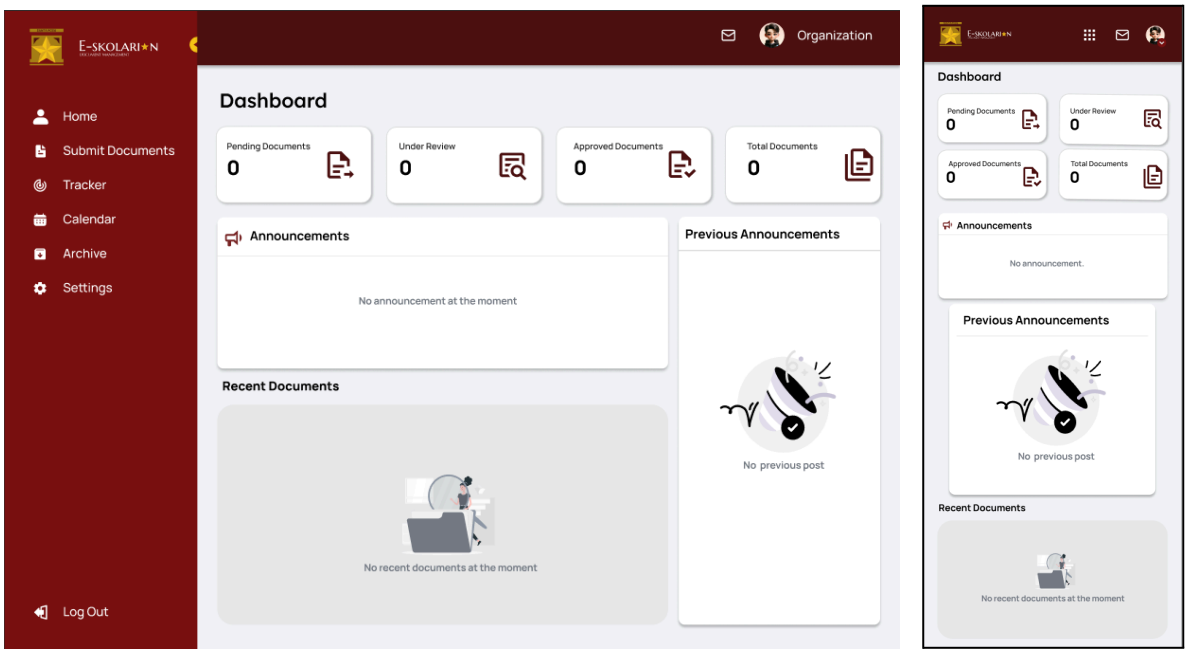


Figure 58: Student Dashboard (Empty State)

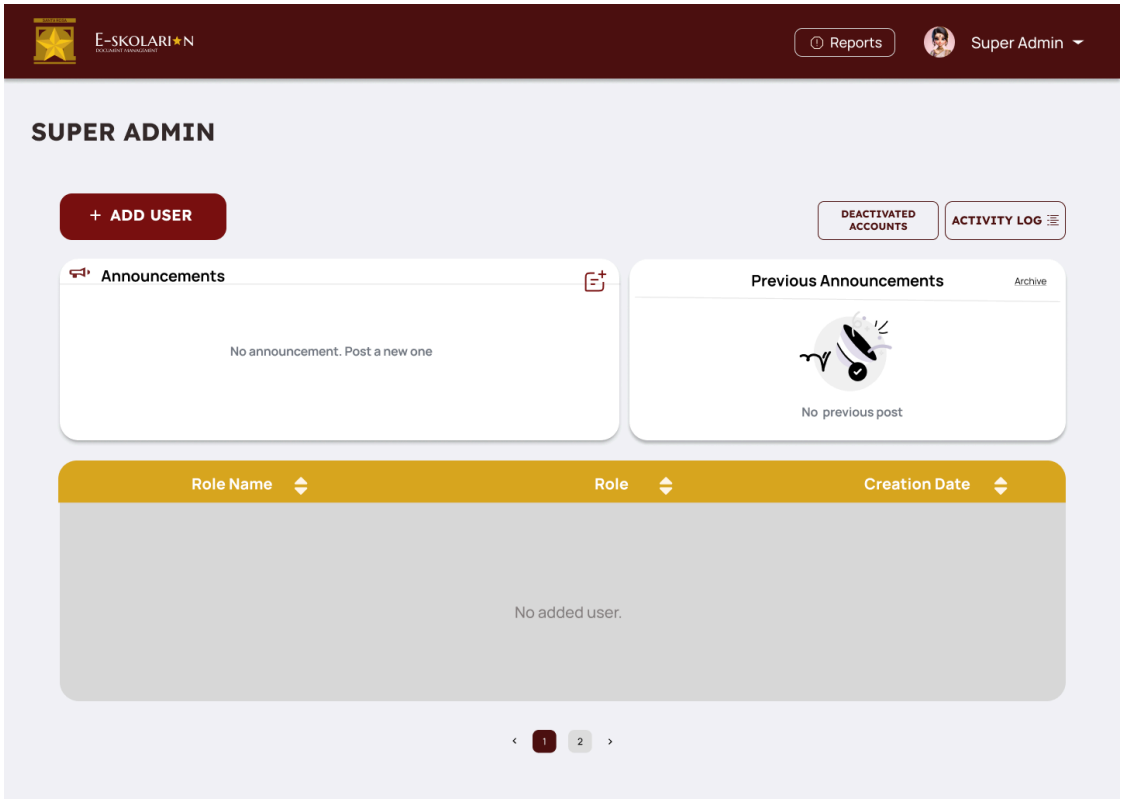


Figure 59: Super Admin Dashboard (Empty State)

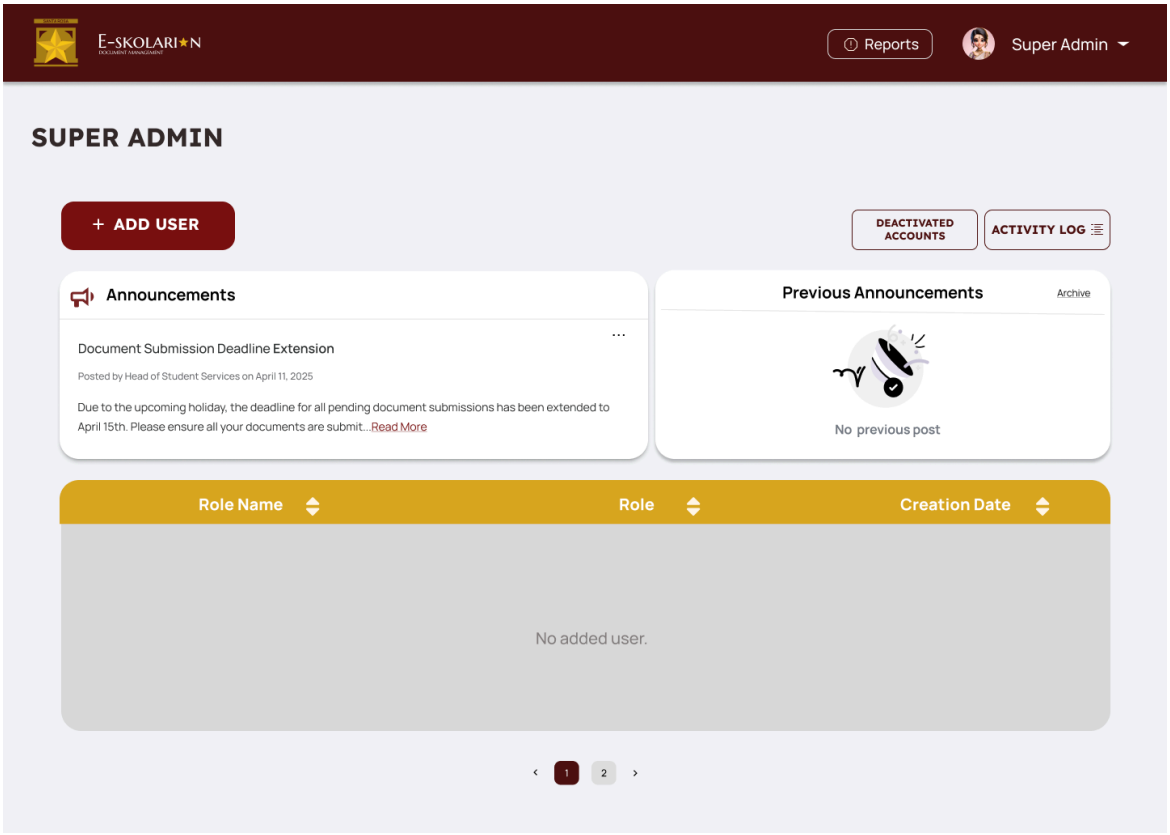


Figure 60: Super Admin Dashboard

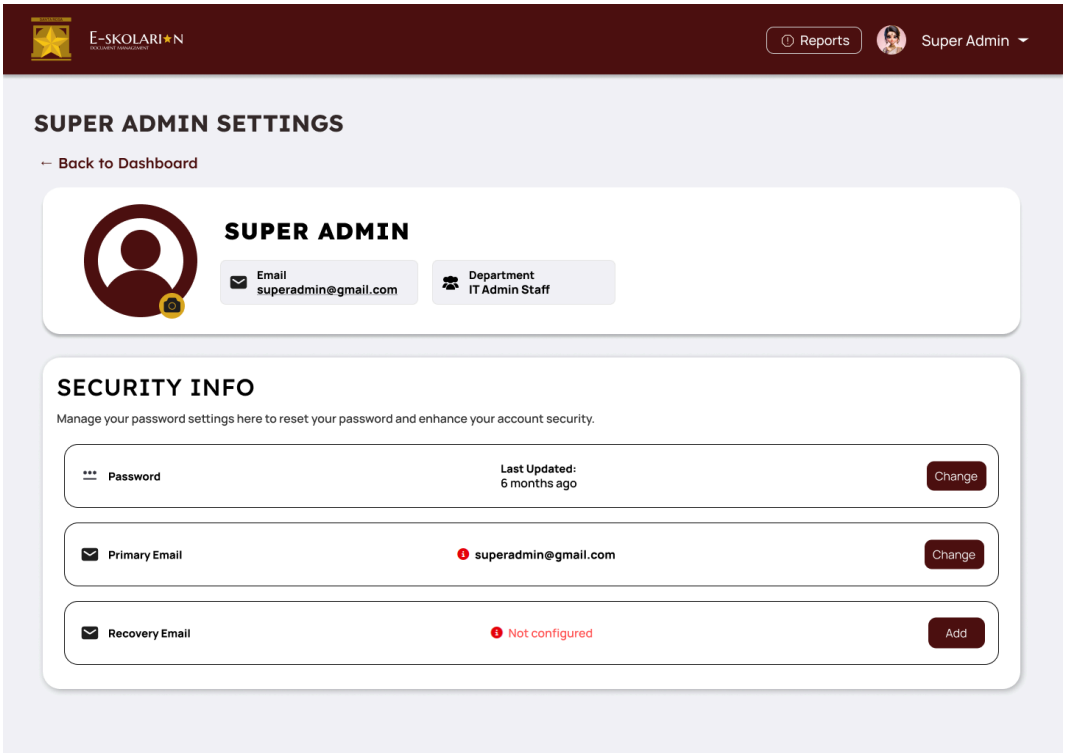
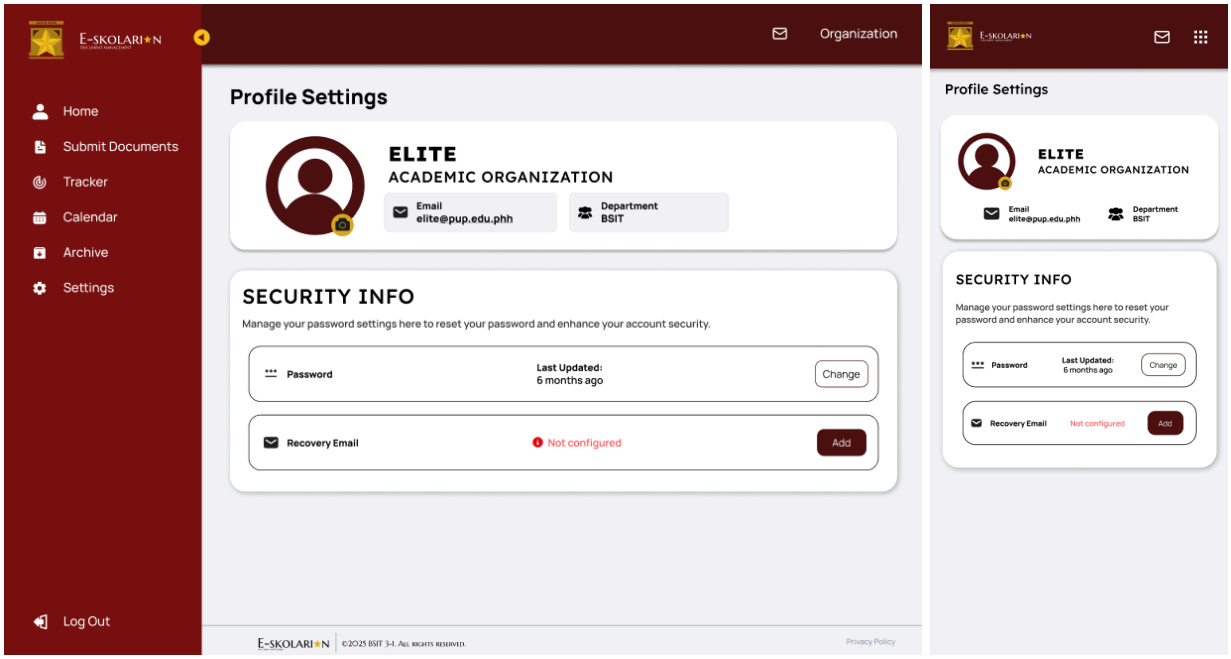


Figure 61: Super Admin Settings



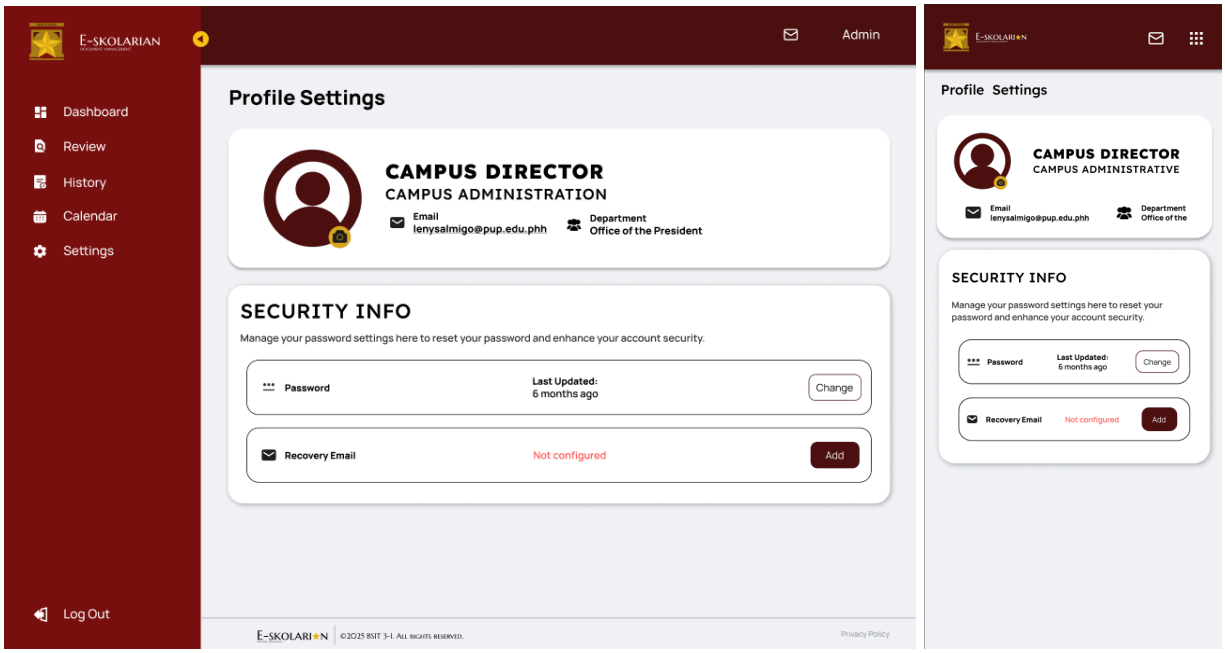


Figure 65: Admin and Student Settings

3.1.4 Functional Requirements

Spec ID	Specification Description	Business Rules/ Data Dependency
FRDB01	The System shall display a document statistics cards (Pending, Under Review, Approved, Total) on the bot student and admin dashboards.	The statistics are limited to their own organization for students. While for admins, statistics reflect overall data across all student organizations.
FRDB02	The system shall show recent document submissions on both the student and admin dashboards.	The visible documents for students are their organizational documents only. Admins view all the submissions from all organizations.
FRDB03	The system shall allow admins to create, edit, and publish announcements that	Announcements include metadata such as the author and timestamp. It will automatically be displayed on all student dashboards.

	will be visible to the student dashboard.	
FRDB04	Both user dashboard shall automatically update to reflect changes in their document statuses.	Data is refreshed in real-time or at defined intervals from the backend submission records.
FRDB05	Only admin users shall have the permission to post or modify announcement.	Role-based access control (RBAC) restricts announcement features to admin users only.
FRDB06	The dashboards shall dynamically render role-specific features and data based on the logged-in user's role.	Upon login, the system identifies the user's role and displays the corresponding dashboard with appropriate access.

3.1.5 Field level specifications

Form Elements

Dashboard & Analytics

Call-out	Field Label	UI Control	Mand?	Editable	Data Type	Value Set	Default Value	Data Example	Data Source
1	Pending Document	Info Card	No	No	Integer	0 -?	0	5	Document DB
2	Under Review	Info Card	No	No	Integer	0 -?	0	5	Document DB
3	Approved Document	Info Card	No	No	Integer	0 -?	0	5	Document DB
4	Total Document	Info Card	No	No	Integer	0 -?	0	5	Document DB
5	Announcements	Text Box	No	Yes	String	None	None	Document Submiss	Created content of admin

								ion Deadline...	
6	Previous Announcements	Static List	No	No	String	None	None	Due to the upcoming holiday …	Announcement DB
7	Post New Announcement	Form Section	Yes	Yes	None	None	None	None	Admin input
7.1	Title	Text Input	Yes	Yes	String	None	None	Document Submission Deadline...	Admin input
7.2	Content	Text area	Yes	Yes	Alphanumeric	None	None	Contentbody of announcement	Admin input
8	Recent Documents Table	Table	No	No	None	None	None	None	Pulled from database
8.1	Tag	Text	No	No	String	Custom format (e.g., IT_001)	None	MM_001	Auto-generated
8.2	Organization	Text	No	No	String	None	None	Junior Marketi	User/org metadata

								ng Association	
8.3	Title	Text	No	No	String	None	None	JMAP Request...	User-provided title
8.4	Date	Text	No	No	Date	YYYY-MM-DD format	NONE	2025-16-2	System Timestamp
8.5	Type	Text	No	No	String	Type of Document	None	Proposal	User-defined

Form Business Rules and Dependencies:

Field Label	Validation / Business Rules	Error Messages	Data Dependencies	Additional Info/ Notes
Title (Announcement - Admin View)	Required	A title is required.	None	Admin Only
Content (Announcement - Admin View)	Required	Content cannot be empty.	None	Admin Only
Tag (Recent Documents)	Auto-generated, unique	Tag must be unique.	Derived from Document Info	Student/Admin View

Buttons, Links and Icons:

Button, Link, Icon Label	OnClick Event	Other Event	Visible	Enabled Vs Disabled	Navigate To	Validation	Dependencies
Post Announcement	Validates if the field is not empty and then proceed to posting	None	Admin Dashboard	Enabled when title & content filled	Stay on Dashboard	Title and Content required	Admin Role

Edit Icon (Announcement)	Open edit modal	Hover effect	Admin Dashboard	Enabled for existing announcements	Edit Popup	None	Admin Role
Cancel Button (Edit Announcement Modal)	OnClick, it will ask a confirmation if the user want to cancel the action	None	Admin Dashboard	Enabled when the modal is active	Stay on Dashboard	None	Admin Role
Save Button (Edit Announcement Modal)	Onclick, it will ask confirmation to the user if they want to save changes	Notification toast for success changes	Admin Dashboard	Enabled when the modal is active	Stay on Dashboard	Title and Content required	Admin Role
Dynamic List	It will redirect to the submission tracker/review	Hover effect	Student and admin dashboard	Enable if there's available content on the list	Redirect to submission tracker/review page	None	Student/Admin Role

4. System Configurations

This section provides an overview of all the steps and setups required to configure the E-skolarian: PUPSRC Organization Document Management System (ODMS) — Dashboard and Analytics Module.It details the requirements for installation, database setup, user account management, security enforcement, application integration, testing, and documentation. These configurations are designed to ensure that the system operates efficiently, securely, and in alignment with the institution's existing technological infrastructure and standards.

Installation and Setting Up

Purpose: Install a browser-accessible web application hosted on Amazon Web Services (AWS) and configure the necessary components to enable the Submission Tracker & Version Control Module within the institution's environment.

Alternatives: Deployment can be adjusted for on-premises servers if needed, using internal infrastructure. The following system supports Windows, MacOS, and Linux environments.

Dependencies: The following system requires the following dependencies:

Web Server: Apache Web Server.
PHP Environment: PHP8.1 or latest version.
PHP Dependency Manager: Composer version 2.8.8 or latest.
Database: MySQL or another Laravel-supported Database.
Frontend Development Tools and Packages: Node.js and npm.

Database Configuration

Purpose: Set up the backend database (MySQL) to manage and access the students' user account, document submissions, feedback comments, and reviewer logs.

Alternatives: The following possible alternatives may include cloud-managed database services like Amazon RDS if scaling or redundancy is necessary.

Customizations: A custom database table will be generated for document types, version control tracking, comment histories and submission categories that reflect student organization workflows.

Dependencies: It is a necessary step to ensure proper indexing for fast querying and data integrity checking mechanisms.

Security Configuration

Purpose: The configuration allows the following module to enhance security measures to protect document data, version records, the students' credentials and the system communication against unauthorized access or data breaches.

Alternatives: In addition to SSL/TLS encryption, optional features like intrusion detection systems, and Web Application Firewalls can be integrated later.

Customizations: Permission levels for document visibility and actions, audit logging for

all document activities, and role-based access control (RBAC) will be enforced. In addition to this, Data encryption at rest and in transit will be strictly implemented.

Dependencies: Mandatory use of SSL certificates (HTTPS), encryption for stored documents and user data, and compliance with the Philippine Data Privacy Act of 2012, and secure AWS configuration.

Application Integration

Purpose: The following module should enable integration with other E-Skolarian modules in the future and prepare the system to interface with potential institutional platforms like the Document Archive and Notification Module.

Alternatives: Use RESTful APIs or middleware services for integration.

Customizations: Configure document metadata exchange, automate submission and status notification, and manage version control tracking during system-by-system integration.

Dependencies: Ensure availability of API endpoints, standard JSON data formats, and consistent system versioning to avoid compatibility issues.

Testing and Validation

Purpose: The following module should be capacitated to test its reliability under different conditions before full deployment. This will be done by the Quality Assurance assigned.

Alternatives: Testing will involve the following:

Unit Testing: This will be done to test the following Laravel back-end logic.

Functional Testing: To test the Submission workflow of the module.

Integration Testing: To test the modules' ability to store documents and its User Interface.

Performance Testing: To test the modules performance under different environments.

User Acceptance Testing: To test the module with actual users and administrators.

Customizations: Custom test cases might include multiple version uploads, review comments, notifications, document resubmissions, and visual tracking of statuses.

Dependencies: Availability of a staging environment on AWS, test datasets, dedicated QA team members, and involvement of student organization representatives for User Acceptance Testing.

Documentation and Testing

Purpose: The documentation should be provided to support users in accordance to the Role-Based Access Control (RBAC).

Alternatives: Documentation delivery can be in also in a form of the following:

- User Manual
- Admin Guides
- Video Walkthroughs
- Frequency Asked Questions (FAQs)

Customizations: Specific instructions will be provided for document uploading, version updating, reviewer feedback viewing, notification management, and troubleshooting common errors.

Dependencies: Ensure materials are mobile-friendly, updated with each new system, feature or patch.

5. Other System Requirements/ Non-Functional Requirements

Requirement Type	Specification Description	Details
Performance	System Response Times	The system should respond to user requests within 2 seconds for most actions and within 5 seconds for heavy operations.

Scalability	Handling Increased Load	The system should be able to handle an increased load (more users, documents, data) without performance degradation.
System Availability	System Uptime	The system should maintain 99.9% uptime, excluding scheduled maintenance periods.
Security	Data Encryption for Sensitive User Data	All sensitive user data and information, including documents and announcements, must be encrypted to ensure data security.
Usability	User Interface Design	The dashboard interface should be intuitive and user-friendly, allowing easy navigation across both desktop and mobile platforms.
Compliance	GDPR Compliance	All data collection and processing must comply with GDPR, including obtaining user consent and adhering to data retention policies.
Data Integrity	Real-Time Data Updates	The system should ensure that document statuses, announcements, and analytics data are accurate and updated in real-time.
Availability	Minimal Downtime	The system should have minimal downtime, ensuring constant availability for document tracking and announcement viewing.

6. Reporting Requirements

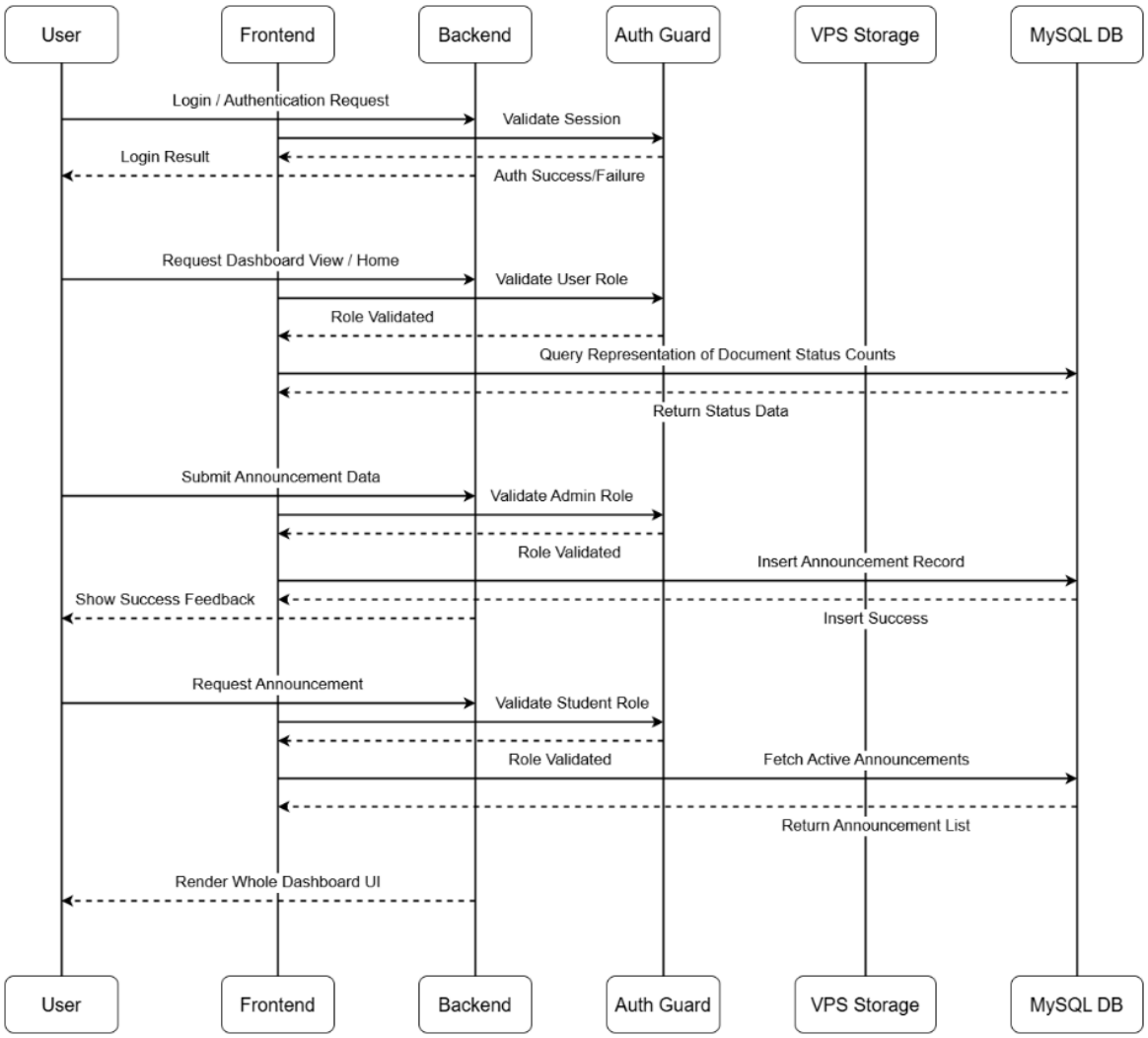
7. Integration Requirements

Purpose:
The Dashboard and Settings Module provides both student and admin users with a visual and interactive overview of document statuses, such as Pending, Under Review, Approved, and Rejected. It also facilitates the creation and management of announcements on the admin side, which are then displayed to student users. The Settings Module enables users to update their profile picture and securely change their password to maintain account security.

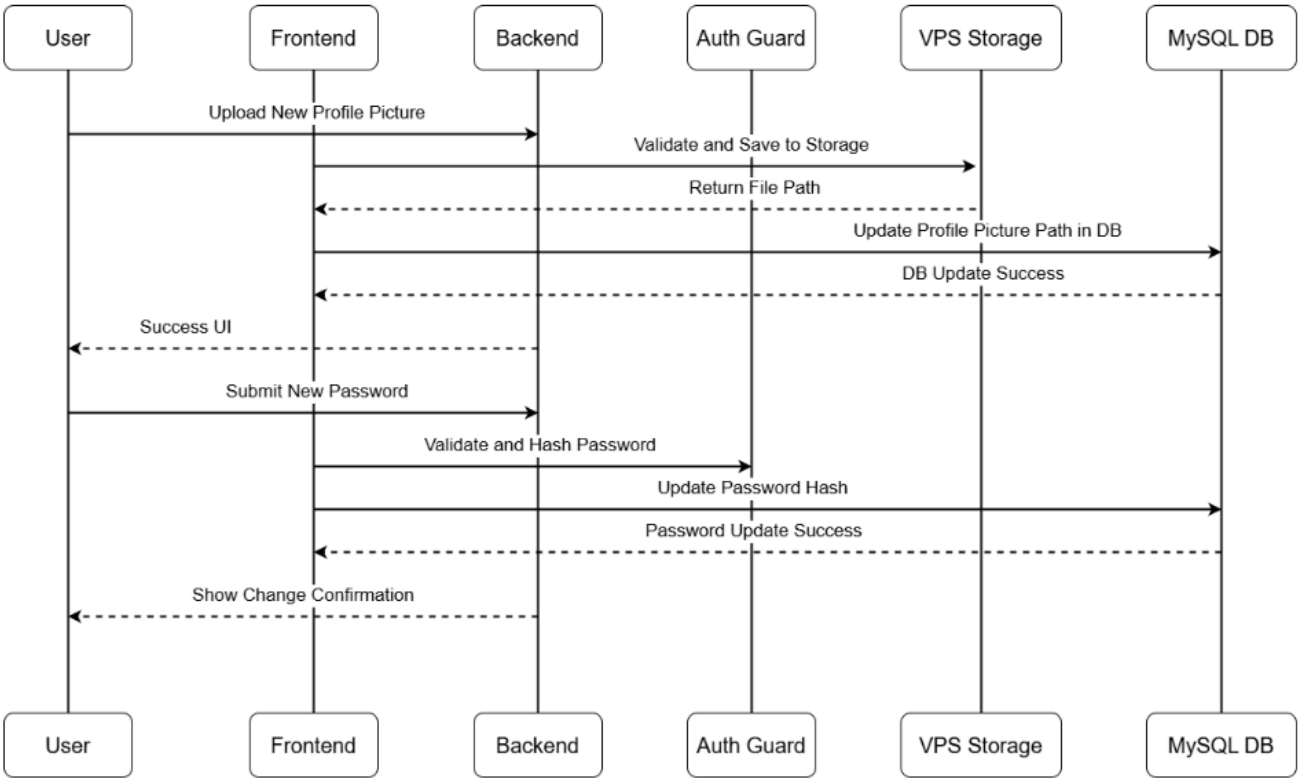
This module is integrated with the following components:

- **Virtual Private Server (VPS) Local Storage** – for storing uploaded profile pictures
- **MySQL Database** – for managing user credentials, metadata, document status data, and announcements

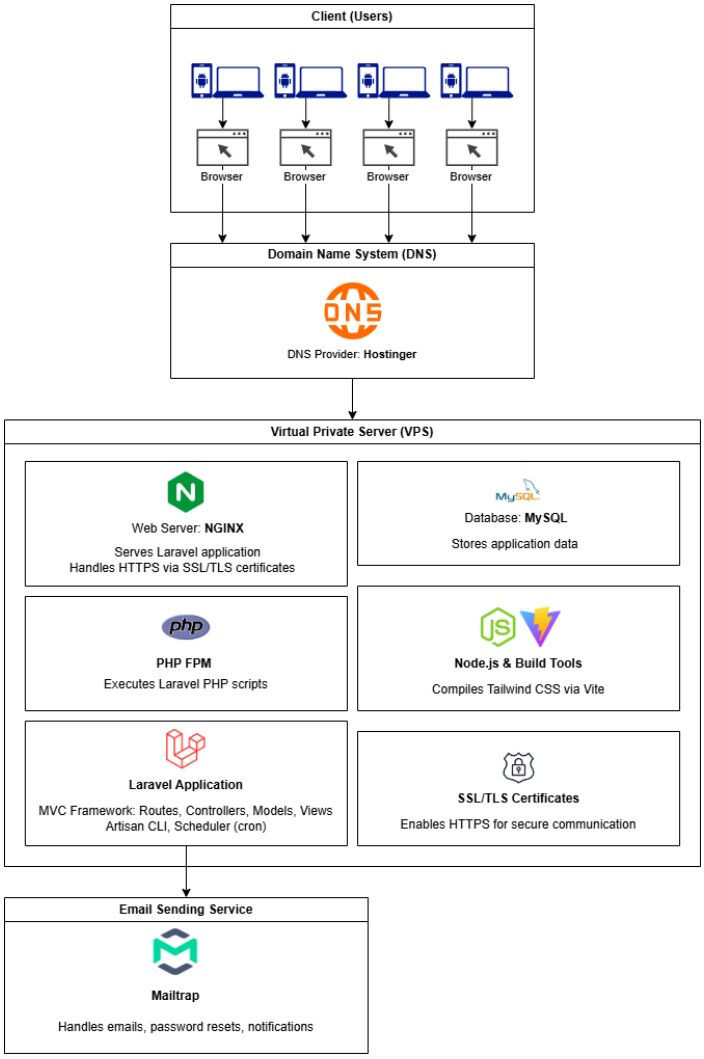
Dashboard's Sequence Diagram



Settings's Sequence Diagram



Architecture Overview Diagram



This integration requirements diagram shows the different integration needs for deploying and operating a Laravel-based web application.

- **Client Interaction:** Users access the application through web browsers on various devices. Their requests are directed through a **Domain Name System (DNS**) managed by **Hostinger**, which resolves domain names to the server's IP address.
- Server Infrastructure: The application is hosted on a **Virtual Private Server (VPS)**, which integrates several core components:
    - **NGINX Web Server:** Serves the Laravel application and manages HTTPS traffic using SSL/TLS certificates.
    - **PHP-FPM:** Executes Laravel's PHP scripts efficiently.

- **Laravel Application:** The core application following the MVC (Model-View-Controller) architecture. It also utilizes Artisan CLI and scheduled tasks (cron jobs).
- **MySQL Database:** Stores and manages all application data securely.
- **Node.js and Build Tools:** Compiles assets such as Tailwind CSS using Vite for frontend optimization.
- **SSL/TLS Certificates:** Ensures all communications are encrypted and secure via HTTPS.
- **Email Services:** For sending emails like password resets and notifications, the application integrates with **Mailtrap**, an external email testing service.

This architecture ensures a seamless, secure, and efficient environment where different systems work together to support the application's functionality and user experience.

Security and Compliance

- Encryption: AES-256-CBC (Laravel Default)
- Access Control: RBAC

7.1 Exception Handling/ Error Reporting

Exception/ Error ID	Error	Cause	Solution Strategy
ERR-001	Profile picture preview failed	Unsupported file format or missing dependencies.	Validate file format and ensure library/services are properly installed.
ERR-002	Error in database connection	Problems with the network or server undergoing maintenance.	Appropriate error messages and double checking of local environment.
ERR-003	Data fetching failed	The database might not contain the requested data, or the linked module might be changed.	Show the relevant error message or explain reason for failure.
ERR-004	Password change validation failed	Current password is invalid; new password	Show particular password requirements and

		does not match strength requirements.	underline failing criteria with examples.
ERR-005	Incorrect display of document numbers	Data synchronization delay between document operations and dashboard metrics.	Implement real-time updates or a manual refresh button.
ERR-006	Failure of form submission in “Post Announcements”	Validation errors or character limit exceeded.	Note specific field errors.

8. Data Migration/ Conversion Requirements

The Data Migration/Conversion Plan for the E-skolarian: PUPSRC Organization Document Management System (ODMS), particularly in the Dashboard and Analytics Module, supports the Polytechnic University of the Philippines Santa Rosa Campus in transitioning from a manual document submission process to a digital approach. The system will minimize the inconvenience and delays frequently found in traditional processes by acting as an organized status tracker and repository for facilitating interactions between administrators and students. To support monitoring and analytics features, this plan calls for digitizing current documents and records. This requires including the availability of necessary records for initial data entry, while risks and constraints include the necessity of user adaptation to the new hybrid system and the possibility of incomplete data during the conversion process. These will be controlled by appropriate training and validation.

8.1 Data Conversion Strategy

The data conversion strategy involves a comprehensive approach to ensure the accurate and efficient migration of existing data. The approach encompasses the extraction, transformation, and loading of data, as well as a conversion schedule and a robust test plan.

Data Extraction

Extract relevant data from existing or manual records, such as physical documents, student organization reports, and others with the help of administrative staff.

Data Transformation

Apply necessary formatting and standardization to align the extracted data with the system’s database structure, ensuring consistency, accuracy, and readiness for analytics and reporting functions.

Data Loading

Import the transformed data into the system, E-skolarian: PUPSRC Organization Document Management System (ODMS).

Phases for planning and preparation, data extraction and transformation, data loading and verification, and testing and validation should all be included in the conversion schedule. This phased strategy will guarantee a smooth transition and reduce interruptions to ongoing administrative duties. Furthermore, User Acceptance Testing (UAT) and Functional and Performance Testing should be part of the test plan for testing converted data.

8.2 Data Conversion Preparation

Several prerequisites must be met before the data conversion can begin to ensure a smooth and secure transition. These include identifying and collecting relevant documents, as well as developing a standardized data entry format. A comprehensive backup plan will be put in place before any data migration starts. This involves creating a full backup of all current data before starting the conversion procedure. While preserving the original data, these backups will serve as reference copies.

If the conversion process encounters errors or failures, a restoration procedure will be in place to return the system to its previous state. This entails restoring the backup files and resetting the system database to its pre-conversion state, ensuring that no data is lost and operations can continue uninterrupted. The team will also document and follow a rollback plan to address issues as quickly and effectively as possible.

8.3 Data Conversion Specifications

Source	Source Data Element	Target	Target Data Element	Conversion Rules	Notes
User Input					
Physical Documents	Event Proposal	E-skolarian Database	Event Proposal	Direct Mapping	Used for planning student-led activities.
Physical Documents	General Plan of Activities	E-skolarian Database	General Plan of Activities	Direct Mapping	Required for outlining planned events.
Physical Documents	Calendar of Activities	E-skolarian Database	Calendar of Activities	Direct Mapping	Details of the schedule of

					planned activities.
Physical Documents	Accomplishment Report	E-skolarian Database	Accomplishment Report	Direct Mapping	Summary of completed organizational tasks or events.
Physical Documents	Constitution and By-Laws	E-skolarian Database	Constitution and By-Laws	Categorization	Used for organization recognition.
Physical Documents	Request Letter	E-skolarian Database	Request Letter	Direct Mapping	Includes requests for facilities, support, etc.
Physical Documents	Off-Campus	E-skolarian Database	Off-Campus	Standardization	Submitted for activities held outside the campus.
Physical Documents	Petition and Concern	E-skolarian Database	Petition and Concern	Standardization	Student concerns or organizational issues.
Review Logs	Review Status	E-skolarian Database (Dashboard View)	Status(Pending, Under Review, Approved)	Direct Mapping	Document counts and analytics display.

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10. Open Issues

Issue ID	Issue	Raised By	Raised On	Solution/ Decision	Resolved By	Resolved On	Status
ISS-01	An incorrect verification code was entered, but a success message was still displayed, and no recovery email was added.	Eurika Bialen	6/12/25	It should show a display message that a verification code is sent to the email and the recovery email is reflected after verifying successfully	Jamil Carapatan	6/16/25	Passed
ISS-02	Super Admin Settings does not open	Eurika Bialen	6/12/25	The Super Admin's settings should be loaded and opens correctly.	Jamil Carapatan	6/18/25	Passed
ISS-03	A confirmation modal opens for undone announcements yet the entered data still retains	Eurika Bialen	6/17/25	The entered data in posting the announcement should not retain or save	Earl Dakila	6/18/25	Passed

Appendix

Figure	Description	Page

